

PARIKSHA365 — STUDY NOTES

Arithmetic — for SSC & RRB

Tier-1/2 focus: arithmetic core + geometry + mensuration + DI; less DI-caselet

SSC

RRB

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

EXAM ALERT**Examiner Blueprint — Arithmetic / Quantitative Aptitude****Top 8 most-tested topics (question share across SSC + Banks):**

Rank	Topic	SSC CGL T - 1	IBPS PO Pre	IBPS PO Mains
1	Percentage, Profit & Loss, Discount	5–8	3–5	5–8
2	Ratio, Proportion & Partnership	3–5	2–4	3–5
3	Time, Work & Pipes	3–5	2–4	3–5
4	Simple & Compound Interest	3–5	3–5	3–5
5	Data Interpretation (Tables, Bar, Line, Pie)	2–3	5–10	15–20
6	Speed-Distance-Time (Trains, Boats)	3–5	2–4	3–5
7	Number System (HCF/LCM, Divisibility, Remainders)	3–5	2–3	2–3
8	Quadratic Equations	0–1	5	5

KEY FACT**Study Priority Order** (return-on-study-time, highest first)

1. **Percentage** — every other topic feeds into it. Master this first; it unlocks Profit-Loss, SI/CI, and DI simultaneously.
2. **Ratio & Proportion** — 2 hours → feeds directly into Partnership, Mixture-Alligation, and speed problems.
3. **SI & CI** — formulas are short; 2 hours of drill makes these 3–5 marks nearly automatic.
4. **Profit-Loss-Discount** — 2 hours once Percentage is solid; examiners love multi-step marked-price chains.
5. **Time-Work-Pipes** — unit-fraction method handles all variants in 10 seconds.
6. **Speed-Distance-Time** — trains, boats, and relative-speed variants all use one framework.
7. **Data Interpretation** — non-negotiable for Banks; 5 hours to master table + bar + pie + line combinations.
8. **Number System** — HCF/LCM + divisibility rules + remainder theorems; mandatory for SSC CGL Tier-1.

How to use this book

This book is different from the static-GK books.

- Here **every topic is broken into a focused set of TYPES**.
- Most topics settle at **8–15 types**.
- Some (Geometry, Input-Output, Puzzles) go higher — **there is no artificial cap**.
- What matters is that **each type unlocks a *distinct* pattern or method**.
- Almost every quantitative question you'll ever see in SSC / Banking / RRB / PSU / State-PSC exams is a **tweak of one of these types**.
- **If you can recognise the type in 5 seconds, you are already halfway to the answer.**

For every TYPE you'll see:

1. **What it looks like** — the recognition cue. Read the stem and ask: "which type is this?"
2. **Core insight / formula** — the single equation or idea that drives it.
3. **Smart / shortcut method FIRST** — solve in your head in under 20 seconds, **with a full derivation** so a twist will not break you.
4. **Traditional method SECOND** — the safe fallback when the shortcut's precondition is not met.
5. **Variations** — 2–3 important tweaks that examiners love.
6. **Self-check** — one mini question to prove you've got the type.

Before each topic you'll find an "Opener" box with:

- **Why the topic matters** — where you meet it in exams.
- **Mental-math arsenal** — the tables / constants you **MUST** know cold (no pen-paper).
- **Memorised formulas** — the 3–6 equations that run the topic.
- **Generic attack plan** — what to identify first, what to compute mentally, when to switch methods.

***Golden rule of arithmetic prep:** the student who knows 10 types cold **beats** the student who has solved 1,000 questions without classifying them.*

How to read this book — your real study load

Total pages: 151.

This is a specialised exam-prep book. **Every page is essential** — including the worked-example drills, mini-mock and trap-cards near the end. Skipping any section costs you marks.

Read end-to-end in order without skipping.

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PART 0 — THE MENTAL-MATH ARSENAL

You cannot be fast at arithmetic without *cached* values in your head. Before any topic, master this core.

0.1 Fraction ↔ Percentage ↔ Decimal table (MUST memorise)

Fraction	Percentage	Decimal
1/1	100 %	1.000
$\frac{1}{2}$	50 %	0.500
$\frac{1}{3}$	33.33 %	0.333...
$\frac{1}{4}$	25 %	0.250
$\frac{1}{5}$	20 %	0.200
$\frac{1}{6}$	16.67 %	0.166...
$\frac{1}{7}$	14.28 %	0.142857...
$\frac{1}{8}$	12.5 %	0.125
$\frac{1}{9}$	11.11 %	0.111...
$\frac{1}{10}$	10 %	0.100
$\frac{1}{11}$	9.09 %	0.0909...
$\frac{1}{12}$	8.33 %	0.0833...
$\frac{1}{13}$	7.69 %	0.0769...
$\frac{1}{14}$	7.14 %	0.0714...
$\frac{1}{15}$	6.66 %	0.0667...
$\frac{1}{16}$	6.25 %	0.0625
$\frac{1}{20}$	5 %	0.050
$\frac{1}{25}$	4 %	0.040
$\frac{1}{40}$	2.5 %	0.025
$\frac{1}{50}$	2 %	0.020

Also the multiples: $\frac{3}{8} = 37.5 \%$, $\frac{5}{8} = 62.5 \%$, $\frac{2}{3} = 66.67 \%$, $\frac{3}{7} = 42.85 \%$, $\frac{4}{7} = 57.14 \%$, $\frac{5}{11} = 45.45 \%$, etc.

0.2 Squares, cubes, roots

Squares of 1-30 (NON-NEGOTIABLE), bonus 31-50.

Cubes of 1-15 (NON-NEGOTIABLE), bonus 16-20.

$\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$, $\sqrt{7} \approx 2.645$, $\sqrt{10} \approx 3.162$.

$\pi \approx 3.1416$, $\pi^2 \approx 9.87$, $e \approx 2.718$.

0.3 Tables 11-20 (yes, all of them)

If you still compute 17×14 by column multiplication, you are losing 15 seconds per question. Drill these.

0.4 Universal shortcuts — S-1 through S-15

These 15 shortcuts each cover an entire **model** (a class of question types), not a single problem. Learn all 15 cold. A worked preview appears here; full application appears at the end of each relevant chapter.

S-1 — The 10%/1%/5% Trio (*%*, *P&L*, *SI/CI*, *Discount*, *T&W wages*, *TSD*)

Move the decimal: 10% → one place left. 1% → two places left. 5% = half of 10%. All other common percentages decompose: 17% = 10% + 5% + 2×1%.

17% of 742 = 74.2 + 37.1 + 14.84 = **126.14** (no pen needed).

S-2 — Net Change Formula: $A + B + \frac{AB}{100}$ (*Successive %*, *CI*, *successive discounts*, *population growth*)

When a value changes by A% then B%, the net % change = $A + B + \frac{AB}{100}$.

Price rises 10% then falls 10%: net = $10 + (-10) + \frac{10 \times (-10)}{100} = -1\%$. **Net loss of 1%.** Two successive discounts of 20% and 25%: net discount = $20 + 25 + \frac{20 \times 25}{100} = 40\%$. **Equivalent single discount = 40%.**

S-3 — LCM = Total Work (*Time & Work*, *Pipes & Cisterns*, *Efficiency*)

Take LCM of all given time values. That is the total work. Each person/pipe's rate = LCM ÷ their time.

A in 12 days, B in 15 days → LCM = 60 units. A's rate = 5/day, B's rate = 4/day. Together = 9/day → $\frac{60}{9} = 6\frac{2}{3}$ days.

S-4 — Component Ratio (*Partnership*, *Mixture*, *Ratio & Proportion*)

Share of first = $\frac{a}{a+b} \times \text{Total}$, where a:b is the ratio.

Profit split 3:5, total ₹640 → First gets $\frac{3}{8} \times 640 = \text{₹}240$.

S-5 — Alligation Cross (*Mixtures & Alligation*, *Average Speed*, *Blending*)

To blend two quantities of costs c_1 and c_2 to get mean price m : Ratio = $(c_2 - m) : (m - c_1)$.

Mix milk at ₹30/L with water (₹0) to get average ₹20/L: Ratio water:milk = $(30-20) : (20-0) = 10 : 20 = 1:2$.
Per 3L mixture: 1L water + 2L milk.

S-6 — Speed–Time Inverse (fixed distance) (TSD, Trains, Boats, Races)

When distance is fixed: $\text{Time} \propto \frac{1}{\text{Speed}}$, so $\frac{T_1}{T_2} = \frac{S_2}{S_1}$.

A drives at 60 km/h and takes 4 h. At 80 km/h it takes $4 \times \frac{60}{80} = 3$ h.

S-7 — Fraction–Percent Table (memorise cold) (ALL percentage topics)

Fraction	%	Fraction	%
$\frac{1}{2}$	50%	$\frac{1}{9}$	11.11%
$\frac{1}{3}$	33.33%	$\frac{1}{10}$	10%
$\frac{1}{4}$	25%	$\frac{1}{11}$	9.09%
$\frac{1}{5}$	20%	$\frac{1}{12}$	8.33%
$\frac{1}{6}$	16.67%	$\frac{2}{3}$	66.67%
$\frac{1}{7}$	14.28%	$\frac{3}{4}$	75%
$\frac{1}{8}$	12.5%	$\frac{4}{5}$	80%

Rule: When you see a %, instantly convert to the fraction above. $\frac{1}{8}$ is far faster to multiply than 12.5%.

S-8 — Digit–Sum Check (Answer verification for any arithmetic)

Cast out 9s: replace every number with its digit-sum, perform the same operation, check against the candidate answer's digit-sum. Wrong answer → digit-sum won't match.

*$37 \times 48 = ?$ Digit-sums: $10 \rightarrow 1$, $12 \rightarrow 3$. Product digit-sum = $1 \times 3 = 3$. Answer 1776: $1+7+7+6 = 21 \rightarrow 3$.
Matches — likely correct.*

S-9 — Base–Deviation Multiplication (Squaring, large products near 100 or 50)

For numbers near base B: $(B+a)(B+b) = B(B+a+b) + ab$.

$97 \times 103 = 100 \times (97+103-100) + (-3)(3) = 100 \times 100 - 9 = 9991$. $48^2 = (50-2)^2 = 2500 - 200 + 4 = 2304$.

S-10 — Reverse Percentage (Finding original price before markup or discount)

"After X% increase, value is V. Find original." → $\text{Original} = \frac{V \times 100}{100 + X}$. Never multiply V by $(100-X)\%$ — that gives the wrong answer.

After 20% rise, price = ₹840. Do NOT compute 840×0.80 . Original = $\frac{840 \times 100}{120} = \frac{84000}{120} = \text{₹700}$.

S-11 — Set Formula (Venn diagrams, survey problems, Probability)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Three-set: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.

100 students: 60 like Cricket, 50 like Football, 30 like both. Neither = $100 - (60 + 50 - 30) = 20$.

S-12 — Percent-Change Symmetry: a% of b = b% of a (Any % computation)

Swap to whichever side is easier to compute.

32% of 50 = 50% of 32 = 16. 4% of 75 = 75% of 4 = 3. No long multiplication needed.

S-13 — Relative Speed (Trains, Boats, Races)

Direction	Relative Speed
Moving towards each other	$S_1 + S_2$
Moving in same direction	$S_1 - S_2$ (faster – slower)
Boat upstream	Boat speed – Current speed
Boat downstream	Boat speed + Current speed

Time to meet/cross = $\frac{\text{Total distance}}{\text{Relative speed}}$.

Train A (100 m) and Train B (150 m) approach each other at 60 km/h and 90 km/h. Relative speed = 150 km/h = $\frac{150}{3.6}$ m/s = $\frac{125}{3}$ m/s. Distance = 250 m. Time = $\frac{250 \times 3}{125} = 6$ seconds.

S-14 — Average Speed for Equal Distances (TSD return trips, two-leg journeys)

Average speed = $\frac{2ab}{a+b}$, where a and b are the two speeds.

Car goes at 40 km/h, returns at 60 km/h. Average = $\frac{2 \times 40 \times 60}{40 + 60} = \frac{4800}{100} = 48$ km/h (not 50 — the arithmetic mean is always wrong here).

S-15 — Partnership Profit Ratio (Partnership, Investment problems)

Profit ratio = $P_1 T_1 : P_2 T_2$ (capital × time).

A invests ₹6,000 for 5 months; B invests ₹4,000 for 9 months. Ratio = 30,000 : 36,000 = 5:6. Total profit ₹22,000 → A gets $\frac{5}{11} \times 22,000 = \text{₹}10,000$.

0.5 Generic problem-solving flow

1. **Read once — label the type.** Don't start solving yet.
2. **Identify the one equation / ratio / insight** the type uses.
3. **Check if values suit the shortcut** (round numbers, known fractions, symmetric setups). If yes, shortcut.
4. **Else traditional method**, but compute mentally where possible (step 0 onwards).
5. **Sanity-check the sign / order of magnitude.** A 120% increase cannot land below the original.

PART 0B — THE SPEED TOOLKIT: SOLVE WITHOUT SOLVING

*In a 60-minute exam with 25 arithmetic questions, you have **144 seconds per question** on average — but some questions will take 3 minutes and others should take 10 seconds. This chapter teaches you how to find the 10-second path first, every time, before committing to the 3-minute path.*

The examiner designs questions that have a long route and a short route. The student who scores 90%+ is not faster at arithmetic — they have trained themselves to see the short route in 3 seconds.

THE 3-SECOND PRE-SCAN (do this before writing a single digit)

When you see a new question, your pencil should be **still** for 3 seconds while you:

1. **Glance at the options.** Are unit digits all different? Are they widely spaced or close? Is one a multiple of 11 when others aren't? Can one be eliminated by sign or magnitude?
2. **Glance at the formula structure.** Does it involve π ? A square root? A power? Does the question ask for "unit digit" or "remainder" — if yes, full computation is never needed.
3. **Ask: Can I plug options backward?** If the answer is a simple number, plugging it into the condition is often faster than solving.

Only after this 3-second scan, choose your path. The ten techniques below give you the path options.

TECHNIQUE 1 — Unit Digit Elimination

What it is: The last digit (unit digit) of a product or power depends **ONLY** on the unit digits of the inputs. You can find the unit digit of $3^{78} \times 7^{43}$ without computing anything beyond the last-digit cycles.

When to use: Any question whose answer involves large powers, large products, or repeated multiplication — and where you can narrow down to one option by unit digit alone.

When NOT to use: When multiple options share the same unit digit. Also when the question asks for the full value (not just unit digit) — the technique gives you unit digit only.

KEY FACT		
Unit digit cycles — memorise cold:		
Last digit of base	Cycle of unit digits	Rule
0	0	Always 0
1	1	Always 1
5	5	Always 5
6	6	Always 6
4	4, 6, 4, 6 ...	Even power → 6 ; Odd power → 4
9	9, 1, 9, 1 ...	Odd power → 9 ; Even power → 1
2	2, 4, 8, 6	Cycle length 4; use (power mod 4) as position
3	3, 9, 7, 1	Cycle length 4; use (power mod 4) as position
7	7, 9, 3, 1	Cycle length 4; use (power mod 4) as position
8	8, 4, 2, 6	Cycle length 4; use (power mod 4) as position

Remainder → position: remainder 1 → 1st element, 2 → 2nd, 3 → 3rd, 0 → 4th (last).

Worked example. Find the unit digit of $2^{84} + 3^{65}$.

2^{84} : cycle {2,4,8,6} length 4. $84 \bmod 4 = 0 \rightarrow$ take last in cycle = **6**.

3^{65} : cycle {3,9,7,1} length 4. $65 \bmod 4 = 1 \rightarrow$ take 1st = **3**.

Sum unit digit: $6 + 3 = 9$. Options with unit digit $\neq 9$ are eliminated immediately.

If options were 156, 213, 249, 284 → only 249 ends in 9 → answer without computing.

TECHNIQUE 2 — Divisibility Elimination (including the $\pi = 22/7$ trick)

What it is: The answer to many geometry problems must be divisible by specific numbers because of the formula structure. You can eliminate options before computing.

When to use: Formula involves $\pi = \frac{22}{7}$: The answer will contain 22 in the numerator → divisible by both **2 and 11**.

If only one option is divisible by 11 and it passes the magnitude check, that option is almost certainly correct.

Formula involves 3 in denominator (cone, pyramid): answer divisible by specific factors after cancellation.

Formula involves $\sqrt{3}$: Answer is irrational.

If all options are integers except one that contains $\sqrt{3}$, that one is correct.

KEY FACT

The $\pi = \frac{22}{7}$ rule:

- Cylinder/Cone/Sphere volumes and surface areas use $\pi = \frac{22}{7}$ in exams.
- Volume of cone = $(\frac{1}{3})(\frac{22}{7})r^2h = \mathbf{22r^2h / 21}$ — the 22 in the numerator means the answer is always divisible by **11** (and by 2) when dimensions are integers.
- Hemisphere TSA = $3\pi r^2 = 3(\frac{22}{7})r^2 = \mathbf{66r^2/7}$ — since $66 = 6 \times 11$, the answer is again divisible by 11.
- **Before computing:** scan the options and eliminate any not divisible by 11. Often only one option remains.

Worked example. A cone has $r = 7$ cm, $h = 6$ cm. Volume = ?

Options:

(a) 280 cm^3 (b) 308 cm^3 (c) 294 cm^3 (d) 315 cm^3

Fast path:

$$\text{Formula} = \left(\frac{22}{21}\right) \times r^2h$$

$$= \left(\frac{22}{21}\right) \times 49 \times 6$$

$$= 22 \times 14 = \mathbf{308}$$

Even faster — divisibility by 11 check:

$308 \div 11 = 28 \checkmark \rightarrow$ option (b) passes

$280 \div 11 = 25.45 \times$

$294 \div 11 = 26.7 \times$

$315 \div 11 = 28.6 \times$

Only (b) is divisible by 11 $\rightarrow$ answer = **(b) 308** in 5 seconds

When NOT to use: When r or h are given as decimals or fractions that don't simplify cleanly. Also when multiple options are divisible by 11 — you'll need to compute the magnitude too.

TECHNIQUE 3 — Magnitude Estimation (the "ballpark" method)

What it is: Round every number to 1 significant figure, compute the rough answer in your head, and eliminate options that are clearly too large or too small.

When to use: Options are spread across different orders of magnitude (e.g., 200, 2000, 20000). One rough calculation kills 3 of 4 options.

When NOT to use: Options are clustered close together (e.g., 1218, 1224, 1230, 1236). A rough calculation will not distinguish them — compute exactly.

Worked example. 3.97% of 2498 + $7.04 \times 39.6 = ?$

Options:

(a) 178 (b) 378 (c) 578 (d) 778

Rough estimate:

$3.97\% \approx 4\%$; $2498 \approx 2500$

4% of 2500 = **100**

$7.04 \approx 7$; $39.6 \approx 40$

$7 \times 40 =$ **280**

Total $\approx 100 + 280 = 380 \rightarrow$ closest option = **(b) 378**

The exam calls this type "Approximation" and always gives options far enough apart that rounding to 1-2% is safe.

TECHNIQUE 4 — Backward Substitution (plug options in)

What it is: Instead of solving the equation algebraically, plug each option into the condition and check which one satisfies it.

When to use:

Quadratic equations (especially when factorisation is hard).

Age problems, profit-loss problems where the unknowns are one of the options.

"Find the number such that..." questions.

When NOT to use:

Options are expressions (like " $2x + 3$ "), not numbers.

The "check" itself takes longer than solving the equation.

The question has infinite solutions — "which of these satisfies" needs every option tested, costing time.

Worked example. $x^2 - 7x + 12 = 0$. Find x . Options:

- (a) 2 (b) 3 (c) 5 (d) 6

Traditional:

Factorise: $(x-3)(x-4) = 0$

Roots: $x = 3$ or $x = 4$

4 is not in the options → **$x = 3$**

Faster — backward substitution:

Plug (b) $x = 3$: $9 - 21 + 12 = 0$ ✓ → **(b) 3** in 4 seconds

Worked example (profit/loss). A shopkeeper sells an item at 20% profit. If he had sold it for ₹12 more, profit would be 30%. Find the cost price.

Options:

- (a) ₹60 (b) ₹80 (c) ₹100 (d) ₹120

Try (d) CP = ₹120:

SP at 20% profit = $120 + 24 = 144$

SP at 30% profit = $120 + 36 = 156$

Difference = $156 - 144 = 12$ ✓ → **CP = ₹120 = (d)**

Try (c) CP = ₹100 (to confirm it fails):

SP at 20% = 120; SP at 30% = 130

Difference = $10 \neq 12$ ✗

TECHNIQUE 5 — Assume Base = 100 (for percentages and ratios)

What it is: In any percentage or profit-loss problem, assume the reference quantity = 100 (or any convenient round number). This turns "express as percentage of original" into pure arithmetic, no variables.

When to use: Percentage change, ratio comparisons, successive change problems, profit/loss/discount chains. Almost always faster than using variables.

When NOT to use: The problem has a specific absolute value that must be preserved (e.g., "find the actual CP if SP = ₹240 and profit = 20%") — base is already fixed by the numbers given.

Worked example. A's salary is 25% more than B's. By what % is B's salary less than A's?

Variable method:

Let $B = x$, then $A = 1.25x$

Difference = $A - B = 1.25x - x = 0.25x$

% less = $\text{Difference} / A \times 100 = 0.25x / 1.25x \times 100$

= $(0.25/1.25) \times 100 = 0.2 \times 100 = 20\%$

Base 100 method (faster):

Let B = 100, then A = 125

Difference = A - B = 125 - 100 = 25

% less = Difference / A × 100 = $\frac{25}{125} \times 100$

= $(\frac{1}{5}) \times 100 = 20\%$

Worked example. A price rises 20%, then falls 20%. Net effect?

Base 100:

Start = 100

After +20 %: $100 \times 1.20 = 120$

After -20 %: $120 \times 0.80 = 96$

Net change = $(96 - 100) / 100 \times 100 = -4\%$

Formula verification: $a + b + ab/100 = 20 + (-20) + (20 \times (-20)) / 100$

= $0 + (-\frac{400}{100})$

= $0 - 4 = -4\% \checkmark$

TECHNIQUE 6 — LCM = Total Work (for Time-Work and Pipes)

What it is: Instead of using fractions (A does $1/a$ of work per day), set total work = LCM of all time values. Then each person's rate per day is an integer. No fractions, no cross-multiplication.

When to use: Every single time-work or pipes problem. This is a universal substitution for this topic — not a special trick.

When NOT to use: Basically always use it. The only exception: if the problem gives fractional portions done ("A completes $\frac{2}{5}$ of a job in 4 days") — then work directly with the fraction.

Worked example. A completes a job in 12 days, B in 18 days. Together, how long?

Fraction method:

A's rate = $\frac{1}{12}$ of the work per day

B's rate = $\frac{1}{18}$ of the work per day

Combined rate = $\frac{1}{12} + \frac{1}{18}$

= $\frac{3}{36} + \frac{2}{36}$ (LCM of 12, 18 = 36) = $\frac{5}{36}$ Time together = $1 / (\frac{5}{36}) = \frac{36}{5} = 7.2 \text{ days}$

LCM method (no fractions):

Total work = LCM(12, 18) = **36 units**

A's rate = $\frac{36}{12}$ = **3 units/day**

B's rate = $\frac{36}{18}$ = **2 units/day**

Combined rate = 5 units/day

Time = $\frac{36}{5}$ = **7.2 days**

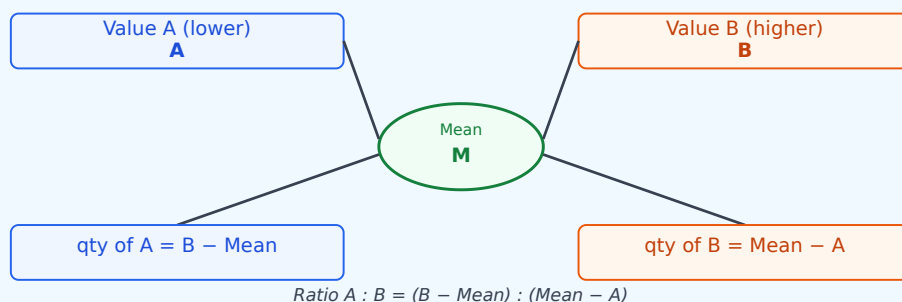
When it saves the most time: Three people, one joins midway. LCM turns $1/a + 1/b + 1/c$ into integer rates; the partial-work calculation becomes subtraction of integers.

TECHNIQUE 7 — The Alligation Cross (the most powerful 2-line shortcut in arithmetic)

What it is: A visual shortcut that finds the ratio in which two components (of known values) must be mixed to achieve a desired average value. Works for: mixture prices, concentrations, average speeds, ratio blending, dilution problems.

KEY FACT

The Alligation Cross:



Ratio A : B = (B - Mean) : (Mean - A)

(The diagonal arrows remind you: each cross-difference belongs to the OPPOSITE side.)

When to use: Any problem mixing two things where you know the individual values and the target mixture value. Average speed problems are a hidden alligation (two speeds → find ratio of distances OR times).

When NOT to use: Three-component mixing (solve in two passes: fix one pair first). When components are not mixed in unknown ratio (ratio is given; you're solving for something else).

Worked example (mixture). A 20-litre solution of 40% alcohol is mixed with a 50-litre solution of 80% alcohol. Find the alcohol % in the mixture.

Step 1 — Find alcohol in each solution. Alcohol in Solution 1 = 40% of 20 litres = $(\frac{40}{100}) \times 20 = 8$ litres

Alcohol in Solution 2 = 80% of 50 litres = $(\frac{80}{100}) \times 50 = 40$ litres

Step 2 — Find total alcohol. Total alcohol = 8 + 40 = **48 litres**

Step 3 — Find total volume of mixture. Total volume = 20 + 50 = **70 litres**

Step 4 — Calculate alcohol % in mixture. Alcohol % = (Total alcohol / Total volume) $\times$ 100

$$= (\frac{48}{70}) \times 100$$

$$= \mathbf{68.57 \%}$$

Alligation verification (cross-check):

Given: Component A = 40 %, Component B = 80 %, Mixture mean = 68.57 %

Ratio formula: (B – Mean) : (Mean – A)

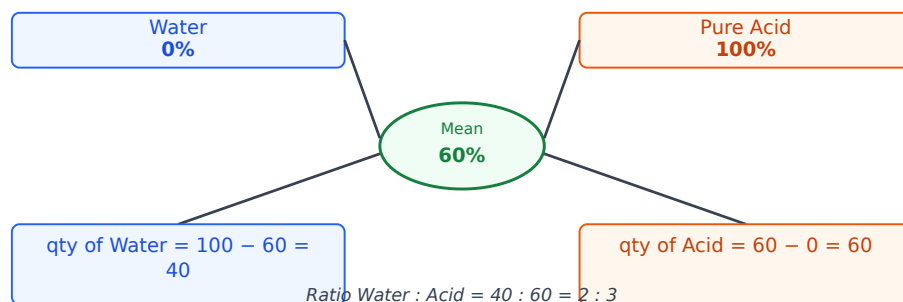
$$= (80 - 68.57) : (68.57 - 40)$$

$$= 11.43 : 28.57$$

$$= \mathbf{2 : 5}$$

Volume ratio given = 20 : 50 = **2 : 5** ✓ (verified — consistent with alligation)

Worked example (find the ratio to achieve a target). Pure acid (100%) is mixed with water (0%) to make 60% acid. Ratio acid:water?



Ratio water : acid = 40 : 60 = 2 : 3

Flip to get acid : water = **3 : 2** ✓

Worked example (average speed, the hidden alligation). A car travels 60 km at 40 km/h and 90 km at 60 km/h. Average speed?

Traditional:

$$\text{Time}_1 = \frac{60}{40} = \mathbf{1.5 \text{ h}}$$

$$\text{Time}_2 = \frac{90}{60} = \mathbf{1.5 \text{ h}}$$

Total distance = 150 km; total time = 3 h

$$\text{Average speed} = \frac{150}{3} = \mathbf{50 \text{ km/h}}$$

Weighted average (equal times, so simple average applies):

$$\begin{aligned}\text{Average speed} &= (S_1 \times T_1 + S_2 \times T_2) / (T_1 + T_2) \\ &= (40 \times 1.5 + 60 \times 1.5) / (1.5 + 1.5) \\ &= (60 + 90) / 3 \\ &= \frac{150}{3} = \mathbf{50 \text{ km/h}} \checkmark\end{aligned}$$

TECHNIQUE 8 — Trigonometric Substitution ($\theta = 45^\circ, 30^\circ$, or 60°)

What it is: Exam questions often give a trigonometric expression and ask "which of these equals it?" or "simplify this." Substituting $\theta = 45^\circ$ makes $\sin \theta = \cos \theta = 1/\sqrt{2}$ and $\tan \theta = 1$ — this simplifies the expression to a number. Match to whichever option also gives that number.

When to use: Identity verification questions ("which option equals $\sin \theta / (1 + \cos \theta)$?"). Expression comparison ("which is equivalent to...?"). Choosing the correct simplification.

When NOT to use: 1. When the question asks you to PROVE an identity (substitution checks a specific case, not a universal proof — a wrong identity can still hold at $\theta = 45^\circ$). Use algebraic steps for proofs. 2. When two or more options give the same value at $\theta = 45^\circ \rightarrow$ also try $\theta = 30^\circ$ or 60° to separate them. 3. When the expression is undefined at $\theta = 45^\circ$ (e.g., involves $\tan 90^\circ$ or $\sec 90^\circ$) — choose a different angle.

Worked example. Which expression equals $\sin \theta / (1 - \cos \theta)$? Options:

- (a) $\operatorname{cosec} \theta + \cot \theta$ (b) $\operatorname{cosec} \theta - \cot \theta$
(c) $\sec \theta + 1$ (d) $\cot \theta + \cos \theta$

Substituting $\theta = 45^\circ$:

$$\begin{aligned}\text{LHS} &= (1/\sqrt{2}) / (1 - 1/\sqrt{2}) \\ \text{Rewrite denominator: } 1 - 1/\sqrt{2} &= (\sqrt{2} - 1)/\sqrt{2} \\ \text{LHS} &= (1/\sqrt{2}) \div ((\sqrt{2} - 1)/\sqrt{2}) = 1 / (\sqrt{2} - 1) \\ \text{Rationalise: } \times (\sqrt{2} + 1)/(\sqrt{2} + 1) &= (\sqrt{2} + 1) / (2 - 1) = \mathbf{\sqrt{2} + 1} \\ \text{Check (a): } \operatorname{cosec} 45^\circ + \cot 45^\circ &= \sqrt{2} + 1 \checkmark \rightarrow \text{matches} \\ \text{No need to check others. Answer} &= \mathbf{(a)}\end{aligned}$$

Algebraic verification (when you have time):

$$\begin{aligned}\text{Multiply numerator and denominator by } (1 + \cos \theta) & \\ &= \sin \theta(1 + \cos \theta) / (1 - \cos^2 \theta) \\ &= \sin \theta(1 + \cos \theta) / \sin^2 \theta \\ &= (1 + \cos \theta) / \sin \theta \\ &= 1/\sin \theta + \cos \theta/\sin \theta = \operatorname{cosec} \theta + \cot \theta \checkmark\end{aligned}$$

COMMON TRAP

Common trap: A question says "simplify $(\sin \theta + \cos \theta)^2$." At $\theta = 45^\circ$, this $= (1/\sqrt{2} + 1/\sqrt{2})^2 = (\sqrt{2})^2 = 2$.

Options:

(a) $1 + 2\sin\theta\cos\theta$

(b) $1 + \sin 2\theta$

(c) $\sin 2\theta + 1$

(d) both (a) and (b). At $\theta = 45^\circ$: (a) $= 1 + 2(\frac{1}{2}) = 2$ ✓; (b) $= 1 + 1 = 2$ ✓. Both match → you cannot distinguish by substitution alone. **The answer is (d).** Lesson: when multiple options match at one angle, try a second angle, or use algebra.

TECHNIQUE 9 — Euler's Totient $\phi(n)$ (co-prime counting)

What it is: $\phi(n)$ = count of integers from 1 to n that are co-prime to n (HCF = 1).

Formula: $\phi(n) = n \times (1 - 1/p_1) \times (1 - 1/p_2) \times \dots$ for each distinct prime factor p of n .

When to use: Questions that ask "how many numbers from 1 to n are co-prime to n ?" or "how many reduced fractions with denominator n exist?" SSC CGL Tier-2, Mathematics optional papers.

When NOT to use: If the question asks "how many are NOT co-prime" — compute $n - 1 - \phi(n)$ (excluding 1 which is always co-prime, and n itself which shares factor n).

Worked example. How many numbers from 1 to 36 are co-prime to 36?

$36 = 2^2 \times 3^2$; distinct prime factors: 2 and 3

$$\begin{aligned}\phi(36) &= 36 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \\ &= 36 \times \frac{1}{2} \times \frac{2}{3} \\ &= 18 \times \frac{2}{3} = 12\end{aligned}$$

Those 12 numbers: 1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35

Worked example. $n = 30 = 2 \times 3 \times 5$. Find $\phi(30)$.

$$\begin{aligned}\phi(30) &= 30 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times (1 - \frac{1}{5}) \\ &= 30 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} \\ &= 15 \times \frac{2}{3} \times \frac{4}{5} \\ &= 10 \times \frac{4}{5} = 8\end{aligned}$$

TECHNIQUE 10 — Casting Out Nines (verify your answer instantly)

What it is: The digital root of a correct calculation must be consistent. The digital root (sum digits until 1 digit) of $A + B$ must equal the digital root of the answer. Catches arithmetic errors in 2 seconds.

When to use: After you compute a long multiplication or addition to verify you haven't made an error.

When NOT to use: Subtraction checks are trickier. Catches only errors that change the digital root — a mistake that moves units digit from 6 to 9 (same digital root mod 9) would pass.

Worked example. $237 \times 43 = 10,191$. Verify.

Digital root of 237: $2 + 3 + 7 = 12$; $1 + 2 = 3$

Digital root of 43: $4 + 3 = 7$

Product digital root: $3 \times 7 = 21$; $2 + 1 = 3$

Digital root of 10,191: $1 + 0 + 1 + 9 + 1 = 12$; $1 + 2 = 3$ ✓

Both sides give 3 → calculation is consistent

ROUGH WORK DISCIPLINE — what to write and what to skip

In a real exam hall, candidates who score 90%+ write **less** rough work than average scorers — not more. They write *decisions*, not computations. Here is the rule:

KEY FACT

Write on rough paper only: ✓ The LCM you chose. ✓ The base-100 assumption. ✓ The alligation cross. ✓ Final step arithmetic.

Never write on rough paper: ✗ Intermediate digit-by-digit multiplications when the unit digit technique can eliminate 3 options first. ✗ Full CI calculation when the 2-year shortcut applies. ✗ Long division when magnitude estimation already narrows to 1 option.

The unit digit pre-check (1 second before any big multiplication):

When a question has four options with *different* unit digits, and your computation involves a multiplication, do the unit digit check FIRST:

Example. Find the value of 123×47 . Options:

(a) 5,733 (b) 5,781 (c) 5,846 (d) 5,922

Unit digit check: $3 \times 7 = 21 \rightarrow$ unit digit = 1. Only option (b) ends in 1 → **answer is (b) 5,781** without any further computation.

EXAM ALERT

Three-step rough-work protocol (do this every question): 1. **Unit digit check** — can I kill 3 options in 2 seconds? If yes, done. 2. **Magnitude estimate** — are remaining options spread apart? If yes, rough calculation kills another. 3. **Exact computation** — only if needed; write the minimum steps to reach the answer.

Most exam candidates skip steps 1 and 2 and go straight to step 3 every time — then run out of time. Train steps 1 and 2 as reflexes.

THE MASTER DECISION TREE

Question arrives

- |— Does it ask for UNIT DIGIT or REMAINDER?
 - |→ Use Technique 1 (unit digit) or remainder cyclicity
- |— Does it involve $\pi = 22/7$? (circle, cone, cylinder, sphere)
 - |→ Check options: eliminate those not divisible by 11
 - |→ Technique 2
- |— Are options far apart ($10\times$ ratio between smallest and largest)?
 - |→ Rough estimate + Technique 3
- |— Are options small numbers and question has one unknown?
 - |→ Try plugging options backwards (Technique 4)
- |— Does it say "x% of something" or "profit/loss/discount"?
 - |→ Assume base = 100 (Technique 5)
- |— Does it say "A does job in m days, B in n days"?
 - |→ LCM = total work (Technique 6)
- |— Does it mix two things to get an average/target value?
 - |→ Alligation cross (Technique 7)
- |— Does it involve a trig expression with θ unspecified?
 - |→ Try $\theta = 45^\circ$ (Technique 8)
- |— Does it ask "how many from 1 to n are co-prime to n"?
 - |→ Euler's Totient $\varphi(n)$ (Technique 9)
- |— None of the above → Solve normally, but use mental math from PART 0 to speed up computation

WHEN SHORTCUTS BACKFIRE — THE PAPER-SETTER'S TRAP

The examiner knows you know the shortcuts. Here is how they defeat them:

1. **Twisted % problem:** "A's salary is 20% more than B's. B's salary is 20% more than C's. A's salary is what % more than C's?" — students use " $20+20=40\%$ " (wrong). Answer: 44% (use base-100 twice, not add). **The shortcut of adding percentages only works for ONE step of change from a FIXED base.**
2. **Alligation with 3 components:** "Milk costing ₹16, ₹20, and ₹24 per litre mixed in some ratio to get ₹19 per litre." Alligation on 3 components simultaneously is ambiguous (infinite valid ratios). The examiner always specifies additional constraints. **Do not force alligation on 3-component mixes.**
3. **Trig substitution gives 2 matching options:** As shown above, $\theta = 45^\circ$ sometimes doesn't separate options. **Always try a second angle before finalising.**
4. **Backward sub when options are large:** If options are 1200, 1400, 1600, 1800 and plugging 1600 takes as long as solving — just solve. Backward sub saves time only when plugging is genuinely faster.
5. **Unit digit when options share it:** If options are 12, 32, 52, 72 — all unit digit 2 — unit digit elimination achieves nothing. **Move directly to magnitude or exact computation.**

PART 1 — PERCENTAGE

1.0 Opener

Why: Percentage is the single highest-frequency topic. It appears *inside* profit-loss, simple/compound interest, data interpretation, DI caselets, ratio problems, population, discount, tax, exam-marks, mixtures, error measurement. If percentage is slow, everything downstream is slow.

Mental-math arsenal:

The fraction-percent table (§ 0.1) — the single most powerful table in arithmetic.

Squares 1-30.

10%, 1%, 5%, 25% as reflexes.

Memorised formulas:

#	Formula	Notes
1	$x\% \text{ of } y = \frac{x}{100} \times y$	Also = $\frac{y}{100} \times x$ (swap trick)
2	$\text{Change \%} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100$	Base is always the OLD value
3	Successive a%, b% → net = $a + b + \frac{ab}{100}$	Signed: decrease = negative
4	A is r% more than B → B is $\frac{r}{100 + r} \times 100\%$ less than A	Reverse percentage
5	After n years = $P \times \left(1 + \frac{r}{100}\right)^n$	Same as CI formula

Generic attack plan: ask "what is the base (100)?" before anything. Then match to a type.

Type 1 — Convert fraction ↔ percentage ↔ decimal

Cue: "Express $\frac{5}{8}$ as a percentage" / "37.5 % = ? fraction".

Insight: The table in § 0.1. Done.

EXAM SHORTCUT

Shortcut: Table lookup. $\frac{5}{8} = 62.5\%$.

Traditional: $\frac{5}{8} = 5 \times \frac{100}{8} = 62.5$.

Variations: Recurring decimals → $0.333... = \frac{1}{3}$; $0.142857... = \frac{1}{7}$ (period 6 — a known pattern).

Self-check: $\frac{7}{8} = ?\%$. → **87.5 %** (50 + 25 + 12.5).

Type 2 — "x % of y" direct calculation

Cue: "Find 23 % of 850".

Insight: Decompose x into multiples of 10, 5, 1.

EXAM SHORTCUT

Shortcut: Break 23 % into parts.

20 % of 850 = **170**

1 % of 850 = 8.5; so 3 % = $3 \times 8.5 = 25.5$

Total = $170 + 25.5 = 195.5$ (done in < 10 seconds mentally)

Traditional: $\frac{23}{100} \times 850 = \frac{19550}{100} = 195.5$.

Variations:

Use the symmetry trick: 36 % of 25 = 25 % of 36 = 9.

Ugly percentages (17.5 %) $\rightarrow 17.5 = 10 + 5 + 2.5$.

Self-check: 36 % of 250 = ? $\rightarrow$ 25 % of 360 = **90**.

Type 3 — "A is what percentage of B?"

Cue: "What percent of 64 is 24?"

Insight: Compute A/B, then map to the table.

EXAM SHORTCUT

Shortcut: $\frac{24}{64} = \frac{3}{8} = 37.5\%$.

Traditional: $(24 \times 100) / 64 = \frac{2400}{64} = 37.5$.

Variations:

"By what percentage is A greater than B?" $\rightarrow (A-B)/B \times 100$ (base is B).

"By what percentage is A less than B?" $\rightarrow (B-A)/B \times 100$ (base is B).

Self-check: 45 is what % of 75? $\rightarrow \frac{45}{75} = \frac{3}{5} = 60\%$.

Type 4 — Percentage increase / decrease (basic)

Cue: "Price rose from 240 to 300. % increase?"

Insight: Change / Original $\times 100$, original (the *old* value) is the base.

EXAM SHORTCUT

Shortcut: $300 - 240 = 60$. $\frac{60}{240} = \frac{1}{4} = 25\%$.

Traditional: $60 \times \frac{100}{240} = 25$.

Variations:

Watch the base direction: "decreased from 300 to 240" $\rightarrow \frac{60}{300} = \frac{1}{5} = 20\%$.

Two different bases give two different answers — the exam exploits this.

Self-check: Salary up from ₹20k to ₹28k $\rightarrow \frac{8}{20} = 40\%$.

Type 5 — Successive percentage change (the single most tested type)

Cue: "A salary is increased by 20 %, then decreased by 10 %. Net change?"

Formula: Net % = $a + b + \frac{a \times b}{100}$ — where a, b are **signed** (decrease = negative).

Example. Salary +20%, then -10%. Net?

► METHOD A — Standard (safe every time)

Step 1: $a = +20$, $b = -10$ ← decrease always enters as negative; treating both as positive is the #1 sign error

Step 2: Net = $20 + (-10) + \frac{20 \times (-10)}{100}$ ← the third term ($ab/100$) is where most marks are lost — never skip it

Step 3: $= 10 + (-2) = +8\%$ ← $20 \times (-10) = -200$; divided by 100 = -2, which subtracts from the +10

Use when: You want a single-line answer without computing a base.

◀ METHOD B — Exam Shortcut

Use when: Numbers feel unfamiliar or you want to double-check your shortcut answer.

Start: base = ₹100

After +20%: $100 \times 1.20 = ₹120$ ← write the multiplier (1.20), not the addition "100+20" — saves one arithmetic step

After -10%: $120 \times 0.90 = ₹108$ ← 0.90 = 10% off MP; do NOT subtract flat 10 from 120

Net: $\frac{108 - 100}{100} \times 100 = +8\%$ ✓

Variations:

Equal-and-opposite: +r% then -r% $\rightarrow$ net = $-\frac{r^2}{100}$ (always a loss). Classic trap.

Discount-on-discount: 10% + 20% successive = $10 + 20 - 2 = 28\%$, NOT 30%.

Self-check: Price +30%, then -20%.

Net change = $a + b + \frac{ab}{100} = 30 + (-20) + \frac{30 \times (-20)}{100}$

$= 10 + \left(-\frac{600}{100}\right) = 10 - 6 = +4\%$

Verify with numbers: $100 \rightarrow \times 1.30 \rightarrow 130 \rightarrow \times 0.80 \rightarrow 104$. Net gain = 4%. ✓

Type 6 — Population / value after n years (compound growth)

Cue: "Population 50,000 grows at 10% annually. Population after 3 years?"

Formula: Final = $P \times (1 + r/100)^n$

Example. $P = 50,000$, $r = 10\%$, $n = 3$.

✂ EXAM SHORTCUT

Shortcut:

Step 1: $(1.1)^3 = 1.331$

Step 2: Final = $50,000 \times 1.331 = 66,550$

Year-by-year (Traditional):

Year 1: $50,000 \times 1.10 = 55,000$ **Year 2:** $55,000 \times 1.10 = 60,500$ **Year 3:** $60,500 \times 1.10 = 66,550$ ✓

Variations:

Declining: $50,000 \times (0.9)^3 = 50,000 \times 0.729 = 36,450$.

Mixed rates: Year 1 +10%, Year 2 +5%, Year 3 -8% → $50,000 \times 1.10 \times 1.05 \times 0.92 = 52,878$.

Find original: 66,550 after 3 years at 10% → $66,550 / 1.331 = 50,000$.

Self-check: ₹8000 at 25 % p.a. CI for 2 years → $8000 \times \left(\frac{5}{4}\right)^2 = 8000 \times \frac{25}{16} = ₹12,500$.

Type 7 — Forward vs reverse percentage relationship

Cue: "A's salary is 25% more than B's. By what % is B's salary less than A's?"

Formula: If A is $r\%$ more than B, then B is $\frac{r}{100 + r} \times 100\%$ less than A.

Base-100 method: Let $B = 100 \rightarrow A = 125$ B is less than A by: $\frac{125 - 100}{125} \times 100 = \frac{25}{125} \times 100 = 20\%$

✂ EXAM SHORTCUT

Formula shortcut: $\frac{25}{100 + 25} \times 100 = \frac{25}{125} \times 100 = 20\%$ (same)

Pair table (cache for exam speed):

A is $r\%$ MORE than B	B is ____% LESS than A
20% more	16.67% less
25% more	20% less
33.33% more	25% less
50% more	33.33% less
100% more	50% less

Type 8 — Income → Expenditure → Savings

Cue: "Raj spends 75% of income. Income +20%, expenditure +10%. % change in savings?"

Method: Fix income = 100. Compute old and new savings. Find % change.

Step 1 — Set up base values. Income = 100, Expenditure = 75, Savings = $100 - 75 = 25$

Step 2 — Find new income and expenditure. New income = $100 \times 1.20 = 120$

New expenditure = $75 \times 1.10 = 82.5$

Step 3 — Find new savings. New savings = $120 - 82.5 = 37.5$

Step 4 — Calculate % change in savings. Change = $37.5 - 25 = 12.5$

% change = $(12.5 \div 25) \times 100 = +50\%$

Self-check. Income +10%, Expense +20%, old expense = 80% of income.

Old: Income = 100, Expense = 80, Savings = $100 - 80 = 20$ **New Income:** $100 \times 1.10 = 110$ **New Expense:** $80 \times 1.20 = 96$ **New Savings:** $110 - 96 = 14$ **% change in savings:** $(14 - 20) \div 20 \times 100 = -\frac{6}{20} \times 100 = -30\%$

Type 9 — Pass / fail marks

Cue: "Pass mark is 40%. A scored 220 and failed by 20 marks. Find max marks."

Step 1: Pass mark = Score + Deficit = $220 + 20 = 240$

Step 2: Pass mark = 40% of Max $\rightarrow 240 = \frac{40}{100} \times M$

Step 3: $M = \frac{240 \times 100}{40} = 240 \times 2.5$

Answer: Max marks = 600

Variations: - Two students, one passes, one fails: set up two equations, subtract to cancel Max. - "Exceeds pass mark by x" vs "falls short by x" — sign flips.

Type 10 — Elections (valid vs invalid votes)

Cue: "10% votes invalid. Winner got 60% of valid votes, exceeding loser by 1800. Total votes?"

Step 1: Let Total votes = T

Step 2: Valid votes = 90% of T = $0.9T$ (10% invalid)

Step 3: Winner = 60% of valid; Loser = 40% of valid (two candidates $\rightarrow$ sum = 100%)

Step 4: Difference = (60% - 40%) of valid = $20\% \times 0.9T = 0.18T$

Step 5: $0.18T = 1800 \rightarrow T = \frac{1800}{0.18} = 10,000$

Type 11 — Mixture-style percentage (alcohol, salt, milk)

Cue: "30 L of a mixture has 20% alcohol. How much water to add to reduce alcohol to 15%?"

Key principle: The PURE SOLUTE (alcohol, milk, acid) quantity stays constant when you add solvent.

Step 1: Pure alcohol = $20\% \times 30 = 6$ L

Step 2: After adding x L water: total = $(30 + x)$ L; alcohol still = 6 L

Step 3: New concentration = $\frac{6}{30 + x} = \frac{15}{100}$

Step 4: $6 \times 100 = 15 \times (30 + x) \rightarrow 600 = 450 + 15x$

Step 5: $15x = 150 \rightarrow x = 10$ L

Self-check. 40 L of 25% milk; add water to make 20% milk.

Step 1 — Find pure milk (stays fixed). Pure milk = $25\% \times 40 = 10$ L

Step 2 — Set up equation for new concentration. $10 / (40 + x) = \frac{20}{100} = \frac{1}{5}$

Step 3 — Solve for x . $5 \times 10 = 40 + x$

$50 = 40 + x$

$x = 10$ L ✓

Type 12 — Price ↔ Quantity inverse (expenditure unchanged)

Cue: "Price of sugar rises 25%. By what % must consumption drop to keep expenditure unchanged?"

Principle: Expenditure = Price × Quantity = constant. So Price ↑ means Quantity must ↓ proportionally.

Step 1 — Set base values. Old: Price = 100, Quantity = 100, Expenditure = $100 \times 100 = 10,000$

Step 2 — Find new quantity at new price. New price = 125 (25 % rise)

Expenditure stays at 10,000

New Quantity = $\frac{10000}{125} = 80$

Step 3 — Find % drop in consumption. Drop in quantity = $(100 - 80) / 100 \times 100 = \frac{20}{100} \times 100 = 20\%$

✂ EXAM SHORTCUT

Shortcut: Price ↑ $r\% \rightarrow$ consumption ↓ by $\frac{r}{100 + r} \times 100\% \rightarrow \frac{25}{125} \times 100 = 20\%$ (same answer, faster)

Self-check. Price up 10% → consumption must drop by $\frac{10}{110} \times 100 = 9.09\%$.

Type 13 — Successive discounts (at shop)

Cue: "Listed ₹2000. Two successive discounts of 20% and 10%. Final selling price?"

Shortcut (successive % formula):

Step 1 — Apply successive % formula. Net discount = $a + b + \frac{ab}{100}$, where $a = -20$, $b = -10$

$$= -20 + (-10) + \frac{(-20) \times (-10)}{100}$$

$$= -30 + \frac{200}{100}$$

$$= -30 + 2 = -28\%$$

Step 2 — Find final selling price. SP = $(100 - 28)\%$ of 2000 = $72\% \times 2000$

$$= 0.72 \times 2000 = \text{₹}1440$$

Step-by-step (Traditional):

After 20% discount: 20% of 2000 = 400

$$\text{Price after 1st discount} = 2000 - 400 = \text{₹}1600$$

After 10% discount: 10% of 1600 = 160

$$\text{Final price} = 1600 - 160 = \text{₹}1440 \checkmark$$

COMMON TRAP

Classic trap: $20\% + 10\% = 30\%$ discount is WRONG. Two successive discounts of 20% and 10% give only 28% effective discount. The second discount applies to the already-reduced price, not the original.

Type 14 — Error in measurement (geometry-meets-percentage)

Cue: "Side of square measured 3 % in excess. Error % in area?"

Insight: Area $\propto$ side². If side is $(1 + r)$, area is $(1 + r)^2 \approx 1 + 2r + r^2$. With $r = 3\% = 0.03$, area error = $6 + 0.09 \approx 6.09\%$.

EXAM SHORTCUT

Shortcut: Successive-% with $a = b = r$. Net = $2r + \frac{r^2}{100}$. $r = 3 \rightarrow 6 + \frac{9}{100} = 6.09\%$.

Traditional: Same algebra explicitly.

Variations:

Cube / volume: three successive $r \rightarrow 3r + \frac{3r^2}{100} + \frac{r^3}{10000} \approx 3r$.

One side $+r\%$, other side $-s\%$: apply signed successive.

Self-check: Radius $+4\% \rightarrow$ area of circle error? $\rightarrow 2(4) + \frac{16}{100} = 8.16\%$.

Type 15 — "x % more than y % of a number" chain problems

Cue: "Find the number whose 25 % is 40 % less than 60 % of 750."

Insight: Work right to left. Compute 60 % of 750, apply 40 % less, then reverse-divide by 25 %.

Step 1 — Compute 60% of 750. $60\% \times 750 = \left(\frac{60}{100}\right) \times 750 = \mathbf{450}$

Step 2 — Take 40% less of 450. "40% less" means 60% of 450
 $= 0.60 \times 450 = \mathbf{270}$

Step 3 — Reverse-divide to find N. 25% of N = 270

$$N = 270 \times \left(\frac{100}{25}\right) = 270 \times 4 = \mathbf{1080}$$

Traditional: Same sequence.

Variations: Add another nesting level; swap "less" for "more"; introduce ratios.

Self-check: 30 % of N = 20 % more than 80 % of 500 $\rightarrow N = 480/0.3 = \mathbf{1600}$.



PART 2 — PROFIT, LOSS, DISCOUNT

2.0 Opener

Why: Shop/market context; 2-4 Qs guaranteed in Banking Prelims and SSC Quantitative.

Mental-math arsenal: percent-fraction table + "CP-SP-MP" vocabulary.

Memorised formulas:

1. Profit = SP – CP
2. Profit % = (Profit / CP) × 100 — **base is always CP** unless explicitly stated
3. Loss % = (Loss / CP) × 100
4. SP when profit: SP = CP × (100 + P%) / 100
5. SP when loss: SP = CP × (100 – L%) / 100
6. Discount = MP – SP
7. Discount % = (Discount / MP) × 100 — **base is MP, not CP**
8. MP = CP × (100 + markup%) / 100
9. Two articles at same SP, one at +x% profit and other at –x% loss → **always a net loss** = $x^2 / 100$ %

Generic attack plan: read "gain/profit/loss" in terms of **CP**; read "discount" in terms of **MP**. Never confuse the bases. If both percentages appear, translate to a common number (set CP = 100).

Type 1 — Basic profit / loss percent

Cue: "CP = 400, SP = 460. Profit %?"

EXAM SHORTCUT

Shortcut: $(460 - 400)/400 = \frac{60}{400} = \frac{3}{20} = 15\%$.

Traditional: $60 \times \frac{100}{400} = 15$.

Variation: SP < CP → loss %, base still CP.

Type 2 — Find SP given CP and profit %

Cue: "CP = 800, wants 12.5 % profit. SP?"

EXAM SHORTCUT

Shortcut: $12.5\% = \frac{1}{8}$. So profit = 100. SP = **900**.

Traditional: SP = $800 \times 112.5\% = 900$.

Variation: Loss instead of profit. 12.5 % loss → SP = $\frac{7}{8} \times 800 = 700$.

Type 3 — Find CP given SP and profit %

Cue: "Sold at 1080 for 20 % profit. CP?"

EXAM SHORTCUT

Shortcut: SP is 120 % of CP $\rightarrow$ $CP = 1080 \times \frac{100}{120} = 900$. The common trap is dividing by 1.2 vs subtracting 20 %.

Traditional: Let CP = x. $1.2x = 1080 \rightarrow x = 900$.

Type 4 — Two articles at same SP, opposite %

Cue: "A man sells two watches at ₹2400 each, gaining 20 % on one, losing 20 % on the other. Net gain/loss %."

Insight: Always a net **LOSS** of $x^2/100$ %. (Constant — cache this.)

EXAM SHORTCUT

Shortcut: $20^2/100 = 4$ % **LOSS**.

Traditional:

Step 1 — Find CP of the article sold at 20 % gain. Gainer's CP = $SP \times 100 / (100 + \text{gain } \%)$

$$= 2400 \times \frac{100}{120} \\ = \text{₹}2,000$$

Step 2 — Find CP of the article sold at 20 % loss. Loser's CP = $SP \times 100 / (100 - \text{loss } \%)$

$$= 2400 \times \frac{100}{80} \\ = \text{₹}3,000$$

Step 3 — Find net loss %. Total CP = 2,000 + 3,000 = **₹5,000**

$$\text{Total SP} = 2,400 + 2,400 = \text{₹}4,800$$

$$\text{Loss} = 5,000 - 4,800 = \text{₹}200$$

$$\text{Loss \%} = \text{Loss} / \text{Total CP} \times 100 = \frac{200}{5000} \times 100 = 4 \%$$

The shortcut saves the whole CP derivation.

Type 5 — Marked price, discount, and profit combined

Cue: "Marked at 25% above CP. Discount 10%. Find profit %."

► **METHOD A — Standard (safe every time)**

Set CP = 100 ← always anchor on CP = 100; the whole chain flows from this single assumption

MP: $100 \times 1.25 = 125$ ← mark-up % is always ON CP

SP: $125 \times 0.90 = 112.5$ ← discount is ON MP, not CP; write the multiplier 0.90 — do NOT subtract flat 10 from 125

Profit %: $\frac{112.5 - 100}{100} \times 100 = 12.5\%$

Use when: Exact values are needed, or MP is given as a direct number, not a %.

✚ **METHOD B — Exam Shortcut**

Use when: Both mark-up % and discount % are given — treat as two successive % changes from CP.

$\text{Net} = a + b + \frac{a \times b}{100}$, where $a = +25$ (mark-up), $b = -10$ (discount) ← discount enters as negative — this is the same formula as successive %

$= 25 - 10 + \frac{25 \times (-10)}{100} = 15 + (-2.5) = +12.5\%$

Skip when: MP is given as a direct number (not "x% above CP") — shortcut won't apply then.

Type 6 — Find discount % to achieve a desired profit

Cue: "Goods marked 40% above CP. Wants 12% profit. Discount %?"

Set CP = 100 **MP:** $100 + 40\% = 140$ **Desired SP:** $100 + 12\% = 112$ **Discount on MP:** $140 - 112 = 28$ **Discount %:** $\frac{28}{140} \times 100 = 20\%$

Type 7 — False weight / dishonest shopkeeper

Cue: "A dealer claims to sell at cost price but uses a 900 g weight for 1 kg. Profit %."

Insight: He charges for 1000 g but gives 900 g. $\text{Gain} = (1000 - 900)/900 \times 100 = 11.11\%$ (base = what he actually gave).

Variation: Uses BOTH a false weight AND marks up the price. Combine via successive %.

Self-check: 950 g instead of 1 kg → profit = $\frac{50}{950} = 5.26\%$.

Type 8 — Dishonest shopkeeper buys too

Cue: "While buying uses 1100 g weight, while selling uses 900 g; charges CP either way. Overall profit %."

✚ **EXAM SHORTCUT**

Shortcut:

On buying, gets 1100 g by paying for 1000 g → effective CP / g is reduced to $\frac{10}{11}$ of claimed.

On selling, delivers 900 g for 1000 g price → SP / g is increased to $\frac{10}{9}$.

$$\text{Ratio SP/CP per gram} = \left(\frac{10}{9}\right) \div \left(\frac{10}{11}\right)$$

$$= \left(\frac{10}{9}\right) \times \left(\frac{11}{10}\right)$$

$$= \frac{11}{9}$$

$$\text{Profit \%} = (\text{SP/CP} - 1) \times 100 = \left(\frac{11}{9} - 1\right) \times 100 = \frac{2}{9} \times 100 = \mathbf{22.22 \%}$$

Type 9 — Cost-price-unknown combined puzzles

Cue: "By selling at ₹810, a man loses 10%. At what price should he sell to gain 10%?"

Step 1: SP = ₹810 represents 90% of CP (because loss = 10%)

Step 2: CP = $\frac{810 \times 100}{90} = \mathbf{₹900}$

Step 3: Desired SP at 10% gain = $\frac{900 \times 110}{100} = \mathbf{₹990}$

Type 10 — Partnership-style gain share (sneaks in)

Covered under Part 9 (Partnership). The link is the same "per unit" approach.

Type 11 — Successive discounts (equivalent single)

Already in Type 5 (percentage). Recapping for SP/MP context: 10 % + 20 % → equivalent 28 %.

10 % + 20 % + 5 % → pairwise: 28 % then with 5 % → 28 + 5 - 1.4 = 31.6 %.

Type 12 — Percentage profit on SP instead of CP (trap)

Cue: "A shopkeeper's profit is 20 % of SP. Find profit % on CP."

Insight: If profit = 20 % of SP, then CP = 80 % of SP. Profit/CP = $\frac{20}{80} = \mathbf{25 \%}$.

Variation: Some exams ask "20 % loss on SP" → loss/CP = $\frac{20}{120} = 16.67 \%$.

PART 3 — SIMPLE & COMPOUND INTEREST

3.0 Opener

Formulas — memorise all 6:

Formula	Expression	Notes
Simple Interest	$SI = \frac{P \times R \times T}{100}$	Base formula
Amount (SI)	$A = P \left(1 + \frac{RT}{100} \right)$	—
Compound Interest	$CI = P \left(1 + \frac{R}{100} \right)^n - P$	Any n
CI - SI (2 years)	$P \times \left(\frac{R}{100} \right)^2$	Very common shortcut
Half-yearly	Rate = R/2, n doubles	$A = P \left(1 + \frac{R}{200} \right)^{2n}$
Quarterly	Rate = R/4, n × 4	$A = P \left(1 + \frac{R}{400} \right)^{4n}$

CI = successive % growth: +R% applied n times → use the successive-% formula from Part 1.

Type 1 — Straight SI calculation

Cue: P, R, T given; find SI.

Example. P = ₹6000, R = 8% p.a., T = 3 years.

$$\text{Step 1: } SI = \frac{P \times R \times T}{100} = \frac{6000 \times 8 \times 3}{100}$$

$$\text{Step 2: } = \frac{144000}{100}$$

Answer: ₹1440

Reverse (find R): Given SI = ₹1440, P = 6000, T = 3. - $R = \frac{SI \times 100}{(P \times T)} = \frac{1440 \times 100}{(6000 \times 3)} = \frac{144000}{18000} = 8\%$

Type 2 — Principal doubling/tripling (SI)

Cue: "At what rate will a sum DOUBLE in 8 years (SI)?"

Key identity: For SI, when Amount = k × P → $RT = 100(k - 1)$.

Doubling (Amount = 2P):

SI = P (the interest earned equals the principal)

$$\frac{P \times R \times T}{100} = P$$

Cancel P: $R \times T = 100$

With $T = 8$: $R = \frac{100}{8} = 12.5\% \text{ p.a.}$

Tripling (Amount = 3P):

SI = 2P $\rightarrow R \times T = 200$

With $T = 8$: $R = \frac{200}{8} = 25\% \text{ p.a.}$

Type 3 — Compound Interest for 2 or 3 years

Cue: "P = ₹10,000, R = 10%, n = 2. Find CI."

Shortcut (successive %):

Step 1: First year growth: 10% $\rightarrow$ Amount = 11,000

Step 2: Second year growth: 10% on 11,000 $\rightarrow$ Amount = 12,100

OR: Net % = $10 + 10 + \frac{10 \times 10}{100} = 21\%$ **CI = 21% of 10,000 = ₹2100**

Formula method:

Step 1: $A = 10000 \times (1.10)^2 = 10000 \times 1.21 = 12100$

Step 2: $CI = 12100 - 10000 = ₹2100 \checkmark$

Type 4 — CI minus SI difference (2-year shortcut)

Formula: $CI - SI = P \times \left(\frac{R}{100}\right)^2$

Example. $P = ₹5000$, $R = 8\%$, $T = 2$ years. Find $(CI - SI)$.

► **METHOD A — Standard (safe every time)**

Step 1: $= 5000 \times \left(\frac{8}{100}\right)^2$ ← *this formula works ONLY when $T = 2$ years; for $T = 3$, use Method B*

Step 2: $= 5000 \times \frac{64}{10000}$ ← $\left(\frac{8}{100}\right)^2 = \frac{64}{10000}$ *not $8^2 = 64$ over 100; squaring the full fraction is the critical step*

Answer: $= \frac{320000}{10000} = ₹32$

Use when: $T = 2$ years exactly.

✂ **METHOD B — Exam Shortcut**

Use when: You want to verify, or $T \neq 2$ years.

Step 1 — Calculate SI over 2 years. $SI = P \times R \times T / 100$

$$= 5,000 \times 8 \times \frac{2}{100}$$

= ₹800

Step 2 — Calculate CI (Amount first). $A = P \times (1 + R/100)^2$

$$= 5,000 \times (1.08)^2$$

$$= 5,000 \times 1.1664$$

= ₹5,832

$$CI = A - P = 5,832 - 5,000 = ₹832$$

Step 3 — Difference. $CI - SI = 832 - 800 = ₹32$ ✓

Why it works: CI earns interest on the first year's interest. That extra $= P \times \left(\frac{R}{100}\right)^2$.

Type 5 — Half-yearly / quarterly compounding

Cue: " $P = ₹8000$, $R = 10\%$ p.a., compounded half-yearly, $T = 1$ year."

Step 1: Half-yearly rate $= \frac{10}{2} = 5\%$

Step 2: Number of half-yearly periods $= 2 \times 1 = 2$

Step 3: $A = 8000 \times (1.05)^2$ $(1.05)^2 = 1.1025$

$$A = 8000 \times 1.1025 = ₹8,820$$

Answer: $CI = 8,820 - 8,000 = ₹820$

COMMON TRAP

Common error: Using the full annual rate for half-yearly compounding. Always halve the rate and double the periods. A 10% annual rate compounded half-yearly gives MORE than 10% effective annual rate (it gives 10.25%).

Type 6 — Equal instalment loan repayment (CI)

Cue: "Loan ₹2500 at 4% CI, repaid in 2 equal annual instalments x. Find x."

Step 1: Present value of instalment 1 = $\frac{x}{1.04}$

Step 2: Present value of instalment 2 = $\frac{x}{(1.04)^2} = \frac{x}{1.0816}$

Step 3: Sum = 2500 $\rightarrow \frac{x}{1.04} + \frac{x}{1.0816} = 2500$

Step 4: $x \left(\frac{1.0816 + 1.04}{1.04 \times 1.0816} \right) = 2500$

Step 5: $x = \frac{2500 \times 1.04 \times 1.0816}{2.1216} \approx \text{₹1325}$

Type 7 — Find rate given CI for 2 years

Cue: "CI on ₹6000 for 2 years = ₹1260. Find rate."

Step 1: $6000 \times [(1 + r)^2 - 1] = 1260$ where $r = R/100$

Step 2: $(1 + r)^2 = \frac{1260}{6000} + 1 = 0.21 + 1 = 1.21$

Step 3: $1 + r = \sqrt{1.21} = 1.1$

Answer: $r = 0.1 \rightarrow R = 10\%$

Type 8 — Find rate from ratio of two CI amounts

Cue: "A sum becomes ₹4840 in 2 years and ₹5324 in 3 years (CI)."

Key insight: In CI, ratio of consecutive year amounts = $(1 + R/100)$.

Step 1: Set up the ratio: $\frac{A_3}{A_2} = 1 + \frac{R}{100}$

Step 2: Substitute values: $\frac{5324}{4840} = 1.1$

So **R = 10%**

Step 3: Find principal: $P = \frac{4840}{(1.1)^2} = \frac{4840}{1.21} = \text{₹4000}$

PART 4 — RATIO, PROPORTION, VARIATION

4.0 Opener

Key formulas — memorise:

Formula	What it means
$a : b = c : d \rightarrow ad = bc$	Cross-multiplication law
$a/b = c/d = e/f = (a+c+e)/(b+d+f)$	Addendo rule (all equal ratios share a common value)
Compound ratio: $(a:b) \times (c:d) = ac:bd$	Multiply numerators, multiply denominators
Direct variation: $y = kx$	More $x \rightarrow$ more y
Inverse variation: $y = k/x$	More $x \rightarrow$ less y

Attack plan: Let the ratio constant = k . Numbers are always $3k$, $5k$ etc. Find k from the constraint, then the actual numbers.

Type 1 — Split a quantity in given ratio

Cue: "Divide ₹2400 among A, B, C in ratio 3 : 5 : 4."

Method: Total parts = $3 + 5 + 4 = 12$. Each part = $2400 \div 12 = 200$.

Example. Divide ₹2400 in 3 : 5 : 4.

Step 1: Total parts = $3 + 5 + 4 = 12$

Step 2: Value of 1 part = $2400 \div 12 = ₹200$

Step 3: $A = 3 \times 200 = ₹600$

Step 4: $B = 5 \times 200 = ₹1000$

Step 5: $C = 4 \times 200 = ₹800$

Check: $600 + 1000 + 800 = 2400 \checkmark$

Type 2 — Ratio changes after adding/subtracting years (ages)

Cue: "A : B = 3 : 5 now. After 6 years, ratio becomes 2 : 3. Find present ages."

Method: Let present ages = $3k$ and $5k$. Write the future-ratio equation and solve for k .

Example. Ratio of A's to B's age = 3 : 5. After 6 years, ratio = 4 : 6 = 2 : 3. Find ages.

Step 1: Let present ages: $A = 3k$, $B = 5k$

Step 2: After 6 years: $(3k + 6) / (5k + 6) = \frac{2}{3}$

Step 3: Cross-multiply: $3(3k + 6) = 2(5k + 6)$

Step 4: $9k + 18 = 10k + 12$

Step 5: $k = 6$

Answer: $A = 3 \times 6 = 18$ years, $B = 5 \times 6 = 30$ years

COMMON TRAP

Common trap: When the ratio is "after 6 years", you add 6 to BOTH the numerator and denominator separately — do NOT multiply the ratio by some factor.

Type 3 — Proportion types

Three sub-types — one table:

Sub-type	Formula	Example
Fourth proportional of a, b, c	$x = \frac{bc}{a}$	3rd prop of 3, 5, 9 $\rightarrow x = 5 \times \frac{9}{3} = 15$
Mean proportional of a, b	$x = \sqrt{ab}$	Mean prop of 4 and 9 $\rightarrow x = \sqrt{36} = 6$
Third proportional of a, b	$x = \frac{b^2}{a}$	Third prop of 2, 6 $\rightarrow x = \frac{36}{2} = 18$

KEY FACT

Quick memory: Fourth prop $\rightarrow$ three numbers given, find the fourth to keep $a:b = c:x$. Mean prop $\rightarrow x$ is the geometric mean. Third prop $\rightarrow x$ comes after a, b in $a:b = b:x$ pattern.

Type 4 — Direct / Inverse variation word problems

Cue: "12 men complete a job in 8 days. How many days will 16 men take?"

Method: More men $\rightarrow$ less time $\rightarrow$ INVERSE variation. Product stays constant: men $\times$ days = constant.

Example. 12 men, 8 days $\rightarrow$ 16 men, ? days.

Step 1: Identify: more men = less days $\rightarrow$ inverse variation

Step 2: $12 \times 8 = 16 \times t$

Step 3: $t = \frac{96}{16} = 6$ days

Example. A machine packs 120 boxes in 4 hours. How many hours for 300 boxes?

Step 1: More boxes $\rightarrow$ more time $\rightarrow$ DIRECT variation

Step 2: Set up proportion: $\frac{120}{4} = 300/t$

Step 3: Solve for t. $t = 300 \times \frac{4}{120}$
 $= \frac{1200}{120}$
 $= 10$ hours

Type 5 — HCF + ratio to find actual numbers

Cue: "Two numbers in ratio 3 : 5 and HCF = 4. Find the numbers."

Insight: If HCF = k and ratio = a : b, then numbers = ak and bk.

Example. Ratio 3 : 5, HCF = 4.

Step 1: Numbers = $3k$ and $5k$ where k = HCF

Step 2: $k = 4$

Answer: Numbers = $3 \times 4 = 12$ and $5 \times 4 = 20$

Verify: $\text{HCF}(12, 20) = 4$ ✓

Type 6 — Investment ratio (bridge to Partnership)

Cue: "A invests ₹4000 for 6 months, B invests ₹5000 for 8 months. Profit ratio?"

Method: Profit ratio = (Capital × Time) for each.

Step 1 — Calculate each investor's weight (Capital × Time). A's weight = $4,000 \times 6 = 24,000$

B's weight = $5,000 \times 8 = 40,000$

Step 2 — Find profit ratio. Ratio = $24,000 : 40,000 = 24 : 40 = 3 : 5$

PART 5 — AVERAGES

5.0 Opener

Two core identities — everything else follows from these:

Identity	Use
Average = Sum $\div$ Count	Find average
Sum = Average $\times$ Count	Find sum (the KEY step in almost every problem)
New avg = Old avg $\pm$ (change $\div$ count)	When one value changes

Attack plan: Never work with averages directly. Convert to SUMS immediately, then operate on sums.

Type 1 — New member joins; average changes

Cue: "A group of n members has average A . A new member joins; average becomes A' . Find the new member's value."

Method: New sum – old sum = new member's value.

Example. 11 students' average = 60. A 12th student joins; average drops by 1 (to 59). Find 12th student's marks.

Step 1: Old sum = $60 \times 11 = 660$

Step 2: New sum = $59 \times 12 = 708$

Step 3: 12th student's marks = $708 - 660 = 48$

EXAM SHORTCUT

Shortcut: New member's score = Old average – (change in average $\times$ new count).

Drop in average = 1; New count = 12

Change in sum = change in average $\times$ count = $1 \times 12 = 12$

New member's score = Old average – Change in sum = $60 - 12 = 48$ ✓

Type 2 — Replacement of a member

Cue: "One person leaves, another joins. Average changes by x . Find the new person's value."

Method: Change in average $\times$ count = change in sum = (new value – old value).

Example. Class of 10, average = 75. Student scoring 50 is replaced. Average rises by 2. Find the new student's score.

Step 1: Rise in average = 2. Count = 10.

Step 2: Rise in sum = $2 \times 10 = 20$

Step 3: New score – Old score = 20

Step 4: New score = $50 + 20 = 70$

KEY FACT

Universal rule: (New value – Old value) = Change in average $\times$ Count. Works for joins, leaves, and replacements.

Type 3 — Weighted average (combining two groups)

Cue: "Group 1 of n_1 people has average A_1 ; Group 2 of n_2 has average A_2 . Combined average?"

Method: Combined avg = $(n_1 \times A_1 + n_2 \times A_2) / (n_1 + n_2)$.

Example. 30 students average 60 marks; 20 students average 75 marks. Overall average?

Step 1: Group 1 sum = $30 \times 60 = 1800$

Step 2: Group 2 sum = $20 \times 75 = 1500$

Step 3: Total sum = $1800 + 1500 = 3300$

Step 4: Total count = $30 + 20 = 50$

Answer: Average = $\frac{3300}{50} = 66$ marks

Alligation check (verify with cross-ratio): - Mean = 66, Group 1 average $A_1 = 60$, Group 2 average $A_2 = 75$
- Ratio = $(A_2 - \text{Mean}) : (\text{Mean} - A_1) = (75 - 66) : (66 - 60) = 9 : 6 = 3 : 2$ - Group sizes = $30 : 20 = 3 : 2$ ✓
(consistent)

Type 4 — Cricket / batting average

Cue: "Batsman averaged 30 in 20 innings. Scores 50 in 21st. New average?"

Step 1: Old sum = $30 \times 20 = 600$

Step 2: New sum = $600 + 50 = 650$

Step 3: New average = $\frac{650}{21} = 30.95$ (≈ 31)

Variation: "He scores 0 in the 21st innings." $\rightarrow$ New avg = $\frac{600}{21} = 28.57$. Averages fall when you score below your average; rise when you score above.

Type 5 — Consecutive integers average

Rule: Average of n consecutive integers = $(\text{First} + \text{Last}) / 2 = \text{middle term}$.

Set	Average
1, 2, 3, 4, 5	3 (middle of 5)
10, 11, 12, 13	11.5 (avg of 11 and 12)
21 to 50	$(21 + 50)/2 = 35.5$

Example. Average of all integers from 15 to 35? $(15 + 35)/2 = 25$. Count = 21 numbers.

Type 6 — Missing value from known average

Cue: "Average of 5 numbers is 40. Four are 32, 38, 45, 50. Find the fifth."

Step 1: Total sum = $40 \times 5 = 200$

Step 2: Sum of four known = $32 + 38 + 45 + 50 = 165$

Step 3: Fifth = $200 - 165 = 35$

PART 6 — TIME & WORK, PIPES & CISTERNS

6.0 Opener

The LCM method — use it every time:

Set total work = LCM of all given time values.

Each person's rate = total work ÷ their solo time = an integer.

Add/subtract rates, then divide total work by combined rate.

Why LCM? It kills fractions. $\frac{1}{12} + \frac{1}{18}$ becomes $3 + 2 = 5$ (integers). Faster, fewer errors.

Type 1 — Two people working together

Cue: "A finishes in a days, B in b days. Together?"

Example. A alone: 12 days. B alone: 18 days. Together?

► METHOD A — Standard (safe every time)

Step 1: LCM(12, 18) = 36 → set as total work ← LCM converts all rates to whole numbers; you never deal with fractions like $\frac{1}{12}$ or $\frac{1}{18}$

Step 2: A's rate = $36 \div 12 = 3$ units/day

Step 3: B's rate = $36 \div 18 = 2$ units/day ← divide by each person's OWN solo time; swapping ($\div 18$ for A, $\div 12$ for B) is the most common error

Step 4: Combined = $3 + 2 = 5$ units/day

Answer: $36 \div 5 = 7.2$ days

Use when: Three or more workers, or someone leaves midway — LCM handles both cleanly.

✂ METHOD B — Exam Shortcut

Use when: Exactly two workers, no interruptions, and you want the fastest single-line answer.

$$\frac{ab}{a+b} = \frac{12 \times 18}{12+18} = \frac{216}{30} = 7.2 \text{ days} \leftarrow \text{works ONLY for two workers; if a third joins, use Method A}$$

Type 2 — B leaves midway; find A's solo time

Cue: "A and B together take 12 days. After 4 days B leaves; A finishes in 10 more days."

Step 1: Let total work = LCM approach; use fraction for this type.

Step 2: Work done in first 4 days (A+B together) = $\frac{4}{12} = \frac{1}{3}$

Step 3: Remaining work = $1 - \frac{1}{3} = \frac{2}{3}$

Step 4: A alone does $\frac{2}{3}$ of work in 10 days

Step 5: A alone does full work in $10 \div (\frac{2}{3}) = 10 \times \frac{3}{2} = 15 \text{ days}$

Type 3 — Efficiency ratio given

Cue: "A is twice as efficient as B. Together they finish in 10 days. How long for A alone?"

Step 1: Efficiency ratio A : B = 2 : 1

Step 2: Set total work = 30 units (choose LCM-friendly number)

Step 3: Together: (2 + 1) = 3 units/day; total time = $\frac{30}{3} = 10 \text{ days}$ ✓ (confirms our setup)

Step 4: A alone: $30 \div 2 = 15 \text{ days}$

Type 4 — Men × Days = constant (manpower scaling)

Cue: "15 men finish in 20 days. How many men for 10 days?"

Rule: Men × Days = constant (for the same total work).

Step 1: Total man-days = $15 \times 20 = 300$

Step 2: $n \times 10 = 300$

Answer: $n = 30 \text{ men}$

Extension: 15 men, 20 days, 8 hours/day. How many men to finish in 12 days, 10 hours/day?

Total man-day-hours = $15 \times 20 \times 8 = 2,400$

New scenario: $n \times 12 \times 10 = 2,400$

$n = 2,400 \div 120 = 20 \text{ men}$

Type 5 — Mixed workforce (men vs women)

Cue: "3 men OR 5 women can finish in 20 days. 2 men + 3 women finish in?"

Step 1: 3 men = 5 women (same output) $\rightarrow 1 \text{ man} = \frac{5}{3} \text{ women}$

Step 2: Convert 2 men to women-equivalent, then add 3 women. $2 \text{ men} = 2 \times \left(\frac{5}{3}\right) = \frac{10}{3} \text{ women-equivalent}$

Total = $\frac{10}{3} + 3 = \frac{10}{3} + \frac{9}{3} = \frac{19}{3} \text{ women-equivalent}$

Step 3: Find total work in woman-days. 5 women take 20 days

Total work = $5 \times 20 = 100 \text{ woman-days}$

Step 4: Find time for $\frac{19}{3}$ women. Time = Total work / Rate

$$= 100 \div \left(\frac{19}{3}\right)$$

$$= 100 \times \frac{3}{19}$$

$$= \frac{300}{19} \approx 15.8 \text{ days}$$

Type 6 — Pipes and leaks

Cue: "Pipe A fills tank in 10 hours. A leak empties it in 15 hours. With both, how long to fill?"

Rule: Fill rate = positive; leak rate = negative. Net rate = sum.

Step 1: Total capacity = LCM(10, 15) = 30 units

Step 2: A fills: $\frac{30}{10} = 3 \text{ units/hour}$

Step 3: Leak empties: $\frac{30}{15} = 2 \text{ units/hour}$

Step 4: Net rate = $3 - 2 = 1 \text{ unit/hour}$

Answer: Time = 30 hours

Type 7 — Alternate day work

Cue: "A works on Day 1, B on Day 2, alternating. A alone 8 days, B alone 12 days."

Step 1: Total work = LCM(8, 12) = 24 units

Step 2: A's rate = $\frac{24}{8} = 3 \text{ units/day}$; B's rate = $\frac{24}{12} = 2 \text{ units/day}$

Step 3: Every 2-day cycle: $3 + 2 = 5 \text{ units completed}$

Step 4: After 4 cycles (8 days): $4 \times 5 = 20 \text{ units done}$; remaining = $24 - 20 = 4 \text{ units}$

Step 5: Day 9 is A's turn: A does 3 units $\rightarrow$ remaining = $4 - 3 = 1 \text{ unit}$

Step 6: Day 10 is B's turn: B needs 1 unit at 2 units/day $\rightarrow$ takes $\frac{1}{2} \text{ day}$

Answer: Total = $9 + 0.5 = 9.5 \text{ days}$

Type 8 — Wages split proportional to work done

Cue: "A alone: 20 days, B alone: 30 days. Both work together; split ₹1800."

Step 1: Total work = LCM(20, 30) = 60 units

Step 2: A's rate = 3 units/day; B's rate = 2 units/day

Step 3: Ratio of work = 3 : 2

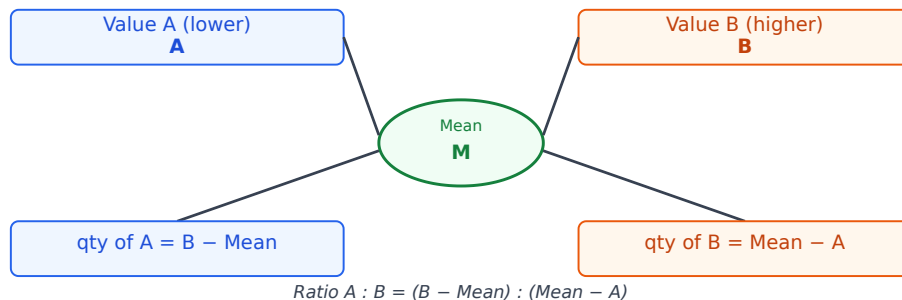
Step 4: A's share = $\left(\frac{3}{5}\right) \times 1800 = \text{₹}1080$

Step 5: B's share = $\left(\frac{2}{5}\right) \times 1800 = \text{₹}720$

PART 7 — MIXTURES & ALLIGATION

7.0 Opener

The Alligation Cross — the only tool you need:



Ratio A : B = (B - Mean) : (Mean - A)

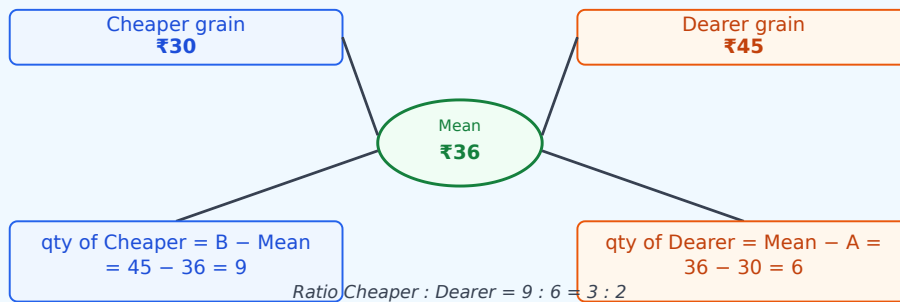
Rule: The cross difference on the RIGHT gives the quantity of the LEFT ingredient, and vice versa — always cross-subtract.

All differences are taken as positive. Works for price, concentration, percentage, speed, or any "average-of-two" scenario.

Type 1 — Two components mixed to achieve a target price

Cue: "Grains at ₹30/kg and ₹45/kg mixed to sell at ₹36/kg. What ratio?"

Step 1: Draw the alligation cross.



Step 2: (B - Mean) = 45 - 36 = **9** → quantity of A (cheaper grain)

Step 3: (Mean - A) = 36 - 30 = **6** → quantity of B (dearer grain)

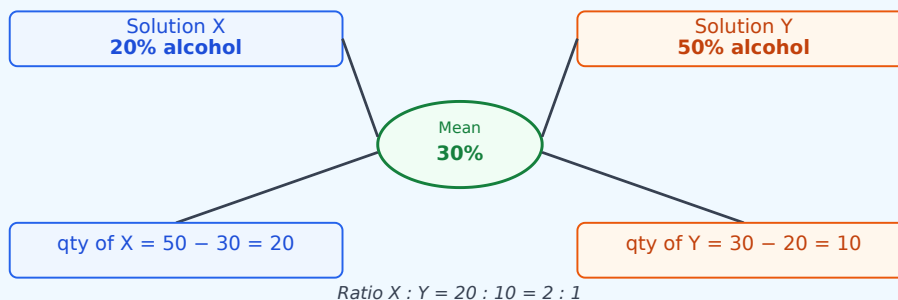
Answer: Ratio A : B = **9 : 6 = 3 : 2**

Interpretation: For every 3 kg of the cheaper grain, mix 2 kg of the costlier grain.

Type 2 — Concentration mixing (milk, alcohol, acid)

Cue: "Solution X has 20% alcohol, solution Y has 50% alcohol. Mix to get 30% alcohol. Ratio X : Y?"

Step 1: Draw the alligation cross.



Step 2: $(B - \text{Mean}) = 50 - 30 = 20 \rightarrow$ quantity of X (weaker solution)

Step 3: $(\text{Mean} - A) = 30 - 20 = 10 \rightarrow$ quantity of Y (stronger solution)

Answer: Ratio X : Y = 20 : 10 = 2 : 1

Type 3 — Repeated replacement (milk-water problem)

Cue: "Vessel has 40 L pure milk. Remove 10 L, add 10 L water. Repeat twice. Final milk?"

Formula: Final pure = Initial $\times (1 - x/V)^n$

where x = quantity replaced each time, V = total volume, n = number of replacements.

Step 1: $x = 10$ L, $V = 40$ L, $n = 2$

Step 2: Fraction of milk remaining per step. Fraction = $(V - x) / V = (40 - 10) / 40 = \frac{30}{40} = \frac{3}{4}$

Step 3: After 2 replacements: milk fraction = $(\frac{3}{4})^2 = \frac{9}{16}$

Step 4: Final milk = $40 \times \frac{9}{16} = 22.5$ L

Step 5 — Find milk % in final mixture. Milk % = (Final milk / Total volume) $\times 100$

$$= (22.5 / 40) \times 100$$
$$= 56.25\%$$

COMMON TRAP

Common mistake: Students compute $(1 - \frac{10}{40})^2 = (\frac{3}{4})^2$ but then forget to multiply by the original volume. The formula gives the *fraction* remaining; multiply by initial volume to get litres.

Type 4 — Profit via adulteration (mixing free component)

Cue: "Grocer buys wheat at ₹20/kg, mixes pebbles (cost ₹0) and sells mixture at ₹20/kg. Find profit %."

Example. Mixes 1 kg pebbles with 4 kg wheat; sells at ₹20/kg.

Step 1: CP = cost of 4 kg wheat = $4 \times 20 = ₹80$ (pebbles are free)

Step 2: SP = 5 kg $\times$ ₹20 = ₹100

Step 3: Profit = $100 - 80 = ₹20$

Answer: Profit % = $\frac{20}{80} \times 100 = 25\%$

Alligation shortcut: Ratio wheat:pebbles = 4:1. Mean price = ₹20 = cost of wheat $\rightarrow$ "gain" = $\frac{20}{16} = \dots$ use the direct approach above; it's faster.

Type 5 — "How much water to add / remove" to achieve target concentration

Cue: "30 L of 40% alcohol solution. Add water to make it 25%. How much water?"

Step 1: Pure alcohol = $40\% \times 30 = 12$ L (stays constant — we only add water)

Step 2: Let water added = x L

Step 3: New concentration: $12 / (30 + x) = \frac{25}{100} = \frac{1}{4}$

Step 4: $30 + x = 12 \times 4 = 48$

Answer: $x = 18$ L of water to add

Key insight: The pure alcohol (or whatever the solute is) does NOT change when you add solvent. Set up the equation on the solute.

PART 8 — TIME, SPEED, DISTANCE (TSD)

8.0 Opener

Core formulas — memorise these cold:

Formula	Notes
$D = S \times T$	The master equation
km/h $\rightarrow$ m/s: multiply by $\frac{5}{18}$	$72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$
m/s $\rightarrow$ km/h: multiply by $\frac{18}{5}$	$15 \text{ m/s} = 15 \times \frac{18}{5} = 54 \text{ km/h}$
Avg speed (equal distances): $2uv/(u+v)$	Harmonic mean — NOT $(u+v)/2$
Avg speed (equal times): $(u+v)/2$	Arithmetic mean
Relative speed (opposite): $u + v$	Approaching each other
Relative speed (same direction): $ u - v $	Chasing

Type 1 — Average speed (equal distance, different speeds)

Cue: "A goes from X to Y at 40 km/h and returns at 60 km/h. Average speed?"

► METHOD A — Standard (safe every time)

Step 1: Equal distances on both legs $\rightarrow$ formula: $\frac{2uv}{u+v}$ $\leftarrow$ "equal distance" is the trigger word; "equal time" is a different case (arithmetic mean applies there)

Step 2: $= \frac{2 \times 40 \times 60}{40 + 60} = \frac{4800}{100} = 48 \text{ km/h}$

Self-check: Answer must be less than $(40+60)/2 = 50$. $\leftarrow$ if your answer is ≥ 50 , you used the wrong formula

Use when: Both legs cover the same distance (typical exam phrasing: "goes and returns").

✂ METHOD B — Exam Shortcut

Use when: You forget the formula or want to verify. Takes ~30 extra seconds.

Assume distance each way = 120 km (LCM of 40 and 60 — gives whole-number times)

Time going: $120 \div 40 = 3$ hours **Time returning:** $120 \div 60 = 2$ hours

Average speed: $\frac{120 + 120}{3 + 2} = \frac{240}{5} = 48 \text{ km/h} \checkmark$

COMMON TRAP

Arithmetic mean trap: $(40 + 60) \div 2 = 50$ is WRONG when distances are equal. The harmonic mean always gives a value **less** than the arithmetic mean. If your answer is ≥ 50 , you fell into the trap.

Type 2 — Trains passing (the "effective length" method)

Rules:

Scenario	Distance covered	Time
Train passes a pole or man	Length of train	L / S
Train passes a platform	$L(\text{train}) + L(\text{platform})$	Sum / S
Two trains cross (opposite)	$L_1 + L_2$	$\text{Sum} / (S_1 + S_2)$
Two trains cross (same direction)	$L_1 + L_2$	$\text{Sum} / S_1 - S_2 $

Example. A 150 m train passes a 250 m platform in 20 seconds. Find the train's speed.

Step 1: Distance = train + platform = $150 + 250 = 400$ m

Step 2: Time = 20 s

Step 3: Speed = $\frac{400}{20} = 20$ m/s

Answer: = $20 \times \frac{18}{5} = 72$ km/h

Type 3 — Boats and streams

Key identities:

KEY FACT

- Downstream speed (D) = Boat speed + Stream speed = $b + s$
- Upstream speed (U) = Boat speed – Stream speed = $b - s$
- Boat speed = $(D + U) / 2$
- Stream speed = $(D - U) / 2$

Example. Downstream: 18 km in 2 hours. Upstream: 12 km in 3 hours. Find boat speed and stream speed.

Step 1: Downstream speed = $\frac{18}{2} = 9$ km/h

Step 2: Upstream speed = $\frac{12}{3} = 4$ km/h

Step 3: Boat speed = $(9 + 4)/2 = 6.5$ km/h

Step 4: Stream speed = $(9 - 4)/2 = 2.5$ km/h

Type 4 — Variable speed; arrive late/early

Cue: "At 4 km/h: 15 min late. At 5 km/h: 15 min early. Find the distance."

Step 1: Let distance = d km

Step 2: Time at 4 km/h – time at 5 km/h = total time difference = $15 + 15 = 30$ min = 0.5 h

Step 3: $d/4 - d/5 = 0.5$

Step 4: $(5d - 4d)/20 = 0.5 \rightarrow d/20 = 0.5$

Answer: $d = 10$ km

Type 5 — Races and head-starts

Cue: "In a 500 m race, A beats B by 100 m." = In the time A runs 500 m, B runs only 400 m.

Speed ratio A : B = 500 : 400 = 5 : 4

Variation — head-start: "A gives B a 50 m head-start in a 500 m race. A wins by 2 s. Find A's speed." $\rightarrow$ B runs 450 m in same time A runs 500 m, or use the time difference.

Type 6 — Clock angle problems

Formulas:

Quantity	Rate
Minute hand	6° per minute
Hour hand	0.5° per minute
Relative speed	5.5° per minute

Angle between hands at H hours M minutes: $|30H - 5.5M|^\circ$

Example. Angle between clock hands at 3:20?

Step 1: $H = 3, M = 20$

Step 2 — Apply the angle formula: $|30H - 5.5M|^\circ$ $30 \times H = 30 \times 3 = 90$

$5.5 \times M = 5.5 \times 20 = 110$

Angle = $|90 - 110| = |-20| = 20^\circ$

PART 9 — PARTNERSHIP

9.0 Opener

Master rule: Profit share $\propto$ (Capital $\times$ Time invested). Compute a weighted "investment-months" for each partner; split profit in that ratio.

Type 1 — Equal time, different capital

Example. A invests ₹5000, B invests ₹8000. Both for 1 year. Profit = ₹6500. Find each share.

Step 1: Ratio = 5000 : 8000 = 5 : 8

Step 2: Total parts = 5 + 8 = 13

Step 3: A's share = $\left(\frac{5}{13}\right) \times 6500 = \text{₹}2500$

Step 4: B's share = $\left(\frac{8}{13}\right) \times 6500 = \text{₹}4000$

Type 2 — Different durations (B joins late)

Example. A invests ₹6000 from January. B invests ₹9000 from April. Profit at year-end = ₹7500. Find shares.

Step 1: A's investment period = 12 months; B's = 12 - 3 = 9 months

Step 2: A's weight = 6000 $\times$ 12 = 72,000

Step 3: B's weight = 9000 $\times$ 9 = 81,000

Step 4: Ratio = 72,000 : 81,000 = 8 : 9

Step 5: A's share = $\frac{8}{17} \times 7500 = \text{₹}3529$

Step 6: B's share = $\frac{9}{17} \times 7500 = \text{₹}3971$

Type 3 — Capital changed midway

Example. A invests ₹4000 for 6 months, then adds ₹2000 more for next 6 months.

Step 1: Period 1: 4000 $\times$ 6 = 24,000

Step 2: Period 2: 6000 $\times$ 6 = 36,000

Step 3: A's effective investment = 24,000 + 36,000 = 60,000 investment-months

Type 4 — Working partner gets salary first

Example. A (working) and B (sleeping) invest ₹5000 each. A gets ₹500/month salary. Annual profit = ₹8400. Find A's total.

Step 1: A's annual salary = $500 \times 12 = ₹6000$

Step 2: Remaining profit = $8400 - 6000 = ₹2400$

Step 3: Capital ratio = $5000 : 5000 = 1 : 1$

Step 4: A's share from remaining = ₹1200; B's share = ₹1200

Answer: A's total = $6000 + 1200 = ₹7200$

Answer: B gets = ₹1200

PART 10 — AGES

Attack plan for every age problem: 1. Let present ages = variables (one unknown usually, two if both unknown). 2. Write the given ratio/sum as an equation. 3. Write the future/past condition as a second equation. 4. Solve. Age can NEVER be negative — check your answer.

Type 1 — Ratio now + ratio later (two-equation system)

Cue: "Ratio of A's age to B's age = 3 : 5. After 8 years, ratio = 5 : 7. Find present ages."

Step 1: Let present ages: $A = 3k$, $B = 5k$

Step 2: After 8 years: $(3k + 8)/(5k + 8) = \frac{5}{7}$

Step 3: Cross-multiply: $7(3k + 8) = 5(5k + 8)$

Step 4: $21k + 56 = 25k + 40$

Step 5: $4k = 16$

Step 6: $k = 4$

Step 7 — Find present ages. $A = 3 \times k = 3 \times 4 = 12$ years

$B = 5 \times k = 5 \times 4 = 20$ years

Type 2 — Sum and difference of ages

Example. The sum of father's and son's ages = 50. Father is $4\times$ the son's age. Find their ages.

Step 1: Let son = x , father = $4x$

Step 2: $x + 4x = 50$

Step 3: $5x = 50$

Step 4: $x = 10$

Answer: Son = 10 years, Father = 40 years

Type 3 — Father-son, n years ago / hence

Example. Father is $3\times$ son's age now. 10 years later, father is $2\times$ son's age.

Step 1: Let son's present age = s ; father's = $3s$

Step 2: After 10 years: $3s + 10 = 2(s + 10)$

Step 3: $3s + 10 = 2s + 20$

Step 4: $s = 10$

Answer: Son = 10 years, Father = 30 years

Type 4 — Average age of group changes

Example. Average age of 4 people = 24. A new person joins; average becomes 22. Find new person's age.

Step 1: Old sum = $24 \times 4 = 96$

Step 2: New sum = $22 \times 5 = 110$

Step 3: New person's age = $110 - 96 = 14$ years

COMMON TRAP

Watch: Average DROPPED when the new person joined. That means the new person is YOUNGER than the current average. Check your answer: $14 < 22$ (new avg) ✓.

PART 10B — NUMBER SYSTEM (HCF, LCM, Divisibility, Remainders, Unit Digit)

SSC CGL Tier-1 gives 3–5 questions from this chapter. IBPS PO Pre gives 2–3. The types below cover >95% of what examiners ask.

10B.0 Opener — types of numbers (memorise the definitions)

Type	Definition	Example
Natural numbers	1, 2, 3, ...	{1, 2, 3, ...}
Whole numbers	0, 1, 2, 3, ...	{0, 1, 2, ...}
Integers	...-2, -1, 0, 1, 2, ...	All of $\mathbb{Z}$
Prime	Exactly 2 factors (1 and itself)	2, 3, 5, 7, 11, 13, ...
Composite	More than 2 factors	4, 6, 8, 9, 10, ...
Co-prime (relative prime)	HCF = 1 (not necessarily prime individually)	(8, 9), (4, 9)
Perfect square	$n = k^2$ for integer k	1, 4, 9, 16, 25, ...
Perfect cube	$n = k^3$	1, 8, 27, 64, 125, ...

KEY FACT

Prime up to 100 (count = 25): 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. Examiners ask "how many primes between X and Y?" — count from this list.

Type 1 — HCF (Highest Common Factor)

Two methods — both exam-valid:

Method A — Prime factorisation: HCF = product of common prime factors at their LOWEST powers. **Method B — Euclid's algorithm:** Divide larger by smaller, replace with remainder; repeat until remainder = 0; last non-zero divisor = HCF.

Example. HCF of 48 and 36.

Method A — Prime factorisation:

$$48 = 2^4 \times 3$$

$$36 = 2^2 \times 3^2$$

Common factors at lowest powers: 2^2 and 3^1

$$\text{HCF} = 4 \times 3 = \mathbf{12}$$

Method B — Euclid's algorithm:

$$48 \div 36 = 1 \text{ remainder } \mathbf{12}$$

$$36 \div 12 = 3 \text{ remainder } \mathbf{0}$$

Last non-zero remainder = HCF = $\mathbf{12}$ ✓

Self-check. HCF of 84 and 126. $\rightarrow 84 = 2^2 \times 3 \times 7$; $126 = 2 \times 3^2 \times 7$. Common: $2^1 \times 3^1 \times 7^1 = \mathbf{42}$.

Type 2 — LCM (Least Common Multiple)

Method — prime factorisation: LCM = product of all prime factors at their HIGHEST powers.

Example. LCM of 48 and 36. All prime factors: 2^4 and 3^2 . LCM = $16 \times 9 = 144$.

Key identity (for 2 numbers only): $\text{HCF} \times \text{LCM} = N_1 \times N_2$. Verify: $12 \times 144 = 1,728 = 48 \times 36 \checkmark$.

Self-check. LCM of 12, 18, 24. $12 = 2^2 \times 3$; $18 = 2 \times 3^2$; $24 = 2^3 \times 3$. LCM = $2^3 \times 3^2 = 8 \times 9 = 72$.

Type 3 — Word problems using HCF / LCM

HCF problems: "Largest tile / rope / measure / group size" → HCF. **LCM problems:** "Smallest number divisible by all / when two events next coincide" → LCM.

Example (HCF). Three ropes 36 m, 48 m, 60 m. Largest equal-length pieces with no waste? $\text{HCF}(36, 48, 60) = 12$. Pieces = $3 + 4 + 5 = 12$ pieces, each **12 m**.

Example (LCM). Traffic lights at intervals of 48 s and 72 s both turn green together. When next? $\text{LCM}(48, 72) = 144 \rightarrow 144$ seconds (**2 min 24 s**).

Example (LCM trick). Smallest number divisible by 4, 6, 9, 15? $\text{LCM}(4,6,9,15) = \text{LCM}(4,6) \times \text{adjust}$: $4=2^2$, $6=2 \times 3$, $9=3^2$, $15=3 \times 5$. LCM = $2^2 \times 3^2 \times 5 = 180$.

Type 4 — Divisibility rules (memorise all 13)

Divisor	Rule
2	Last digit even (0, 2, 4, 6, 8)
3	Digit sum divisible by 3
4	Last 2 digits form a number divisible by 4
5	Last digit 0 or 5
6	Divisible by both 2 AND 3
7	Double the last digit; subtract from remaining number; repeat. If result divisible by 7, so is original.
8	Last 3 digits divisible by 8
9	Digit sum divisible by 9
10	Last digit 0
11	Alternating digit sum (from right) divisible by 11
12	Divisible by both 3 AND 4
13	Triple the last digit; add to remaining number; repeat. If divisible by 13, so is original.
25	Last 2 digits are 00, 25, 50, or 75

Example. Is 1,729 divisible by 7?

Last digit = 9; double it = 18

Remaining number = 172; subtract: $172 - 18 = 154$

Repeat: last digit = 4; double = 8; remaining = 15; subtract: $15 - 8 = 7$

7 is divisible by 7 → **YES**

Type 5 — Remainders

Basic remainders: $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$. Same for products.

Example. Remainder when 73×54 is divided by 6. $73 \bmod 6 = 1$; $54 \bmod 6 = 0$. Product mod 6 = $0 \times 1 = 0$.

Cyclicity-based remainders: Find the pattern of $r^n \bmod m$.

Example. Remainder when 2^{100} is divided by 7.

$$2^1 \bmod 7 = 2$$

$$2^2 \bmod 7 = 4$$

$$2^3 \bmod 7 = 8 \bmod 7 = 1 \rightarrow \text{cycle restarts}$$

Cycle: {2, 4, 1}, length = 3

$$100 \bmod 3 = 1 \rightarrow \text{take 1st element of cycle} = 2$$

Type 6 — Unit digit / last digit of x^n

Memorise the unit-digit cycles:

Unit digit of base	Cycle	Length
0, 1, 5, 6	0, 1, 5, 6 (themselves)	1
4, 9	$4 \rightarrow 6 \rightarrow 4 \rightarrow 6 \dots$ and $9 \rightarrow 1 \rightarrow 9 \rightarrow 1 \dots$	2
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

Example. Unit digit of 3^{47} . Cycle {3, 9, 7, 1}, length 4. $47 \bmod 4 = 3 \rightarrow$ 3rd position = **7**.

Example. Unit digit of 4^{31} . Cycle {4, 6}, length 2. $31 \bmod 2 = 1 \rightarrow$ 1st = **4**.

Type 7 — Number of factors (divisors)

Formula. If $N = p_1^a \times p_2^b \times p_3^c \times \dots$, number of factors = $(a+1)(b+1)(c+1)\dots$

Example. Number of factors of 360. $360 = 2^3 \times 3^2 \times 5^1$. Factors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = \mathbf{24}$.

Even vs odd factors:

Even factors: reduce power of 2 by one choice (min power = 1, not 0). Even factors of 360 = $3 \times 3 \times 2 = \mathbf{18}$.

Odd factors: include only odd prime bases (2^0). Odd factors of 360 = $(2+1)(1+1) = \mathbf{6}$.

Verify: $18 + 6 = 24 \checkmark$.

Sum of all divisors formula. If $N = p_1^a \times p_2^b \times p_3^c$, then:

$$\text{Sum of all divisors} = \frac{p_1^{a+1} - 1}{p_1 - 1} \times \frac{p_2^{b+1} - 1}{p_2 - 1} \times \frac{p_3^{c+1} - 1}{p_3 - 1}$$

Example. Sum of divisors of 12. $12 = 2^2 \times 3^1$.

$$\text{Sum} = \frac{2^3 - 1}{2 - 1} \times \frac{3^2 - 1}{3 - 1} = \frac{7}{1} \times \frac{8}{2} = 7 \times 4 = \mathbf{28}.$$

Verify by listing: $1 + 2 + 3 + 4 + 6 + 12 = 28 \checkmark$.

Type 8 — Advanced number theory (Wilson, Fermat, CRT)

These appear in SSC CGL Tier-2 and occasionally in Banks Mains. Knowing the theorems converts 2-minute problems into 10-second eliminations.

Wilson's Theorem. For any prime p : $(p - 1)! \equiv -1 \pmod{p}$.

Equivalently: $(p - 1)! + 1$ is divisible by p .

Example. Find remainder when $6!$ is divided by 7. Here $p = 7$ (prime). Wilson: $(7-1)! = 6! \equiv -1 \equiv 6 \pmod{7}$.

Remainder = 6.

Quick check: $6! = 720$. $720 \div 7 = 102$ remainder **6** ✓.

Fermat's Little Theorem. If p is prime and $\gcd(a, p) = 1$, then:

$$a^{(p-1)} \equiv 1 \pmod{p}$$

Example. Find remainder when 2^{100} is divided by 101. Since 101 is prime and $\gcd(2, 101) = 1$: $2^{100} \equiv 1 \pmod{101}$. **Remainder = 1.**

Why it matters for exams: Whenever the exponent is $p-1$ and the modulus is prime p , the remainder is 1 instantly — no need to find cycles.

Chinese Remainder Theorem (CRT) — basic exam version.

When a number N leaves remainder r_1 when divided by m_1 , and remainder r_2 when divided by m_2 (where $\gcd(m_1, m_2) = 1$), there exists a unique N in $[0, m_1 m_2)$ satisfying both conditions.

SSC exam pattern: "A number when divided by 5 leaves remainder 3, and when divided by 7 leaves remainder 2. Find the smallest such number."

Solve: Need $N \equiv 3 \pmod{5}$ and $N \equiv 2 \pmod{7}$.

List numbers $\equiv 3 \pmod{5}$: 3, 8, 13, 18, 23, 28, ...

Check which is $\equiv 2 \pmod{7}$: $23 \bmod 7 = 2$ ✓. Smallest $N = \mathbf{23}$.

(General solution: $N = 23 + 35k$ for any integer $k \geq 0$.)

Type 9 — "Find the missing digit for divisibility" problems

Pattern: "The number 3_6 is divisible by 9. Find the missing digit."

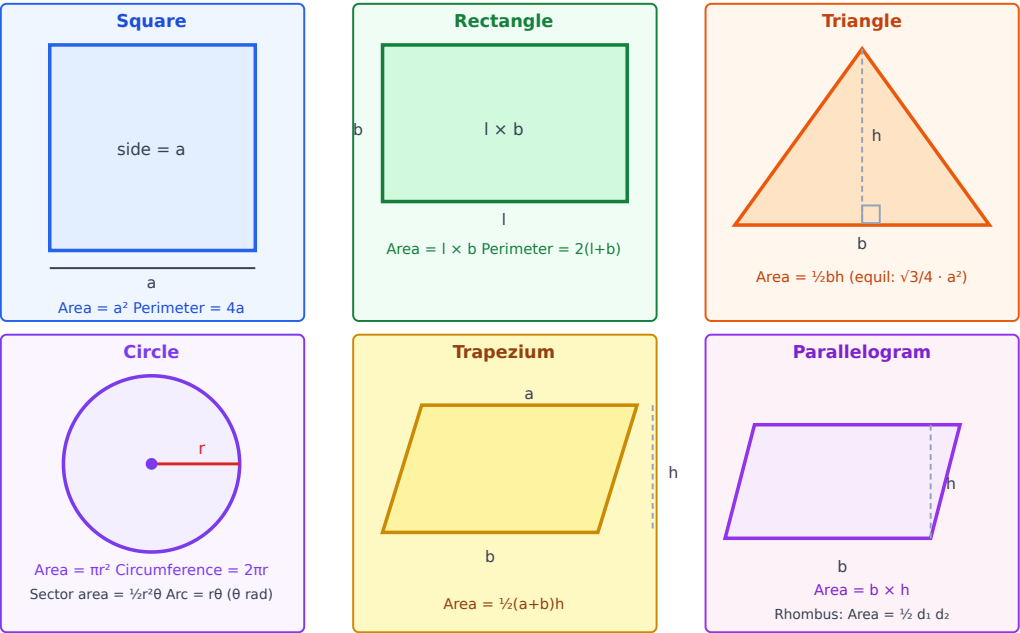
Method: Digit sum must be divisible by 9. $3 + \square + 6 = 9 + \square$. For divisibility by 9: $\square = \mathbf{0}$ (sum = 9) or $\square = \mathbf{9}$ (sum = 18).

Example. 5_2 is divisible by 6. Divisible by 6 = divisible by 2 AND 3. Already ends in 2 (even ✓). Digit sum = $5 + \square + 2 = 7 + \square$ must be divisible by 3 $\rightarrow \square = \mathbf{2}$ (sum 9) or $\square = \mathbf{5}$ (sum 12). Both work; examiners usually expect the smallest: $\square = \mathbf{2}$.

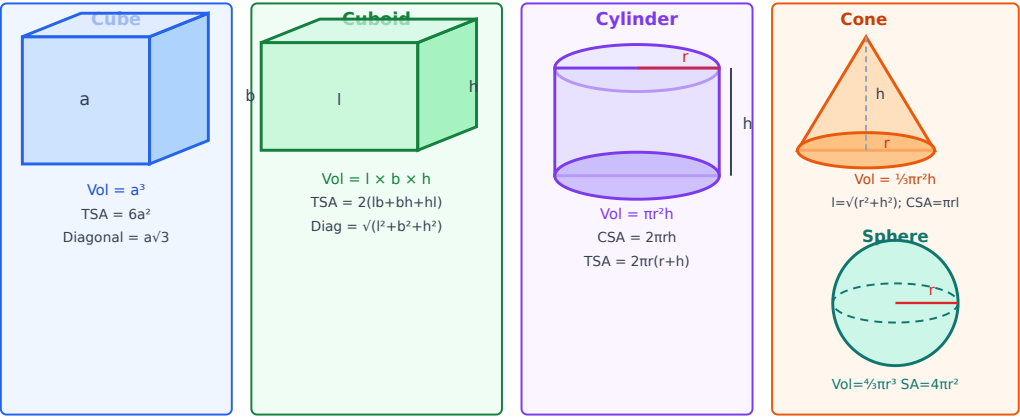
PART 11 — MENSURATION (2D & 3D)

11.0 Opener — Visual formula reference

All 2D shapes at a glance:



All 3D solids at a glance:



Quick-recall table — 2D:

Shape	Area	Perimeter
Square (side a)	a^2	$4a$
Rectangle ($l \times b$)	$l \times b$	$2(l + b)$
Triangle (base b, height h)	$\frac{1}{2}bh$	$a + b + c$
Equilateral triangle (side a)	$(\frac{\sqrt{3}}{4})a^2$	$3a$
Circle (radius r)	πr^2	$2\pi r$
Sector (r, angle θ°)	$(\frac{\theta}{360})\pi r^2$	$2r + r\theta$ (θ rad: $\frac{1}{2}r^2\theta$)
Trapezium (parallel a, b; height h)	$\frac{1}{2}(a + b)h$	$a + b + \text{leg}_1 + \text{leg}_2$
Parallelogram (base b, height h)	bh	$2(\text{adj. sides})$

Shape	Area	Perimeter
Rhombus (diagonals d_1, d_2)	$\frac{1}{2}d_1d_2$	$4 \times \text{side}$

Quick-recall table — 3D:

Solid	Volume	CSA	TSA
Cube (side a)	a^3	$4a^2$	$6a^2$
Cuboid ($l \times b \times h$)	lbh	$2h(l + b)$	$2(lb + bh + hl)$
Cylinder (r, h)	$\pi r^2 h$	$2\pi rh$	$2\pi r(r + h)$
Cone ($r, h, l = \sqrt{r^2 + h^2}$)	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r + l)$
Sphere (r)	$\frac{4}{3}\pi r^3$	—	$4\pi r^2$
Hemisphere (r)	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
Frustum ($R, r, h, l = \sqrt{h^2 + (R - r)^2}$)	$\frac{1}{3}\pi h(R^2 + r^2 + Rr)$	$\pi l(R + r)$	$\pi l(R + r) + \pi(R^2 + r^2)$
Pyramid (sq. base a , slant l , ht h)	$\frac{1}{3}a^2 h$	$\frac{1}{2} \times 4a \times l = 2al$	$a^2 + 2al$
Prism (base area A_b , perimeter p , ht h)	$A_b \times h$	$p \times h$ (lateral)	$2A_b + ph$

Type 1 — Direct substitution (area / perimeter)

Recognition cue: Straight shape given $\rightarrow$ find area or perimeter. No trick — just formula + arithmetic.

Example. A triangle has base 12 cm, height 9 cm. Area = $\frac{1}{2} \times 12 \times 9 = 54 \text{ cm}^2$.

Example. Rectangle 15×8 . Perimeter = $2(15 + 8) = 46 \text{ cm}$. Area = $15 \times 8 = 120 \text{ cm}^2$.

Self-check. Equilateral triangle side 6 cm. Area = $(\sqrt{\frac{3}{4}}) \times 36 = 9\sqrt{3} \text{ cm}^2$. Perimeter = 18 cm.

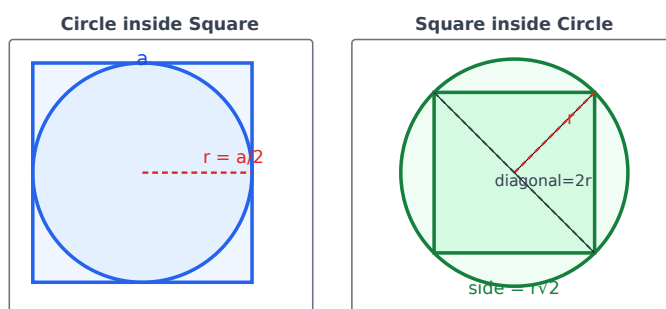
Type 2 — Optimisation (largest/smallest shape inscribed in another)

Recognition cue: "Largest circle inscribed in a square," "largest square inscribed in a circle" — relate the dimensions.

Key rules: Largest circle in a square of side $a \rightarrow$ radius $r = a/2$.

Largest square inscribed in a circle of radius $r \rightarrow$ side = $r\sqrt{2}$; diagonal = $2r$.

Largest equilateral triangle in a circle of radius $r \rightarrow$ side = $r\sqrt{3}$.



Example. Largest circle carved from a square tile of side 14 cm. Find its area.

Given: Side of square = 14 cm

Step 1: Largest inscribed circle has radius = side $\div 2 = 14 \div 2 = 7 \text{ cm}$

Step 2: Area = $\pi \times r^2 = (\frac{22}{7}) \times 7^2 = (\frac{22}{7}) \times 49 = 22 \times 7$

Answer: Area = 154 cm^2

Self-check. Side of the largest square that fits in a circle of radius 5 cm $\rightarrow$ side = $5\sqrt{2} \approx 7.07$ cm.

Type 3 — Path / road around a rectangle or garden

Recognition cue: "A path of width w surrounds a garden of $l \times b$. Find area of path."

Method: Outer rectangle dimensions = $(l + 2w) \times (b + 2w)$. Area of path = Outer – Inner.

Example. Garden 40×30 m, path 2 m wide. Outer = $44 \times 34 = 1,496$. Inner = $40 \times 30 = 1,200$. Path area = **296 m²**.

Watch out: Width w is added on BOTH sides — total increase in each dimension = $2w$, not w .

Type 4 — Shaded-region problems

Recognition cue: A shape has something cut out (circle from rectangle, square from circle). Area = Large – Small.

Example. Square of side 14 cm has the largest possible circle cut out. Shaded area = $14^2 - \pi(7^2) = 196 - 154 =$ **42 cm²**.

Example. Ring: outer radius 7 cm, inner radius 3.5 cm. Find its area.

Given: R (outer) = 7 cm, r (inner) = 3.5 cm

Step 1: Area of ring = $\pi(R^2 - r^2) = \pi \times (7^2 - 3.5^2) = \pi \times (49 - 12.25) = \pi \times 36.75$

Step 2: Substitute $\pi = \frac{22}{7} = \left(\frac{22}{7}\right) \times 36.75 = 22 \times 5.25$

Answer: Area = **115.5 cm²**

Type 5 — Combined solids (volume and surface area)

Recognition cue: Two solids joined (cone + hemisphere = ice-cream cone; cylinder + cone = tent).

Volume: Add volumes. **Surface area:** Add CSA only (not base of cone + flat circle of hemisphere when they share a face).

Example. A toy is a cone ($r = 3.5$ cm, $h = 4$ cm) placed on top of a hemisphere (same $r = 3.5$ cm).

Step 1 — Find slant height of cone.

$$l = \sqrt{(r^2 + h^2)} = \sqrt{(3.5^2 + 4^2)} = \sqrt{(12.25 + 16)} = \sqrt{28.25} \approx \mathbf{5.315 \text{ cm}}$$

Step 2 — Volume of cone.

$$V_1 = \left(\frac{1}{3}\right) \times \pi \times r^2 \times h = \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 12.25 \times 4 = \mathbf{51.33 \text{ cm}^3}$$

Step 3 — Volume of hemisphere.

$$V_2 = \left(\frac{2}{3}\right) \times \pi \times r^3 = \left(\frac{2}{3}\right) \times \left(\frac{22}{7}\right) \times 3.5^3 = \left(\frac{2}{3}\right) \times \left(\frac{22}{7}\right) \times 42.875 = \mathbf{89.83 \text{ cm}^3}$$

Step 4 — Total volume.

$$V = V_1 + V_2 = 51.33 + 89.83 = \mathbf{141.16 \text{ cm}^3}$$

Step 5 — Total surface area (CSA of cone + curved surface of hemisphere; shared flat circle is interior).

$$\text{TSA} = \pi r l + 2\pi r^2 = \left(\frac{22}{7}\right) \times 3.5 \times 5.315 + 2 \times \left(\frac{22}{7}\right) \times 3.5^2 \approx 58.47 + 77 = \mathbf{135.47 \text{ cm}^2}$$

Type 6 — Similar figures (ratio of sides $\rightarrow$ ratio of areas/volumes)

Rule. If ratio of corresponding sides = $k : 1$, then ratio of areas = $k^2 : 1$, ratio of volumes = $k^3 : 1$.

Example. Two similar triangles with sides 4 cm and 6 cm. Ratio of areas = $4^2 : 6^2 = 16 : 36 = 4 : 9$.

Example. Two spheres with radii 3 cm and 6 cm. Ratio of volumes = $3^3 : 6^3 = 27 : 216 = 1 : 8$.

| Type 7 — Frustum of cone (truncated cone = bucket / flower pot shape)

Formulae. R = bottom radius, r = top radius, h = height, l = slant height = $\sqrt{h^2 + (R-r)^2}$.

$$\text{Volume} = (\pi h/3)(R^2 + Rr + r^2)$$

$$\text{CSA} = \pi l(R + r)$$

$$\text{TSA} = \pi l(R + r) + \pi(R^2 + r^2)$$

Example. Bucket: R = 21 cm, r = 14 cm, h = 24 cm.

$$\text{Slant height: } l = \sqrt{h^2 + (R - r)^2} = \sqrt{576 + 49} = \sqrt{625} = 25 \text{ cm}$$

$$R + r = 21 + 14 = 35 \text{ cm}$$

$$\text{CSA} = \pi \times l \times (R + r) = \left(\frac{22}{7}\right) \times 25 \times 35 = 22 \times 25 \times 5 = 2,750 \text{ cm}^2$$

| Type 8 — Rolling / rotation (wheel, cylinder rolling)

Rule. Distance covered in one revolution = circumference = $\pi D = 2\pi r$.

Example. Wheel radius 35 cm makes 500 revolutions. Distance = $500 \times 2\pi \times 35 = 500 \times \frac{220}{7} = 500 \times 31.43 \approx 110,000 \text{ cm} = 1.1 \text{ km}$.

| Type 9 — Effect of changing one dimension on area / volume

Rule. If radius increases by r%, area increases by $[2r + r^2/100]\%$ (i.e., successive percentage of TWO sides).
Volume has THREE dimensions.

Example. Each side of a cube increases by 20%. New volume = $1.2^3 = 1.728$ times $\rightarrow$ volume increases by **72.8** %.

Example. Radius of cylinder +10%, height -10%. New volume = $\pi(1.1r)^2(0.9h) = 1.089\pi r^2 h \rightarrow$ increases by **8.9** %.

| Type 10 — Triangle-inside-triangle, midpoint theorem

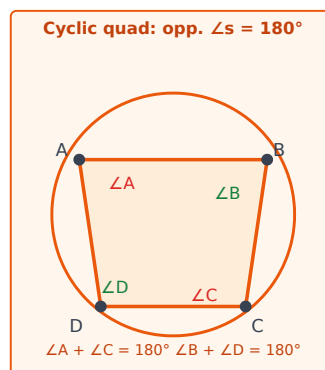
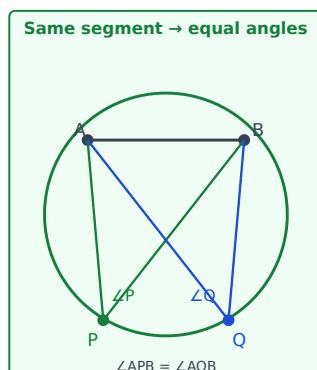
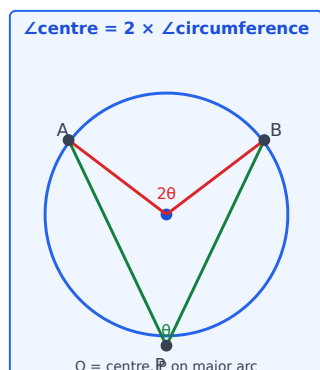
Midpoint theorem. Line joining midpoints of two sides of a triangle is parallel to the third side and HALF its length.

Example. In $\triangle ABC$, D and E are midpoints of AB and AC. If BC = 24 cm, then DE = **12 cm**.

Medians and centroid. Centroid divides each median in 2:1 from vertex. Area of each of the 6 sub-triangles formed by medians = total area / 6.

PART 12 — GEOMETRY

Circle theorems — visual reference



Type 1 — Triangle basics

Key rules to memorise: Angle sum = 180° . Exterior angle = sum of two non-adjacent interior angles.

Triangle inequality: each side < sum of other two sides.

Example. In $\triangle ABC$, $\angle A = 65^\circ$, $\angle B = 55^\circ$. Find the exterior angle at C.

$$\angle C = 180^\circ - 65^\circ - 55^\circ = 60^\circ$$

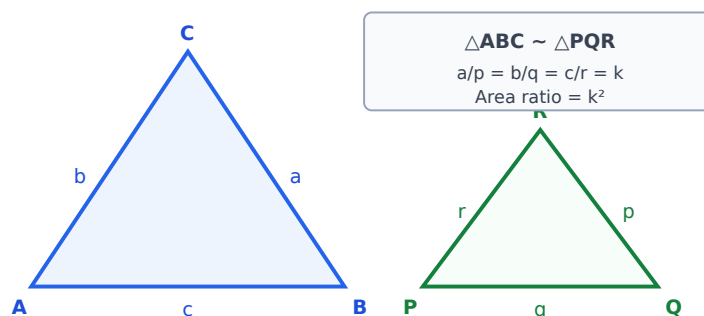
$$\text{Exterior angle at C} = 180^\circ - 60^\circ = 120^\circ$$

EXAM SHORTCUT

Shortcut: Exterior angle = sum of the two non-adjacent interior angles = $65^\circ + 55^\circ = 120^\circ$ ✓

Type 2 — Similarity & congruence

Similarity criteria: AA (two angles equal), SAS (ratio of two sides equal + included angle), SSS (all three side ratios equal).



Key property: If $\triangle ABC \sim \triangle DEF$ with ratio $k:1$, then:

Corresponding sides in ratio $k:1$.

Areas in ratio $k^2:1$.

Example. $\triangle ABC \sim \triangle PQR$. $AB = 6$ cm, $PQ = 9$ cm. Area of $\triangle ABC = 24$ cm². Area of $\triangle PQR$?

$$\text{Ratio of sides} = 6 : 9 = 2 : 3$$

$$\text{Ratio of areas} = 2^2 : 3^2 = 4 : 9$$

$$24 / \text{area(PQR)} = \frac{4}{9}$$

$$\text{area(PQR)} = 24 \times \frac{9}{4} = \mathbf{54 \text{ cm}^2}$$

Type 3 — Basic proportionality theorem (Thales) + Midpoint

BPT (Thales): A line parallel to one side of a triangle divides the other two sides in equal ratios.

Example. In $\triangle ABC$, $DE \parallel BC$. $AD = 4 \text{ cm}$, $DB = 6 \text{ cm}$. $AE = 5 \text{ cm}$. Find EC .

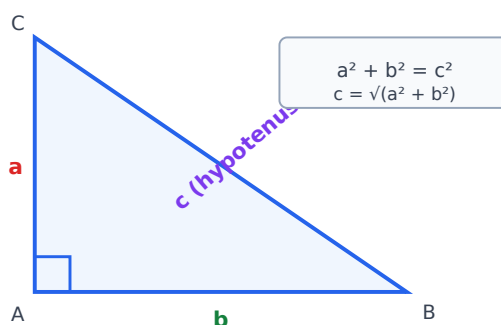
By BPT: $AD/DB = AE/EC$

Substitute: $\frac{4}{6} = 5/EC$

Cross-multiply: $EC = 5 \times \frac{6}{4} = \mathbf{7.5 \text{ cm}}$

Type 4 — Pythagorean theorem + Pythagorean triples

Theorem: $a^2 + b^2 = c^2$ (where c = hypotenuse).



Triples to recognise on sight (saves computation):

Triple	Multiples to watch
3-4-5	6-8-10, 9-12-15, 15-20-25
5-12-13	10-24-26
8-15-17	—
7-24-25	—
20-21-29	—

Example. Ladder 13 m leans against a wall; foot is 5 m from wall. How high does it reach? 5-12-13 triple → height = **12 m** (no calculation needed).

Type 5 — Centroid, circumcentre, incentre, orthocentre

Point	Where	Key property
Centroid	Intersection of medians	Divides each median 2:1 from vertex
Circumcentre	Perpendicular bisectors of sides	Equidistant from all 3 vertices
Incentre	Angle bisectors	Equidistant from all 3 sides (= inradius)
Orthocentre	Altitudes	Angle with circumcentre + centroid are collinear (Euler line)

Example. Centroid of triangle with vertices $(0,0)$, $(6,0)$, $(3,9)$ = $((0+6+3)/3, (0+0+9)/3) = \mathbf{(3, 3)}$.

Type 6 — Circle theorems

Three theorems examiners test most:

1. Angle at centre = $2 \times$ angle at circumference (subtended by same arc).
2. Angles in the same segment are equal.
3. Opposite angles of a cyclic quadrilateral sum to 180° .

Example. Arc AB subtends $\angle AOB = 140^\circ$ at the centre. What angle does it subtend at the circumference (on the major arc)? Angle at circumference = $140^\circ/2 = 70^\circ$.

Example. In cyclic quadrilateral ABCD, $\angle A = 95^\circ$. Find $\angle C$. $\angle A + \angle C = 180^\circ \rightarrow \angle C = 85^\circ$.

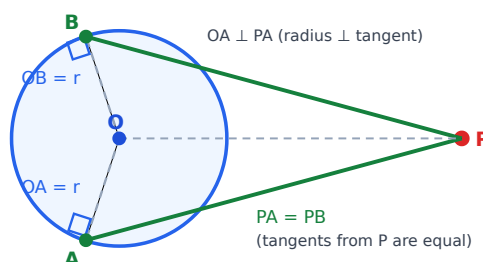
Type 7 — Tangent properties

Rules: Tangent to a circle is perpendicular to the radius at the point of tangency.

Two tangents drawn from an external point are equal in length.

If PT and PS are tangents from P to circle centre O, then $PT = PS$.

$\angle TPO = \angle SPO$ (the external point lies on the angle bisector of the two radii).

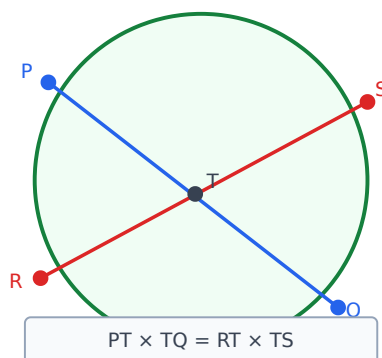


Example. From an external point P, two tangents PA and PB touch a circle at A and B. $PA = 8$ cm. What is PB? $\rightarrow PB = 8$ cm (tangents from external point are equal).

Example. $OP = 10$ cm, radius $OA = 6$ cm. Length of tangent $PA = \sqrt{OP^2 - OA^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm.

Type 8 — Intersecting chords and secants

Chord intersection (inside circle): If two chords PQ and RS intersect at T inside the circle, then $PT \times TQ = RT \times TS$.



Secant-secant from external point: $PA \times PB = PC \times PD$ (where A, B and C, D are the near and far intersection points on each secant).

Example. Two chords intersect inside a circle. One chord is divided 4 and 6 (= 24). Other chord has one segment 3 cm. Find the other segment: $3 \times x = 24 \rightarrow x = 8 \text{ cm}$.

Type 9 — Polygon angle sums

Interior angle sum = $(n - 2) \times 180^\circ$. Exterior angle sum = always 360° (for any convex polygon).

Regular polygon: Each interior angle = $(n-2) \times 180^\circ / n$. Each exterior angle = $360^\circ / n$.

Example. How many sides does a regular polygon have if each interior angle = 150° ?

Each exterior angle = $180^\circ - 150^\circ = 30^\circ$

Number of sides = $360^\circ \div 30^\circ = 12$

Type 10 — Coordinate geometry

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Mid-point: $((x_1 + x_2)/2, (y_1 + y_2)/2)$

Section formula (internal division in $m:n$): $((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$

Slope: $m = (y_2 - y_1)/(x_2 - x_1)$. Parallel lines: equal slope. Perpendicular lines: $m_1 \times m_2 = -1$.

Example. Slope of line joining (2, 5) and (6, 13) = $(13-5)/(6-2) = \frac{8}{4} = 2$.

PART 13 — TRIGONOMETRY (for SSC-CGL / RRB NTPC level)

13.0 Opener

Standard angle values — the non-negotiable table:

θ	sin	cos	tan
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	∞

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Complementary: $\sin(90^\circ - \theta) = \cos \theta$, and similarly for all ratios.

Type 1 — Standard-angle substitution

Attack: Replace θ with $30^\circ/45^\circ/60^\circ/90^\circ$ and evaluate using the table above.

Example. Find: $2\sin 60^\circ - \tan 45^\circ + \cos 30^\circ = 2(\frac{\sqrt{3}}{2}) - 1 + (\frac{\sqrt{3}}{2}) = \sqrt{3} - 1 + \frac{\sqrt{3}}{2} = 3\frac{\sqrt{3}}{2} - 1 \approx 1.598$.

Type 2 — Complementary-angle simplifications

Rule: $\sin(90^\circ - \theta) = \cos \theta$, $\cos(90^\circ - \theta) = \sin \theta$, $\tan(90^\circ - \theta) = \cot \theta$.

Example. $\sin 35^\circ / \cos 55^\circ = \sin 35^\circ / \sin(90^\circ - 55^\circ) = \sin 35^\circ / \sin 35^\circ = 1$.

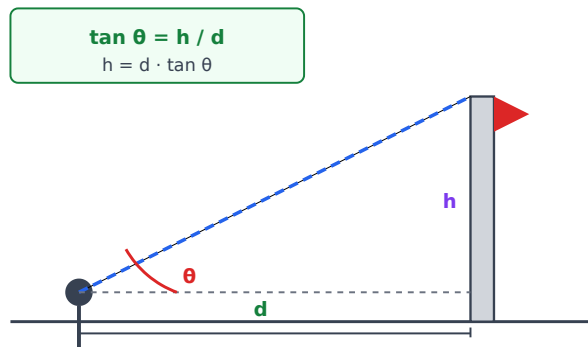
Type 3 — Identity-based simplifications

Tip: Convert all to sin/cos, or use: $\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$.

Example. $(\sin\theta + \cos\theta)^2 - 1 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta - 1 = 1 + 2\sin\theta\cos\theta - 1 = 2\sin\theta\cos\theta = \sin 2\theta$.

Type 4 — Height & Distance (angle of elevation / depression)

Method: Draw the right triangle. Label the known angle (angle of elevation from ground, or depression from height). Use tan for opposite/adjacent.



Example. A tower is observed at an angle of elevation of 60° from a point 20 m away (on the ground). Find the height of the tower.

$$\tan 60^\circ = \text{height} / 20$$

$$\sqrt{3} = \text{height} / 20$$

$$\text{height} = 20 \times \sqrt{3} = 20\sqrt{3} \approx 34.64 \text{ m}$$

Example. From the top of a cliff 30 m high, the angle of depression of a boat is 30° . Find the horizontal distance of the boat from the cliff base.

Angle of depression = 30° (measured from the horizontal downward)

$$\tan 30^\circ = 30 / \text{distance}$$

$$1/\sqrt{3} = 30 / \text{distance}$$

$$\text{distance} = 30 \times \sqrt{3} = 30\sqrt{3} \approx 51.96 \text{ m}$$

Watch out: "Angle of depression" is measured FROM the horizontal DOWNWARD. The angle inside the right triangle (from the horizontal to the line of sight) equals the angle of elevation seen from the boat.

Type 5 — Two observers / two angles

Setup: Tower of height h . Points A and B on the ground on the same side, at distances d_1 and d_2 . Angles of elevation α and β .

$\tan \alpha = h/d_1$ and $\tan \beta = h/d_2$. Eliminate h or d to find the unknown.

Example. From A and B (100 m apart), angles of elevation of a tower are 60° and 30° . Find tower height.

Let: h = height of tower, d = distance from A (nearer point) to base of tower

Step 1: From point A (angle = 60°): $\tan 60^\circ = h / d$ $\sqrt{3} = h / d \therefore d = h / \sqrt{3}$

Step 2: From point B (angle = 30° , which is 100 m behind A): $\tan 30^\circ = h / (d + 100)$ $1/\sqrt{3} = h / (d + 100) \therefore d + 100 = h\sqrt{3}$

Step 3: Substitute $d = h/\sqrt{3}$ into Step 2: $h/\sqrt{3} + 100 = h\sqrt{3}$ $100 = h\sqrt{3} - h/\sqrt{3}$ $100 = h(\sqrt{3} - 1/\sqrt{3})$ $100 = h \times (3 - 1)/\sqrt{3}$ $100 = 2h/\sqrt{3}$

Step 4: Solve for h : $h = 100\sqrt{\frac{3}{2}}$ $h = 50\sqrt{3}$

Answer: $h = 50\sqrt{3} \approx 86.6$ m

Type 6 — Compound angles, double angles, and max/min of trigonometric expressions

These appear in SSC CGL Tier-2 and Banks PO Mains. Memorise the six formulas below — most SSC compound-angle questions reduce to direct substitution.

Compound angle formulas:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Double angle formulas:

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

t-substitution ($t = \tan \theta/2$ — converts trig expressions to rational form):

$$\sin \theta = \frac{2t}{1 + t^2}; \cos \theta = \frac{1 - t^2}{1 + t^2}; \text{ where } t = \tan(\theta/2)$$

When to use: Equations of the form $a \sin \theta + b \cos \theta = c$ — substitute t to convert to a quadratic in t .

Maximum and minimum of $a \sin \theta + b \cos \theta$:

$$\text{Maximum value} = +\sqrt{a^2 + b^2}$$

$$\text{Minimum value} = -\sqrt{a^2 + b^2}$$

SSC exam pattern: "Find the maximum of $3 \sin \theta + 4 \cos \theta$."

$$\text{Maximum} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

$$\text{Minimum} = -5.$$

Example (compound angle). Evaluate $\sin 75^\circ$.

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$
$$= \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Watch out: Students write $\sin(A + B) = \sin A + \sin B$ (wrong — this confuses linearity of addition with trig addition; $\sin$ is NOT a linear function). The correct expansion always involves products of sines and cosines.

PART 14 — ALGEBRA

14.0 Opener — Why algebra rules SSC CGL Tier-2

Algebra is the **highest-yield** topic in SSC CGL Tier-2: 8–12 questions / paper. It is also the topic where students lose the most marks, because most problems hinge on **memorised identities** rather than calculation. If you cannot reproduce the 12 core identities below in 5 seconds, you cannot score here.

The 12 identities you MUST cache (write them out from memory daily for 7 days):

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $a^2 - b^2 = (a + b)(a - b)$
4. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
5. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$
6. $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)^3 - 3ab(a + b)$
7. $a^3 - b^3 = (a - b)(a^2 + ab + b^2) = (a - b)^3 + 3ab(a - b)$
8. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
9. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
10. $a^2 + b^2 = (a + b)^2 - 2ab = (a - b)^2 + 2ab$
11. $a + 1/a \rightarrow (a + 1/a)^2 = a^2 + 1/a^2 + 2; (a + 1/a)^3 = a^3 + 1/a^3 + 3(a + 1/a)$
12. $a - 1/a \rightarrow (a - 1/a)^2 = a^2 + 1/a^2 - 2; (a - 1/a)^3 = a^3 - 1/a^3 - 3(a - 1/a)$

Examiner trap: when given $a + 1/a = 3$, students freeze. Square both sides $\rightarrow a^2 + 1/a^2 + 2 = 9 \rightarrow a^2 + 1/a^2 = 7$. That is the entire mechanic of half the SSC algebra paper.

14.1 Linear equations — one, two, three variables

Theory

A linear equation: $ax + b = 0$ (one variable) or $ax + by = c$ (two variables) — degree 1 in every variable.

Solving simultaneous equations — three methods:

- **Substitution** — express one variable from one equation, plug into the other.
- **Elimination** — multiply equations to make one variable's coefficient match, then subtract.
- **Cross-multiplication** — for $ax + by + c = 0$ and $px + qy + r = 0$: $x / (br - qc) = y / (cp - ar) = 1 / (aq - bp)$

Worked Example 1 (EASY)

Q. Solve: $3x + 5y = 11$ and $2x - y = 7$.

Step 1. From eq.2: $y = 2x - 7$

Step 2. Substitute into eq.1: $3x + 5(2x - 7) = 11$

Step 3. Expand: $3x + 10x - 35 = 11$

Step 4. Combine: $13x = 46$

Step 5. $x = \frac{46}{13}$. Hmm, not clean — check work.

Re-check eq.1: $3x + 5y = 11$. Substitute $x = \frac{46}{13}$ $y = 2(\frac{46}{13}) - 7 = \frac{92}{13} - \frac{91}{13} = \frac{1}{13}$. Verify: $3(\frac{46}{13}) + 5(\frac{1}{13}) = \frac{138}{13} + \frac{5}{13} = \frac{143}{13} = 11 \checkmark$

Answer: $x = \frac{46}{13}$ $y = \frac{1}{13}$.

(Examiner lesson — even "ugly" fractions can be correct. Always verify by substitution.)

Worked Example 2 (MEDIUM)

Q. Father's age is 4 times son's age. After 5 years, father will be 3 times son's age. Find their present ages.

Step 1. Let son = x . Father = $4x$.

Step 2. After 5 years: son = $x + 5$, father = $4x + 5$.

Step 3. Given: $4x + 5 = 3(x + 5)$

Step 4. Expand: $4x + 5 = 3x + 15$

Step 5. Solve: $x = 10$

Answer: Son = 10 years, Father = 40 years.

Worked Example 3 (HARD — three variables)

Q. Solve: $x + y + z = 6$, $x - y + z = 2$, $x + y - z = 0$.

Step 1. Add eq.1 and eq.2: $2x + 2z = 8 \rightarrow x + z = 4$

Step 2. Add eq.1 and eq.3: $2x + 2y = 6 \rightarrow x + y = 3$

Step 3. From eq.1: $y + z = 6 - x$

Step 4. From Step 1: $z = 4 - x$. From Step 2: $y = 3 - x$.

Step 5. Substitute into eq.1: $x + (3 - x) + (4 - x) = 6 \rightarrow 7 - x = 6 \rightarrow x = 1$

Answer: $x = 1$, $y = 2$, $z = 3$. Verify: $1 + 2 + 3 = 6 \checkmark$; $1 - 2 + 3 = 2 \checkmark$; $1 + 2 - 3 = 0 \checkmark$

14.2 Quadratic equations — full toolkit

Theory

Standard form: $ax^2 + bx + c = 0$ ($a \neq 0$).

Three solution methods:

- **Factorisation** — split the middle term.
- **Quadratic formula** — $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Completing the square** — used to derive the formula; rarely needed for SSC.

Discriminant $D = b^2 - 4ac$:

D value	Roots
$D > 0$	Two distinct real roots
$D = 0$	Two equal real roots (one repeated)
$D < 0$	No real roots (complex conjugates)
D is a perfect square	Roots are rational

Vieta's formulas — quadratic (sum/product of roots) — MUST MEMORISE:

- Sum of roots: $\alpha + \beta = -b / a$
- Product of roots: $\alpha \cdot \beta = c / a$
- Sum of squares: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- Sum of cubes: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]$
- $1/\alpha + 1/\beta = (\alpha + \beta) / \alpha\beta$
- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \leftrightarrow |\alpha - \beta| = \sqrt{D} / |a|$
- $\alpha/\beta + \beta/\alpha = (\alpha^2 + \beta^2) / \alpha\beta$

Forming a quadratic from given roots p and q:

$$x^2 - (p + q)x + pq = 0$$

Vieta's formulas — cubic $ax^3 + bx^2 + cx + d = 0$ (SSC CGL Tier-2 / Banks Mains):

If α, β, γ are the three roots:

- Sum: $\alpha + \beta + \gamma = -b / a$
- Sum of products taken two at a time: $\alpha\beta + \beta\gamma + \gamma\alpha = c / a$
- Product: $\alpha\beta\gamma = -d / a$

Exam pattern: "If α, β, γ are roots of $2x^3 - 3x^2 + x - 5 = 0$, find $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$."

$$\alpha + \beta + \gamma = -(-3)/2 = 3/2; \alpha\beta\gamma = -(-5)/2 = 5/2.$$

Worked Example 1 (EASY — factorisation)

Q. Solve: $x^2 - 7x + 12 = 0$.

Step 1. Find two numbers whose product = 12 and sum = 7. $\rightarrow$ 3 and 4.

Step 2. Split: $x^2 - 3x - 4x + 12 = 0$

Step 3. Factor: $x(x - 3) - 4(x - 3) = 0$

Step 4. $(x - 3)(x - 4) = 0$

Answer: $x = 3$ or 4 .

Worked Example 2 (MEDIUM — Vieta's application)

Q. If α and β are roots of $x^2 - 5x + 3 = 0$, find $\alpha^3 + \beta^3$.

Step 1. Sum $\alpha + \beta = -(-5)/1 = 5$. Product $\alpha\beta = 3$.

Step 2. Identity: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Step 3. Substitute: $\alpha^3 + \beta^3 = 5^3 - 3(3)(5)$

Step 4. Compute: $= 125 - 45 = 80$

Answer: $\alpha^3 + \beta^3 = 80$.

Worked Example 3 (MEDIUM — discriminant analysis)

Q. For what value of k does $(k - 1)x^2 + (k + 4)x + (k + 7) = 0$ have equal roots?

Step 1. Equal roots $\Rightarrow D = 0 \Rightarrow (k + 4)^2 - 4(k - 1)(k + 7) = 0$

Step 2. Expand $(k + 4)^2 = k^2 + 8k + 16$

Step 3. Expand $4(k - 1)(k + 7) = 4(k^2 + 6k - 7) = 4k^2 + 24k - 28$

Step 4. $D = k^2 + 8k + 16 - 4k^2 - 24k + 28 = 0$

Step 5. Combine: $-3k^2 - 16k + 44 = 0 \rightarrow 3k^2 + 16k - 44 = 0$

Step 6. Quadratic formula: $k = [-16 \pm \sqrt{(256 + 528)}] / 6 = [-16 \pm \sqrt{784}] / 6 = [-16 \pm 28]/6$

Answer: $k = \frac{12}{6} = 2$ or $k = -\frac{44}{6} = -\frac{22}{3}$.

Worked Example 4 (HARD — forming quadratic + relation)

Q. If α and β are roots of $x^2 + px + q = 0$, form a quadratic whose roots are $\alpha + 1/\beta$ and $\beta + 1/\alpha$.

Step 1. From given: $\alpha + \beta = -p$; $\alpha\beta = q$.

Step 2. Sum of new roots: $(\alpha + 1/\beta) + (\beta + 1/\alpha) = (\alpha + \beta) + (1/\alpha + 1/\beta) = (\alpha + \beta) + (\alpha + \beta)/\alpha\beta = (\alpha + \beta)(1 + 1/\alpha\beta) = -p(1 + 1/q) = -p(q + 1)/q$.

Step 3. Product of new roots: $(\alpha + 1/\beta)(\beta + 1/\alpha) = \alpha\beta + \alpha/\alpha + \beta/\beta + 1/(\alpha\beta) = \alpha\beta + 2 + 1/(\alpha\beta) = q + 2 + 1/q = (q^2 + 2q + 1)/q = (q + 1)^2/q$.

Step 4. New quadratic: $x^2 - (\text{sum})x + (\text{product}) = 0 \Rightarrow x^2 + p(q + 1)x/q + (q + 1)^2/q = 0 \Rightarrow$ Multiply through by q :
 $qx^2 + p(q + 1)x + (q + 1)^2 = 0$

Worked Example 5 (HARD — quadratic comparison, Banks favourite)

Q. Solve and find relation between x and y : I: $x^2 - 7x + 12 = 0$ II: $y^2 - 9y + 20 = 0$

Step 1. Equation I: factor $\rightarrow (x - 3)(x - 4) = 0 \rightarrow x = 3$ or 4 .

Step 2. Equation II: factor $\rightarrow (y - 4)(y - 5) = 0 \rightarrow y = 4$ or 5 .

Step 3. Compare every x with every y :

x	y	Relation
3	4	$x < y$
3	5	$x < y$
4	4	$x = y$
4	5	$x < y$

Step 4. All cases: $x \leq y$.

Answer: $x \leq y$.

(Banks PO trick: never just compare maximums or minimums — compare every pair, then take the most-constraining single relation.)

14.3 Algebraic identities — application toolkit

These are NOT theory — they are **direct exam questions**. SSC asks 2–3 identity-substitution Qs every paper.

Worked Example 1 — Direct identity ($a + 1/a$)

Q. If $a + 1/a = 5$, find $a^2 + 1/a^2$ and $a^3 + 1/a^3$.

Step 1. Square both sides: $(a + 1/a)^2 = a^2 + 1/a^2 + 2 = 25$

Step 2. $\therefore a^2 + 1/a^2 = 23$

Step 3. Cube: $(a + 1/a)^3 = a^3 + 1/a^3 + 3(a + 1/a) = 125$

Step 4. $\Rightarrow a^3 + 1/a^3 = 125 - 3(5) = 125 - 15 = 110$

Worked Example 2 — Three-variable identity

Q. If $a + b + c = 6$, $ab + bc + ca = 11$, find $a^2 + b^2 + c^2$.

Step 1. Identity: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

Step 2. $36 = (a^2 + b^2 + c^2) + 2(11)$

Step 3. $a^2 + b^2 + c^2 = 36 - 22 = 14$

Worked Example 3 — $a^3 + b^3 + c^3 - 3abc$

Q. If $a + b + c = 0$, find $a^3 + b^3 + c^3$.

Step 1. Identity: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Step 2. Since $a + b + c = 0$, $\text{RHS} = 0$.

Step 3. $\therefore a^3 + b^3 + c^3 = 3abc$

(One of the most-tested SSC results. Memorise as a one-liner: $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$.)

Worked Example 4 — Substitution chain

Q. If $x + y = 5$ and $xy = 6$, find $(x - y)^2$ and $x^3 + y^3$.

Step 1. $(x - y)^2 = (x + y)^2 - 4xy = 25 - 24 = 1$.

Step 2. $x^3 + y^3 = (x + y)^3 - 3xy(x + y) = 125 - 3(6)(5) = 125 - 90 = 35$.

Worked Example 5 (HARD)

Q. If $x + 1/x = \sqrt{3}$, find $x^6 + 1/x^6$.

Step 1. $(x + 1/x)^2 = x^2 + 1/x^2 + 2 = 3 \Rightarrow x^2 + 1/x^2 = 1$.

Step 2. Cube the result of Step 1: $(x^2 + 1/x^2)^3 = x^6 + 1/x^6 + 3(x^2 + 1/x^2) = 1$

Step 3. $\Rightarrow x^6 + 1/x^6 = 1 - 3(1) = -2$.

14.4 Polynomials — factor & remainder theorem

Theory

Remainder Theorem. If polynomial $p(x)$ is divided by $(x - a)$, the remainder is $p(a)$.

Factor Theorem. $(x - a)$ is a factor of $p(x)$ **if and only if** $p(a) = 0$.

Synthetic division can speed up dividing by $(x - a)$ — but for SSC, evaluating $p(a)$ directly is fastest.

Worked Example 1

Q. Find the remainder when $p(x) = x^3 - 4x^2 + 5x + 7$ is divided by $(x - 2)$.

Step 1. Remainder = $p(2)$

Step 2. $p(2) = 2^3 - 4(2^2) + 5(2) + 7 = 8 - 16 + 10 + 7 = 9$

Worked Example 2

Q. If $(x - 3)$ is a factor of $x^3 + ax^2 + bx - 18$ and the remainder when divided by $(x - 1)$ is -12 , find a and b .

Step 1. Factor theorem at $x = 3$: $27 + 9a + 3b - 18 = 0 \Rightarrow 9a + 3b = -9 \Rightarrow 3a + b = -3 \dots (i)$

Step 2. Remainder at $x = 1$: $1 + a + b - 18 = -12 \Rightarrow a + b = 5 \dots (ii)$

Step 3. Subtract (ii) from (i): $2a = -8 \Rightarrow a = -4$

Step 4. From (ii): $b = 5 - (-4) = 9$

14.5 Surds, indices, and rationalisation

Theory — Laws of Indices

- $a^m \times a^n = a^{m+n}$

- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^m = a^m \times b^m$
- $a^0 = 1$ (for $a \neq 0$)
- $a^{-n} = 1/a^n$
- $a^{(1/n)} = \sqrt[n]{a}$

Theory — Surds

A surd is an irrational root, e.g., $\sqrt{2}$, $\sqrt[3]{5}$, $\sqrt{2 + \sqrt{3}}$.

Rationalisation — eliminate the surd from a denominator by multiplying by the **conjugate**:

Denominator	Multiply by	Result
$\sqrt{a}$	$\sqrt{a}$	a
$a + \sqrt{b}$	$a - \sqrt{b}$	$a^2 - b$
$\sqrt{a} + \sqrt{b}$	$\sqrt{a} - \sqrt{b}$	$a - b$
$1 / (a + \sqrt{b})$	$(a - \sqrt{b}) / (a - \sqrt{b})$	$(a - \sqrt{b}) / (a^2 - b)$

Worked Example 1 (MEDIUM)

Q. Rationalise: $(3 + 2\sqrt{2}) / (3 - 2\sqrt{2})$

Step 1. Multiply numerator and denominator by conjugate of denominator: $(3 + 2\sqrt{2})$:

Step 2. Numerator: $(3 + 2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$

Step 3. Denominator: $(3)^2 - (2\sqrt{2})^2 = 9 - 8 = 1$

Answer: $17 + 12\sqrt{2}$

Worked Example 2 (HARD — surds substitution)

Q. If $x = (\sqrt{3} + 1)/(\sqrt{3} - 1)$ and $y = (\sqrt{3} - 1)/(\sqrt{3} + 1)$, find $x^2 + y^2 + xy$.

Step 1. Rationalise x : $x = (\sqrt{3} + 1)^2 / ((\sqrt{3})^2 - 1^2) = (3 + 2\sqrt{3} + 1)/2 = (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$

Step 2. By symmetry: $y = 2 - \sqrt{3}$

Step 3. $x + y = 4$. $xy = (2)^2 - (\sqrt{3})^2 = 4 - 3 = 1$.

Step 4. $x^2 + y^2 = (x + y)^2 - 2xy = 16 - 2 = 14$.

Step 5. $x^2 + y^2 + xy = 14 + 1 = 15$.

14.6 Logarithms

Theory

Definition. $\log_a N = x \Leftrightarrow a^x = N$ ($a > 0$, $a \neq 1$, $N > 0$).

Standard bases: - Common log: base 10, written $\log N$. - Natural log: base $e \approx 2.718$, written $\ln N$.

Properties (memorise cold):

- $\log(MN) = \log M + \log N$
- $\log(M/N) = \log M - \log N$
- $\log(M^k) = k \cdot \log M$
- $\log_a a = 1$

- $\log_a 1 = 0$
- Change of base: $\log_a b = \log b / \log a$ (any consistent base)
- $\log_a b \cdot \log_b a = 1$
- $\log_{(a^k)} M = (1/k) \cdot \log_a M$

Worked Example 1 (EASY)

Q. Find $\log_2 64$.

Step 1. $64 = 2^6$.

Step 2. $\log_2 64 = \log_2 2^6 = 6 \times \log_2 2 = 6$.

Worked Example 2 (MEDIUM — change of base)

Q. If $\log_2 3 = a$, find $\log_8 9$ in terms of a .

Step 1. $9 = 3^2$ and $8 = 2^3$.

Step 2. $\log_8 9 = \log_{2^3} 3^2 = \left(\frac{2}{3}\right) \cdot \log_2 3 = 2a/3$.

Worked Example 3 (HARD)

Q. Solve: $\log_2 x + \log_2 (x - 2) = 3$.

Step 1. $\log_2 [x(x - 2)] = 3$

Step 2. $x(x - 2) = 2^3 = 8$

Step 3. $x^2 - 2x - 8 = 0 \rightarrow (x - 4)(x + 2) = 0 \rightarrow x = 4$ or -2

Step 4. Reject $x = -2$ (log of negative is undefined).

Answer: $x = 4$.

14.7 Sequences and series — AP, GP, HP

Theory — Arithmetic Progression (AP)

Form: $a, a + d, a + 2d, \dots$

- nth term: $T_n = a + (n - 1)d$
- Sum of n terms: $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$ where l is the last term
- Three numbers in AP: $a - d, a, a + d$ (use this when the *sum* is given)
- Five numbers in AP: $a - 2d, a - d, a, a + d, a + 2d$

Sum of first n natural numbers $= n(n + 1)/2$

Sum of first n squares $= n(n + 1)(2n + 1)/6$

Sum of first n cubes $= [n(n + 1)/2]^2 = (\text{sum of first } n \text{ naturals})^2$

Theory — Geometric Progression (GP)

Form: $a, ar, ar^2, ar^3, \dots$

- nth term: $T_n = a \cdot r^{(n-1)}$
- Sum of n terms: $S_n = a(r^n - 1)/(r - 1)$ (for $r \neq 1$)
- Sum of infinite GP ($|r| < 1$): $S_\infty = a / (1 - r)$
- Three numbers in GP: $a/r, a, ar$ (use this when the *product* is given)

Theory — Harmonic Progression (HP)

A sequence whose **reciprocals** form an AP. To work with an HP, take reciprocals → AP → solve → reciprocate back.

Worked Example 1 (EASY — AP)

Q. Find the sum of the AP: 3, 7, 11, ..., up to 20 terms.

Step 1. $a = 3, d = 4, n = 20$.

Step 2. $S_n = (n/2)[2a + (n - 1)d] = (\frac{20}{2})[6 + 19(4)] = 10 \times (6 + 76) = 10 \times 82 = \mathbf{820}$

Worked Example 2 (MEDIUM — three-term AP)

Q. The sum of three numbers in AP is 24 and their product is 440. Find the numbers.

Step 1. Let the three numbers be $(a - d), a, (a + d)$.

Step 2. Sum: $3a = 24 \Rightarrow a = 8$.

Step 3. Product: $(8 - d)(8)(8 + d) = 440 \Rightarrow 8(64 - d^2) = 440 \Rightarrow 64 - d^2 = 55 \Rightarrow d^2 = 9 \Rightarrow d = \pm 3$.

Step 4. Numbers: **5, 8, 11** (or 11, 8, 5).

Worked Example 3 (MEDIUM — GP)

Q. First term of GP = 2, common ratio = 3. Find sum of first 5 terms.

Step 1. $S_5 = a(r^n - 1)/(r - 1) = 2(3^5 - 1)/(3 - 1) = 2(243 - 1)/2 = \mathbf{242}$

Worked Example 4 (HARD — infinite GP)

Q. Sum of an infinite GP is 5 and the sum of its squares is 15. Find first term and common ratio.

Step 1. $S_\infty = a/(1 - r) = 5 \Rightarrow a = 5(1 - r) \dots (i)$

Step 2. Squares form a GP with first term a^2 and ratio r^2 . Their sum: $a^2/(1 - r^2) = 15 \Rightarrow a^2 = 15(1 - r)(1 + r) \dots (ii)$

Step 3. From (i): $a^2 = 25(1 - r)^2$. Equate to (ii): $25(1 - r)^2 = 15(1 - r)(1 + r)$

Step 4. Cancel $(1 - r)$: $25(1 - r) = 15(1 + r) \Rightarrow 25 - 25r = 15 + 15r \Rightarrow 40r = 10 \Rightarrow \mathbf{r = \frac{1}{4}}$.

Step 5. $a = 5(1 - \frac{1}{4}) = 5 \times \frac{3}{4} = \frac{15}{4}$.

14.8 Inequalities

Theory

- Adding/subtracting same value: inequality preserved.
- Multiplying / dividing by **positive** number: inequality preserved.
- Multiplying / dividing by **negative** number: **flip the inequality**.
- Squaring is dangerous when sign of variable unknown — use only on positive expressions.

For quadratic inequality $ax^2 + bx + c > 0$ ($a > 0$): 1. Find roots α, β . 2. Solution: $x < \alpha$ or $x > \beta$ (outside the roots). 3. For $ax^2 + bx + c < 0$: $\alpha < x < \beta$ (between the roots).

(Reverse if $a < 0$.)

AM $\geq$ GM $\geq$ HM inequality (high-frequency SSC/Banks tool):

For positive numbers a and b :

$$AM = \frac{a + b}{2}; GM = \sqrt{ab}; HM = \frac{2ab}{a + b}$$

AM $\geq$ GM $\geq$ HM always. Equality holds when and only when $a = b$.

Exam applications: - If sum $(a + b)$ is fixed $\rightarrow$ product (ab) is **maximum** when $a = b$. Maximum product = $(\text{sum}/2)^2$. - If product (ab) is fixed $\rightarrow$ sum $(a + b)$ is **minimum** when $a = b$. Minimum sum = $2\sqrt{ab}$. - Standard SSC form: "If $x > 0$, find the minimum of $x + 9/x$." $\rightarrow x + 9/x \geq 2\sqrt{x \cdot 9/x} = 2\sqrt{9} = 6$. Equality at $x = 3$. - For n positive variables: arithmetic mean $\geq$ geometric mean. If $x_1 + x_2 + x_3 = 30$, $\max(x_1 x_2 x_3) = 10^3 = 1000$ (all equal).

Cauchy-Schwarz inequality (SSC CGL Tier-2 / Banks Mains):

$$(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$$

Exam use: Find the maximum of $(ax + by)$ given $a^2 + b^2$ and $x^2 + y^2$ are known.

$$\text{Maximum of } (ax + by) = \sqrt{a^2 + b^2} \times \sqrt{x^2 + y^2}$$

Example. If $a^2 + b^2 = 25$ and $x^2 + y^2 = 9$, maximum of $ax + by = \sqrt{25} \times \sqrt{9} = 5 \times 3 = 15$.

Worked Example 1 (EASY)

Q. Solve: $3x - 7 \leq 5x + 1$.

Step 1. Move x -terms to one side: $-7 - 1 \leq 5x - 3x$

Step 2. $-8 \leq 2x \Rightarrow x \geq -4$.

Answer: $x \in [-4, \infty)$.

Worked Example 2 (MEDIUM — quadratic inequality)

Q. Solve: $x^2 - 5x + 6 < 0$.

Step 1. Factor: $(x - 2)(x - 3) < 0$.

Step 2. Roots are 2 and 3. Since $a > 0$ and we want < 0 , solution is **between** the roots.

Answer: $2 < x < 3$.

Worked Example 3 (HARD — AM-GM application)

Q. If $x, y > 0$ and $x + y = 10$, find the maximum value of xy .

Step 1. AM $\geq$ GM: $(x + y)/2 \geq \sqrt{xy}$

Step 2. $5 \geq \sqrt{xy} \Rightarrow xy \leq 25$.

Step 3. Equality when $x = y = 5$.

Answer: Maximum $xy = 25$.

PART 14B — PERMUTATIONS & COMBINATIONS

14B.0 Opener

The first decision rule: Order matters → **Permutation**. Order doesn't matter → **Combination**.

Question type	Permutation or Combination?
In how many ways can letters be arranged?	Permutation
Arranging people in a row, around a table	Permutation
Forming numbers from digits	Permutation
Choosing a committee	Combination
Picking 4 cards from 52	Combination
Selecting 2 fruits from 5	Combination
Drawing a line through 2 of 10 points	Combination (line is the same regardless of order)
Forming triangle from 3 of n points	Combination

14B.1 Fundamental counting principle

FCP. If task A can be done in **m** ways and task B in **n** ways, then both can be done in **m × n** ways.

Worked Example. Number of 3-digit numbers using digits 1, 2, 3, 4, 5 with **no repetition**:

Step 1. Hundreds place: 5 choices.

Step 2. Tens place: 4 remaining choices.

Step 3. Units place: 3 remaining choices.

Step 4. Total = $5 \times 4 \times 3 = 60$.

Same question, with repetition allowed: $5 \times 5 \times 5 = 125$.

14B.2 Factorial — the building block

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

- $0! = 1$ (definition)
- $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$

Memorise factorials up to 8! cold — you need them on every P&C question.

14B.3 Permutations $P(n, r)$

Theory

$P(n, r) = n! / (n - r)!$ = number of ways to arrange r items chosen from n distinct items.

- $P(n, n) = n!$ (arranging all n)
- $P(n, 1) = n$
- $P(n, 0) = 1$ (one way to arrange nothing)

Worked Example 1 (EASY)

Q. In how many ways can 5 people sit in 3 chairs?

Step 1. $P(5, 3) = 5!/(5 - 3)! = 5!/2! = \frac{120}{2} = 60$.

Worked Example 2 (MEDIUM — restrictions)

Q. How many ways can the letters of "DELHI" be arranged so that vowels are always together?

Step 1. "DELHI" has letters D, E, L, H, I — 5 distinct letters. Vowels: E, I (2 vowels).

Step 2. Treat the two vowels as one block. So you arrange [DLHI] + [block] = 4 units.

Step 3. Arrangements of 4 units = $4! = 24$.

Step 4. Within the vowel block, E and I can swap = $2! = 2$.

Step 5. Total = $24 \times 2 = 48$.

Worked Example 3 (HARD — never together / gaps method)

Q. In how many ways can 5 boys and 3 girls sit in a row such that no two girls sit together?

Step 1. Arrange 5 boys first: $5! = 120$ ways.

Step 2. Now there are 6 gaps (left of B_1 , between each pair, right of B_5) where girls can sit.

Step 3. Choose 3 of 6 gaps for the 3 girls: $C(6, 3) = 20$.

Step 4. Arrange 3 girls in those 3 gaps: $3! = 6$.

Step 5. Total = $5! \times C(6, 3) \times 3! = 120 \times 20 \times 6 = 14,400$.

14B.4 Permutations with repeated objects

If among n items, p are alike of one kind, q alike of another, etc., the number of distinct arrangements = $n! / (p! \cdot q! \cdot \dots)$.

Worked Example 1 (MEDIUM)

Q. How many distinct arrangements of letters in "MISSISSIPPI"?

Step 1. Count letters: M = 1, I = 4, S = 4, P = 2. Total $n = 11$.

Step 2. Arrangements = $11! / (1! \cdot 4! \cdot 4! \cdot 2!)$

Step 3. Compute: $11! = 39,916,800$. Denominator: $1 \times 24 \times 24 \times 2 = 1,152$.

Step 4. $= \frac{39916800}{1152} = 34,650$.

14B.5 Circular permutations

n distinct objects around a circular table:

- $(n - 1)!$ if rotations are considered the same.
- $(n - 1)! / 2$ if reflections are also considered the same (e.g., a necklace).

Worked Example

Q. In how many ways can 7 people sit around a round table?

Answer: $(7 - 1)! = 6! = 720$.

Variation: with 2 specific people always together?

- Treat the pair as one block $\rightarrow$ 6 units around the table $\rightarrow (6 - 1)! = 5! = 120$ arrangements
- The pair can internally swap: $\times 2$

- Answer: $120 \times 2 = 240$.

14B.6 Combinations $C(n, r)$

Theory

$C(n, r) = \frac{n!}{[r! \cdot (n - r)!]}$ — number of ways to choose r items from n distinct, where order doesn't matter.

Properties: - $C(n, r) = C(n, n - r)$ - $C(n, 0) = C(n, n) = 1$ - $C(n, 1) = n$ - Sum of all $C(n, k)$ for $k = 0$ to $n = 2^n$ (number of subsets of an n -set)

Worked Example 1 (EASY)

Q. From 10 students, in how many ways can we form a team of 4?

Step 1. $C(10, 4) = \frac{10!}{(4! \cdot 6!)} = \frac{(10 \times 9 \times 8 \times 7)}{(4 \times 3 \times 2 \times 1)} = \frac{5040}{24} = 210$.

Worked Example 2 (MEDIUM — committee with restriction)

Q. A committee of 5 from 6 men and 4 women, with at least 2 women.

Step 1. "At least 2 women" means 2W + 3M, OR 3W + 2M, OR 4W + 1M.

Step 2. Count each case: - 2W + 3M: $C(4, 2) \times C(6, 3) = 6 \times 20 = 120$ - 3W + 2M: $C(4, 3) \times C(6, 2) = 4 \times 15 = 60$ - 4W + 1M: $C(4, 4) \times C(6, 1) = 1 \times 6 = 6$

Step 3. Total = $120 + 60 + 6 = 186$.

Worked Example 3 (HARD — geometry of selection)

Q. 12 points are on a plane. No 3 collinear. Find: (a) Number of straight lines drawn through pairs of points. (b) Number of triangles formed.

Step 1. Lines need 2 points: $C(12, 2) = 66$.

Step 2. Triangles need 3 non-collinear points: $C(12, 3) = 220$.

Answer: (a) 66 lines, (b) 220 triangles.

Worked Example 4 (HARD — handshakes / diagonals)

Q. A convex polygon has 20 sides. How many diagonals?

Step 1. Number of straight segments through any 2 of 20 vertices = $C(20, 2) = 190$.

Step 2. Subtract the 20 sides (which are not diagonals): $190 - 20 = 170$ diagonals.

EXAM SHORTCUT

Formula shortcut: Diagonals of n -gon = $\frac{n(n - 3)}{2} = 20 \times \frac{17}{2} = 170$ ✓

14B.7 Number formation problems

Worked Example 1

Q. How many 4-digit numbers can be formed from 1, 2, 3, 4, 5 (no repetition)?

Step 1. First digit: 5 choices. 2nd: 4. 3rd: 3. 4th: 2.

Step 2. Total = $5 \times 4 \times 3 \times 2 = 120$.

Worked Example 2 (MEDIUM — including 0)

Q. How many 4-digit numbers can be formed from 0, 1, 2, 3, 4, 5 with no repetition?

Step 1. First digit cannot be 0 (else it's a 3-digit number). So 5 choices (any non-zero).

Step 2. Remaining 3 digits: pick 3 from the remaining 5 (since 1 was used). That's $P(5, 3) = 60$.

Step 3. Total = $5 \times 60 = 300$.

Worked Example 3 (HARD — divisibility)

Q. How many 4-digit numbers from 1, 2, 3, 4, 5 (no repetition) are divisible by 5?

Step 1. Divisibility by 5 $\Rightarrow$ last digit must be 5.

Step 2. Fix last digit = 5. Remaining 3 digits from $\{1, 2, 3, 4\}$.

Step 3. Arrangements = $P(4, 3) = 24$.

Answer: 24 numbers.

PART 14C — PROBABILITY

14C.0 Opener — the universal formula

P(event) = (favourable outcomes) / (total outcomes)

This works only when all outcomes are **equally likely**.

Three sanity checks before any calculation:

1. Are outcomes equally likely? (Coin yes; weighted die no.)
2. Does the question ask for "with replacement" or "without"? (Changes the denominator on each draw.)
3. Is the event "AND" (multiplication) or "OR" (addition)?

14C.1 Coin tosses

Single coin: $P(H) = P(T) = \frac{1}{2}$.

n coins (or 1 coin tossed n times): sample space = 2^n .

Worked Example 1 (EASY)

Q. Probability of getting exactly 2 heads when 3 coins are tossed.

Step 1. Sample space = $2^3 = 8$ outcomes.

Step 2. Outcomes with exactly 2 heads: HHT, HTH, THH = 3 outcomes.

Step 3. $P = \frac{3}{8}$.

Worked Example 2 (MEDIUM)

Q. Probability of at least 1 head in 4 tosses.

Step 1. Easier via complement: $P(\text{at least 1 H}) = 1 - P(\text{no H}) = 1 - P(\text{all T})$.

Step 2. $P(\text{all T}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$.

Step 3. $P(\text{at least 1 H}) = 1 - \frac{1}{16} = \frac{15}{16}$.

14C.2 Dice

One die: outcomes 1–6. $P(\text{any specific number}) = \frac{1}{6}$.

Two dice: sample space = 36 (6×6). Sum can range from 2 to 12.

Sum frequency table — cache this:

Sum	Outcomes	Count	P
2	(1,1)	1	$\frac{1}{36}$
3	(1,2),(2,1)	2	$\frac{2}{36}$
4	(1,3),(3,1),(2,2)	3	$\frac{3}{36}$
5	4 outcomes	4	$\frac{4}{36}$

Sum	Outcomes	Count	P
6	5 outcomes	5	$\frac{5}{36}$
7	6 outcomes (most likely)	6	$\frac{6}{36} = \frac{1}{6}$
8	5	5	$\frac{5}{36}$
9	4	4	$\frac{4}{36}$
10	3	3	$\frac{3}{36}$
11	2	2	$\frac{2}{36}$
12	(6,6)	1	$\frac{1}{36}$

Worked Example 1 (EASY)

Q. Two dice rolled. Probability of sum = 9.

Step 1. Outcomes giving sum 9: (3,6), (4,5), (5,4), (6,3) = 4 outcomes.

Step 2. $P = \frac{4}{36} = \frac{1}{9}$.

Worked Example 2 (MEDIUM)

Q. Two dice rolled. Probability that the sum is a prime number.

Step 1. Possible sums: 2, 3, 4, ..., 12. Primes: 2, 3, 5, 7, 11.

Step 2. Counts (from table): $1 + 2 + 4 + 6 + 2 = 15$ favourable outcomes.

Step 3. $P = \frac{15}{36} = \frac{5}{12}$.

14C.3 Cards (52-card deck)

Cache the deck composition:

- Total: 52 cards
- 4 suits: ♠ Spades (black), ♣ Clubs (black), ♥ Hearts (red), ♦ Diamonds (red)
- 26 red, 26 black
- Each suit: 13 cards (A, 2, 3, ..., 10, J, Q, K)
- Face cards: J, Q, K → 4 of each = 12 face cards total
- Aces: 4 (one per suit)
- Numbered cards (2–10): $9 \times 4 = 36$

Worked Example 1 (EASY)

Q. Probability of drawing a King from a shuffled deck.

Answer: $\frac{4}{52} = \frac{1}{13}$.

Worked Example 2 (MEDIUM — without replacement)

Q. Two cards drawn one after another (without replacement). Probability both are aces.

Step 1. $P(\text{1st ace}) = \frac{4}{52} = \frac{1}{13}$.

Step 2. Given 1st was ace, $P(2\text{nd ace}) = \frac{3}{51} = \frac{1}{17}$.

Step 3. Joint $P = \left(\frac{1}{13}\right) \times \left(\frac{1}{17}\right) = \frac{1}{221}$.

Worked Example 3 (HARD)

Q. Probability that a 5-card hand has exactly 2 aces.

Step 1. Total 5-card hands = $C(52, 5)$.

Step 2. Favourable: choose 2 aces out of 4 $\times$ choose 3 non-aces out of 48 = $C(4, 2) \times C(48, 3)$.

Step 3. $P = [C(4, 2) \times C(48, 3)] / C(52, 5) = [6 \times 17,296] / 2,598,960 = \frac{103776}{2598960} \approx \mathbf{0.0399}$ (about 4%).

14C.4 Balls in urn — with and without replacement

Worked Example 1 (EASY — single draw)

Q. A bag has 5 red, 4 blue, 3 green. Probability of drawing a blue ball.

Step 1. Total = $5 + 4 + 3 = 12$.

Step 2. $P(\text{blue}) = \frac{4}{12} = \frac{1}{3}$.

Worked Example 2 (MEDIUM — without replacement)

Q. Same bag. Two balls drawn without replacement. Probability both are red.

Step 1. $P(1\text{st red}) = \frac{5}{12}$.

Step 2. Given 1st red, bag now has 4 red, 4 blue, 3 green = 11 total. $P(2\text{nd red}) = \frac{4}{11}$.

Step 3. Joint $P = \left(\frac{5}{12}\right) \times \left(\frac{4}{11}\right) = \frac{20}{132} = \frac{5}{33}$.

Worked Example 3 (HARD — at least one)

Q. Same bag. Three balls drawn (without replacement). Probability that at least one is green.

Step 1. Easier via complement. $P(\text{none green}) =$ pick all 3 from non-green ($5 + 4 = 9$ balls).

Step 2. $P(\text{all non-green}) = C(9, 3) / C(12, 3) = \frac{84}{220} = \frac{21}{55}$.

Step 3. $P(\text{at least 1 green}) = 1 - \frac{21}{55} = \frac{34}{55}$.

14C.5 Conditional probability and independence

Conditional: $P(A | B) = P(A \cap B) / P(B)$.

Independent events: $P(A \cap B) = P(A) \cdot P(B)$. Equivalently, $P(A | B) = P(A)$.

Worked Example 1 (MEDIUM)

Q. Two cards drawn without replacement. Given that the first is a King, what is the probability the second is also a King?

Step 1. After first King, deck has 51 cards, 3 Kings.

Step 2. $P(2\text{nd King} | 1\text{st King}) = \frac{3}{51} = \frac{1}{17}$.

Worked Example 2 (HARD — Bayes-style)

Q. A factory has two machines: A produces 60% of items, B produces 40%. 2% of A's items are defective; 5% of B's are. A randomly chosen item is defective. What's the probability it came from machine B?

Step 1. $P(A) = 0.6$, $P(B) = 0.4$. $P(D | A) = 0.02$, $P(D | B) = 0.05$.

Step 2. $P(D) = P(A) \cdot P(D | A) + P(B) \cdot P(D | B) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.020 = 0.032$.

Step 3. $P(B | D) = P(B) \cdot P(D | B) / P(D) = (0.4)(0.05) / 0.032 = 0.020 / 0.032 = \frac{5}{8} = \mathbf{0.625}$.

14C.6 Bayes' theorem, Expected value, and Binomial distribution

These three appear in Banks PO Mains and occasionally SSC CGL Tier-2. One worked example each is enough to recognise the pattern.

Bayes' Theorem (2-event version):

$$P(H_1 | E) = \frac{P(H_1) \cdot P(E|H_1)}{P(H_1) \cdot P(E|H_1) + P(H_2) \cdot P(E|H_2)}$$

Reading: "Given that event E occurred, what is the probability it came from hypothesis H_1 ?"

The Worked Example 2 in 14C.5 above is a direct application (machine A vs machine B). The general steps are always: (i) write $P(H_1)$, $P(H_2)$; (ii) write $P(E|H_1)$, $P(E|H_2)$; (iii) compute total $P(E) = P(H_1) \cdot P(E|H_1) + P(H_2) \cdot P(E|H_2)$; (iv) apply Bayes.

Watch out: Students forget to compute the denominator $P(E)$ and instead use just $P(E|H_1)$ — this gives the wrong numerator-only answer. Always build the full denominator first.

Expected Value $E[X] = \sum x \cdot P(x)$:

For a random variable X taking values $x_1, x_2, \dots$ with probabilities $p_1, p_2, \dots$:

$$E[X] = x_1p_1 + x_2p_2 + x_3p_3 + \dots$$

Exam pattern: "A player wins ₹5 if a die shows 6, and loses ₹1 for any other outcome. What is the expected profit per throw?"

$$E[X] = 5 \times (1/6) + (-1) \times (5/6) = 5/6 - 5/6 = \mathbf{0} \text{ (fair game).}$$

A biased version: win ₹6 on a 6 (prob $1/6$), lose ₹1 otherwise. $E[X] = 6/6 - 5/6 = 1/6 \approx \mathbf{₹0.17}$ expected gain per throw.

Binomial Distribution (brief intro for exams):

When an experiment is repeated n independent times, each with probability p of success:

$$P(X = r) = C(n, r) \cdot p^r \cdot (1 - p)^{(n - r)}$$

When to use: "n independent trials, each with the same probability p of success — find the probability of exactly r successes."

Example. A fair coin is tossed 5 times. Probability of exactly 3 heads.

$$n = 5, p = 1/2, r = 3.$$

$$P(X = 3) = C(5, 3) \times (1/2)^3 \times (1/2)^2 = 10 \times (1/32) = \mathbf{10/32 = 5/16 \approx 0.3125}.$$

Mean of binomial: $E[X] = np$. **Variance:** $np(1 - p)$.

For the above: mean = $5 \times 1/2 = 2.5$; variance = $5 \times 1/2 \times 1/2 = 1.25$.

14C.7 Mini-mock (10 Qs — full solutions in next section)

1. Three coins tossed. P(at least 2 heads).
2. Two dice rolled. P(both even).
3. From a deck, P(face card or red).
4. 4 boys, 3 girls. Form a committee of 3 with at least 1 girl.
5. Letters of WORLD arranged. P(starts with W).
6. From 1 to 100, pick a number at random. P(divisible by 3 or 5).
7. Two cards drawn (with replacement). P(both spades).
8. P(2 dice sum > 9).
9. Box has 6 white, 4 black balls. Pick 2 without replacement. P(same colour).
10. 5 letters posted into 5 envelopes randomly. P(at least one in correct envelope) — derangement variant.

Solutions sketch

1. $P(\geq 2 H) = (3 + 1)/8 = \frac{1}{2}$.
2. $P(\text{both even}) = \left(\frac{3}{6}\right)^2 = \frac{1}{4}$.
3. Face cards = 12. Red = 26. Red face = 6. Union = $12 + 26 - 6 = 32$. $P = \frac{32}{52} = \frac{8}{13}$.
4. Total committees = $C(7, 3) = 35$. All-boys = $C(4, 3) = 4$. At least one girl = $35 - 4 = 31$. $P = \frac{31}{35}$.
5. Total arrangements $5! = 120$. Starting with W: fix W $\rightarrow 4! = 24$. $P = \frac{24}{120} = \frac{1}{5}$.
6. Multiples of 3: 33. Multiples of 5: 20. Multiples of 15: 6. Union = $33 + 20 - 6 = 47$. $P = \frac{47}{100}$.
7. $P(\text{spade}) = \frac{13}{52} = \frac{1}{4}$. With replacement: $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$.
8. Sums 10, 11, 12 $\rightarrow 3 + 2 + 1 = 6$ outcomes. $P = \frac{6}{36} = \frac{1}{6}$.
9. $P(\text{both white}) = C(6,2)/C(10,2) = \frac{15}{45} = \frac{1}{3}$. $P(\text{both black}) = C(4,2)/C(10,2) = \frac{6}{45} = \frac{2}{15}$. Sum = $\frac{7}{15}$.
10. Number of derangements of 5 = 44. $P(\text{at least one correct}) = 1 - \frac{44}{120} = \frac{76}{120} = \frac{19}{30}$.



PART 15 — DATA INTERPRETATION (DI)

DI isn't a separate topic — it's an assembly of percent, ratio, average applied to visuals.

| Types (question *styles*):

1. Table DI.
2. Bar graph DI (single, double, stacked).
3. Line graph DI.
4. Pie chart DI.
5. Caselet / paragraph DI (text-only).
6. Mixed chart (table + bar + line).
7. Missing DI (some cells blank; solve with given %/avg clues).

Each DI set has 4-5 sub-questions that translate into the core types. The speed gain is **reading the chart once, storing totals mentally, answering 5 Qs with that one pass.**

APPENDIX A — The 30-Second Diagnostic

Before solving any arithmetic question, run this 30-second classifier:

1. Is there a **percentage or ratio** word? → Parts 1-4.
2. Is there a **₹, cost, SP, MP, profit, discount**? → Part 2.
3. **Rate / time / interest / years / compound**? → Part 3.
4. **Days, hours, pipes, men, work, job**? → Part 6.
5. **Speed, distance, train, boat, stream, escalator, clock**? → Part 8.
6. **Mixture, alloy, milk, water, alcohol, ratio**? → Part 7.
7. **Partners, investment, capital, months**? → Part 9.
8. **Ages, father, son, years hence, years ago**? → Part 10.
9. **Area, perimeter, volume, surface**? → Parts 11-12.
10. **Triangle, circle, angle, tangent, chord**? → Part 12.
11. **sin, cos, tan, height, angle of elevation**? → Part 13.
12. **Letters arranged, committee, dice rolled, cards picked**? → Part 14.
13. Has a **chart / table / pie / bar**? → Part 15 + routing to 1-5.

This is the discipline. **Classify → shortcut → verify**. Every time.

Use this book alongside the Reasoning and English books for complete skill coverage.

EXAM-FOCUS APPENDIX — BANKS (PO, Clerk, RRB Bank, RBI)

Use this appendix when reading this same book under the title "**Quantitative Aptitude — for Banks**".
Banks weight DI + Caselet + Approximation + Quadratic comparison far more than SSC; geometry / mensuration / trigonometry are NEAR-ZERO weight.

Banks-priority chapter map

Topic in this book	Banks weight	Drill priority
Part 1 — Percentage	High	● Day 1
Part 2 — Profit & Loss	Mid	● Day 2
Part 3 — Simple + Compound Interest	High	● Day 3
Part 4 — Ratio + Average	High	● Day 3
Part 5 — Number system + LCM/HCF	Low	● skim
Part 6 — Time & Work + Pipes	Mid	● Day 4
Part 7 — Mixture & Alligation	Mid	● Day 4
Part 8 — Time, Speed, Distance + Boats + Trains	Mid	● Day 5
Part 9 — Partnership	Low	● skim
Part 10 — Ages	Low	● skim
Part 11 — Mensuration 2D	Low	● skim
Part 12 — Geometry	Negligible	○ skip
Part 13 — Trigonometry / Heights	Negligible	○ skip
Part 14 — Permutation / Probability	Mid	● Day 6
Part 15 — DI + Caselet	Highest	● Day 7-15
Part 17 (this) — Approximation + Quadratic + Quantity comparison	High	● Day 16-20

NEW Part — Approximation (Banks-only chapter)

What it tests: find the closest integer / "roughly equal to" value of a complex arithmetic expression in 30 seconds.

Method (drill): 1. Round each number to the nearest "easy" value (2-significant-figure, half, quarter, etc.). 2. Apply BODMAS on rounded values. 3. Compare with options.

Worked example. $? \approx 39.97 \% \text{ of } 1899.98 + 24.04 \times 17.97$.

Round: $40 \% \text{ of } 1900 = 760$. $24 \times 18 = 432$. Sum ≈ 1192 . Pick option closest to 1192.

Hot-list of round-able quantities:

Difficult	Round to
14.97	15
24.04	24
39.97	40
11.95	12
16.66 %	$\frac{1}{6}$
33.33 %	$\frac{1}{3}$

Difficult	Round to
12.5 %	$\frac{1}{8}$
14.28 %	$\frac{1}{7}$
$\sqrt{26}$	≈ 5.1
$\sqrt{48}$	≈ 6.93
12 % of 250	30

Drill: 50 sums daily for a week. Time-per-sum should drop from 60 s $\rightarrow$ 25 s.

NEW Part — Quadratic comparison (Banks specialty)

What it tests: two quadratics in x and y ; solve both, choose: $x > y$ / $x \geq y$ / $x < y$ / $x \leq y$ / $x = y$ / no relation.

Method: 1. Factor or use formula on each. 2. Get all 4 roots: x_1, x_2, y_1, y_2 . 3. Compare the **smallest x** with the **largest y** , and **largest x** with the **smallest y** . 4. Decision rules: If $\text{smallest}_x \geq \text{largest}_y \rightarrow x \geq y$ If $\text{largest}_x \leq \text{smallest}_y \rightarrow x \leq y$ Otherwise $\rightarrow$ **No relation**

Worked example. $x^2 - 7x + 12 = 0 \Rightarrow x \in \{3, 4\}$. $y^2 - 9y + 20 = 0 \Rightarrow y \in \{4, 5\}$.

$\text{smallest}_x = 3, \text{largest}_y = 5 \rightarrow 3 < 5$ (so NOT $x \geq y$). $\text{largest}_x = 4, \text{smallest}_y = 4 \rightarrow 4 = 4$ (so NOT $x < y$). $\rightarrow$ **No relation can be established.**

NEW Part — Quantity comparison

Given Quantity I and Quantity II as expressions; choose: $I > II$ / $I \geq II$ / $I < II$ / $I \leq II$ / $I = II$ / no relation.

Method: evaluate both quantities; compare directly. Often one is a rate problem and the other a percentage / formula.

DI deep-drill (high yield — re-read Part 15)

Banks DI types reorder: 1. **Tabular** — straight-up tables with multi-row, multi-column. 2. **Bar chart** — single, double, stacked. 3. **Pie chart** — % distribution; total often given. 4. **Line graph** — trend across years. 5. **Mixed combo** — bar + line, table + pie. 6. **Caselet** — DI in pure paragraph form, no chart. **Banks Mains specialty.** 7. **Missing-value DI** — table with one cell missing; deduce from constraints. 8. **Radar chart** — RBI & SBI Mains.

For each DI set (5 questions): 1. Read the chart caption + axis labels — 30 s. 2. Compute totals / row sums / column sums in margin if helpful — 30 s. 3. Then attack each of the 5 questions in 60-90 s.

Total budget per DI set: 5-7 min. With 4 sets in IBPS PO Mains DI section, that's 25-30 min for 20 marks — high ROI.

EXAM-FOCUS APPENDIX — SSC + RRB

Use this appendix when reading this same book under the title "**Arithmetic — for SSC & RRB**". SSC + RRB weight Geometry + Mensuration + Trigonometry HEAVILY (15-20 marks combined), and DI is shorter / simpler.

SSC + RRB priority chapter map

Topic in this book	SSC weight	RRB weight	Drill priority
Part 1 — Percentage	High	High	● Day 1
Part 2 — Profit, Loss, Discount	High	High	● Day 1
Part 3 — SI + CI	Mid	Mid	● Day 2
Part 4 — Ratio + Proportion + Average	High	High	● Day 2
Part 5 — Number system + LCM/HCF + Divisibility	High	High	● Day 3
Part 6 — Time & Work + Pipes	Mid	Mid	● Day 4
Part 7 — Mixture / Alligation	Mid	Mid	● Day 4
Part 8 — Time-Speed-Distance + Boats + Trains	High	High	● Day 5
Part 9 — Partnership	Mid	Low	● Day 6
Part 10 — Ages	Mid	Mid	● Day 6
Part 11 — Mensuration 2D	High	High	● Day 7
Part 12 — Geometry (incl. Triangles, Circles)	Highest	High	● Day 8-10
Part 13 — Trigonometry + Heights	Highest	Mid	● Day 11-13
Part 14 — Permutation / Probability	Low	Low	● skim
Part 15 — DI	Mid (smaller table/bar)	Low	● Day 14
Algebra (basic identities)	High	Mid	● Day 15

Key SSC algebra identities to memorise

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$(a - b)^2 = a^2 + b^2 - 2ab$$

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{if } a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$a + 1/a = k \Rightarrow a^2 + 1/a^2 = k^2 - 2 ; a^3 + 1/a^3 = k^3 - 3k$$

$$a^2 + b^2 \geq 2ab \text{ (AM } \geq \text{ GM)}$$

Trigonometry: identities you MUST recall in <2 s

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\sin(90 - \theta) = \cos \theta$$

$$\cos(90 - \theta) = \sin \theta$$

$$\tan(90 - \theta) = \cot \theta$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \cdot \tan B)$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

$$\text{standard angles: } \sin 0 = 0, \sin 30 = \frac{1}{2}, \sin 45 = \frac{1}{\sqrt{2}}, \sin 60 = \frac{\sqrt{3}}{2}, \sin 90 = 1$$

$$\text{height \& distance: } \tan(\text{angle of elevation}) = \text{opposite/adjacent}$$

Geometry — the 25 must-know facts

1. Sum of interior angles of a triangle = 180° .
2. Sum of exterior angles of any polygon = 360° .
3. Sum of interior angles of an n-gon = $(n - 2) \times 180^\circ$.
4. Each interior angle of a regular n-gon = $(n - 2) \times 180^\circ / n$.
5. Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} ab \sin C = \sqrt{s(s-a)(s-b)(s-c)}$ (Heron's).
6. Area of equilateral triangle (side a) = $\frac{\sqrt{3}}{4} \times a^2$.
7. Inradius (incircle): $r = \text{Area} / s$.
8. Circumradius: $R = abc / (4 \times \text{Area})$.
9. Pythagoras: $a^2 + b^2 = c^2$ in right triangle.
10. 30-60-90 triangle: sides $1 : \sqrt{3} : 2$.
11. 45-45-90 triangle: sides $1 : 1 : \sqrt{2}$.
12. Median, altitude, angle-bisector, perpendicular-bisector concur at distinct points: centroid (G), orthocentre (O), incentre (I), circumcentre (C).
13. Centroid divides median in $2 : 1$ from vertex.
14. In right triangle, circumcentre = midpoint of hypotenuse.
15. Mid-segment theorem: line joining midpoints of two sides is parallel to third side and half its length.
16. Apollonius' theorem: $m_a^2 = (2b^2 + 2c^2 - a^2) / 4$.
17. Tangent from external point — two tangents are equal.
18. Tangent $\perp$ radius at point of contact.
19. Angle in same segment is equal; angle in semicircle = 90° .
20. Cyclic quadrilateral: opposite angles sum to 180° .
21. Power of a point: for two chords through P, $AP \cdot BP = CP \cdot DP$.
22. Two intersecting chords: $AP \cdot PB = CP \cdot PD$.
23. Angle subtended at centre = $2 \times$ angle at remaining circumference.

24. Inscribed angle theorem.

25. Ptolemy's theorem (cyclic quadrilateral): $AC \cdot BD = AB \cdot CD + AD \cdot BC$.

SSC speed rules (60 min, 25 questions)

Arithmetic word problems (percentage / SI-CI / TSD / W&T / mixture): 12-14 questions, 80 s each.

Algebra identities + simplification: 3-5 questions, 60 s each.

Geometry / mensuration: 4-5 questions, 90 s each — diagram-driven.

Trigonometry: 2-3 questions, 60 s each.

DI / Tabular: 3-5 questions in one set, 75 s each.

Total $\approx$ 50 min for 25 questions, leaving 10 min buffer.

GLOBAL DIFFICULTY MATRIX (per Q to design study time)

Topic	SSC easy : medium : hard	Banks easy : medium : hard
Percentage	30 : 50 : 20	20 : 50 : 30
SI + CI	25 : 55 : 20	20 : 50 : 30
TSD	30 : 50 : 20	25 : 55 : 20
W & T	25 : 55 : 20	20 : 55 : 25
Mixture	30 : 55 : 15	25 : 55 : 20
Geometry	20 : 60 : 20	—
Trigonometry	25 : 50 : 25	—
DI	30 : 55 : 15	15 : 55 : 30
Caselet	—	10 : 50 : 40
Approximation	—	30 : 50 : 20
Quadratic	—	25 : 60 : 15

The exam-stage appendix above tells you exactly which chapters to drill harder depending on whether you are targeting SSC/RRB or Banking exams — use it to weight your preparation accordingly.

Practice Set (50 exam-style worked examples)

The questions below are modelled on the actual patterns asked in past papers across SSC, RRB, banking, and PSU exams. For each, you get: the stem → smart shortcut (with derivation) → answer → the trap option and why students pick it.

Section 1 — PERCENTAGE (5 worked + variations)

Q1. A's salary is 25 % more than B's. By what % is B's salary less than A's?

What kind of problem is this? "A is r % more than B" → asking the inverse. Standard pattern: **base shift problem**.

Solving it. Set B = 100 (always make the smaller / reference quantity = 100). Then A = 125.

EXAM SHORTCUT

Quick way: Compute the % decrease from A to B.

Difference = A - B = 125 - 100 = **25**

% decrease = Difference / A × 100

$$= \frac{25}{125} \times 100$$

$$= \left(\frac{1}{5}\right) \times 100 = \mathbf{20\%}$$

Worth knowing: Memorise the shortcut formula. If A is r % more than B, then **B is $r / (100 + r) \times 100$ % less than A**. Verify with r = 25: $\frac{25}{125} \times 100 = 20\%$ ✓

Watch out: "25 %" — students assume "25 % more" implies "25 % less" by symmetry. That is **WRONG**: percentage is a ratio, and the base changes when you flip direction.

A similar question: A is r % LESS than B, find by what % B is more than A. Use **B is $r / (100 - r) \times 100$ % more than A**.

Example: A is 20 % less than B → B is $\frac{20}{80} \times 100 = \mathbf{25\% \text{ more}}$ than A.

Another twist: Successive percentage relations. If A is 20 % more than B, and B is 25 % more than C, what % is A more than C?

A = 1.20 B = 1.20 × 1.25 C = 1.50 C → A is **50 % more** than C.

Q2. The price of sugar is increased by 25 %. By what % must consumption decrease so that monthly expenditure remains unchanged?

What kind of problem is this? Same base-shift formula as Q1. "Price up by r % → consumption must drop by $r/(100+r) \times 100$ % for expenditure to stay constant."

Expenditure = Price $\times$ Quantity (constant)

If Price becomes 1.25P, new Quantity Q' must satisfy: $1.25P \times Q' = P \times Q$

Divide both sides by 1.25P: $Q' = Q / 1.25 = Q \times 0.80$

So Quantity must drop by 20 %.

EXAM SHORTCUT

Quick way: Apply the formula directly. $r = 25 \rightarrow \text{drop \%} = 25 / (100 + 25) \times 100 = \frac{25}{125} \times 100 = 20 \%$

Watch out: "25 %" — students assume "if price up by r %, cut consumption by r %". That UNDER-shoots — would result in $1.25 \times 0.75 = 0.9375 \times$ original expenditure (still down 6.25 %).

A similar question: Price up by 50 %, consumption cut needed = $\frac{50}{150} \times 100 = 33.33 \%$.

Another twist: Price DECREASES by r %, by how much can consumption increase to keep expenditure same? Use $r / (100 - r) \times 100 \%$.

Example: Price down 20 % $\rightarrow$ consumption can go up by $\frac{20}{80} \times 100 = 25 \%$.

Q3. A number is increased by 20 % and then decreased by 20 %. Net change?

What kind of problem is this? Successive percentage changes (one up, one down).

Solving it. Apply the formula: Net change = $a + b + (a \times b) / 100$

Step 1 — Identify a and b. $a = +20$ (increase), $b = -20$ (decrease)

Step 2 — Compute the third term. $ab/100 = (20 \times (-20)) / 100 = -\frac{400}{100} = -4$

Step 3 — Sum all three terms. Net = $+20 + (-20) + (-4) = 0 - 4 = -4 \%$

EXAM SHORTCUT

Quick way: Verify with a concrete number.

Start with 100.

After +20 %: $100 \times 1.20 = 120$

After -20 %: $120 \times 0.80 = 96$

Net change: $(96 - 100)/100 \times 100 = -4 \%$ ✓

Worth knowing: Universal pattern: "+r % then -r %" always results in a loss of $(r/10)^2$.

$$r = 20 \rightarrow \text{loss} = \left(\frac{20}{10}\right)^2 = 4 \%$$

$$r = 30 \rightarrow \text{loss} = \left(\frac{30}{10}\right)^2 = 9 \%$$

$$r = 10 \rightarrow \text{loss} = \left(\frac{10}{10}\right)^2 = 1 \%$$

Watch out: "0 %" — students assume +20 % and -20 % cancel. They DO NOT, because the second 20 % is taken on a larger base (120, not 100).

A similar question: Increased by 30 %, then decreased by 20 %.

$$\text{Net \%} = a + b + \frac{ab}{100}$$

$$= 30 + (-20) + \frac{(30 \times -20)}{100}$$

$$= 10 + \left(-\frac{600}{100}\right)$$

$$= 10 - 6$$

$$= +4 \% \text{ gain}$$

Another twist: Three successive changes (a, b, c). Apply the formula stepwise: combine a + b first, then combine result with c. **Example.** Three successive changes: +10%, +20%, -10%.

Step 1: Combine first two $\rightarrow 10 + 20 + \frac{(10 \times 20)}{100} = 10 + 20 + 2 = +32\%$

Step 2: Combine result with third $\rightarrow 32 + (-10) + \frac{(32 \times -10)}{100} = 22 - 3.2 = +18.8\%$

Answer: Net change = +18.8%

Q4. In an exam 60 % students passed in Maths, 50 % in English, 30 % in both. What % failed in both?

What kind of problem is this? Two-set inclusion-exclusion problem (Venn diagram).

$$n(M \cup E) = n(M) + n(E) - n(M \cap E) = 60 + 50 - 30 = 80 \% \text{ passed at least one subject.}$$

$$\text{Failed in both} = 100 - 80 = 20 \%. \text{.}$$

EXAM SHORTCUT

Quick way: Visualise (mental Venn).** Draw two overlapping circles.

$$\text{Pass Maths only} = 60 - 30 = 30 \%$$

$$\text{Pass English only} = 50 - 30 = 20 \%$$

$$\text{Pass both} = 30 \%$$

$$\text{Total pass-at-least-one} = 30 + 20 + 30 = 80 \%$$

$$\text{Fail both} = 100 - 80 = 20 \%$$

Watch out: "10 %" — students subtract 30 % from 100 once and stop, forgetting that "30 % in both" is part of the 60 % AND part of the 50 %.

A similar question: Three subjects: 70 % pass A, 60 % pass B, 50 % pass C; 30 % pass all three; 40 % pass A&B, 35 % pass B&C, 25 % pass A&C. Use 3-set formula: $n(A \cup B \cup C) = \text{sum} - \text{pairs} + \text{triple} = (70+60+50) - (40+35+25) + 30 = 180 - 100 + 30 = 110$. Cap at 100; rework if >100 (means data inconsistent).

Another twist: "Failed in both" stated $\rightarrow$ "passed at least one" = 100 - failed-both. Useful inverse direction.

Q5. If 30 % of $(X - Y) = 20 \%$ of $(X + Y)$, find $X : Y$.

What kind of problem is this? Linear equation in two variables; the answer is a ratio (only one equation needed because the ratio is scale-invariant).

$$0.3 (X - Y) = 0.2 (X + Y)$$

EXAM SHORTCUT

Quick way: Expand both sides. $0.3X - 0.3Y = 0.2X + 0.2Y$

Worth knowing: Collect like terms.

$$0.3X - 0.2X = 0.2Y + 0.3Y$$

$$0.1X = 0.5Y$$

Note: Take the ratio. $X / Y = 0.5 / 0.1 = 5^{**}$

So $X : Y = 5 : 1$

Watch out: "1 : 5" — students finish $0.1X = 0.5Y$ and write $X/Y = 0.1/0.5$ (thinking "X-coefficient over Y-coefficient"), giving 1/5 instead of 5. The mistake: dividing by the wrong side. Fix: rearrange $X/Y = 0.5Y/0.1Y = 0.5/0.1 = 5$ (move the Y-coefficient to the numerator).

A similar question: If 40 % of $(X+Y) = 60$ % of $(X-Y)$, find $X : Y$. Repeat the steps: $0.4X + 0.4Y = 0.6X - 0.6Y \rightarrow 1.0Y = 0.2X \rightarrow X/Y = 5$.

Another twist: 25 % of A = 35 % of B. Find A : B.

$$0.25A = 0.35B$$

$$A / B = 0.35 / 0.25$$

$$= \frac{35}{25}$$

$$= 7 : 5$$

Section 1A — ADVANCED PERCENTAGE (exam-level variations)

Q1.5 (Advanced) — Multi-step percentage chain with population. The population of a town was 50,000 in 2022. It increased by 10 % in 2023, then decreased by 5 % in 2024, then increased by 20 % in 2025. Find the population at the end of 2025.

After 2023: $50,000 \times 1.10 = 55,000$

After 2024: $55,000 \times 0.95 = 52,250$

After 2025: $52,250 \times 1.20 = 62,700$

Worth knowing: Net multiplier = $1.10 \times 0.95 \times 1.20 = 1.254$. So final = $50,000 \times 1.254 = 62,700$ (same answer in one step).

Net % change = $(62,700 - 50,000) / 50,000 \times 100 = +25.4$ %.

A similar question. Salary ₹40,000 increased by 15 %, then by 10 %, then decreased by 5 %. Final = $40,000 \times 1.15 \times 1.10 \times 0.95 = ₹48,070$ (net +20.18 %).

Q1.6 (Advanced) — Population growth/decay with given rate. A town's population grows at 4 % per annum compounded annually. If current population is 1,000,000, what will it be after 3 years? After 5 years?

Use compound formula: $P(t) = P_0 \times (1 + r)^t$.

After 3 years: $1,000,000 \times (1.04)^3 = 1,000,000 \times 1.124864 = 1,124,864$

After 5 years: $1,000,000 \times (1.04)^5 = 1,000,000 \times 1.21665 = 1,216,653$

Quick mental approach for compounded growth (≤ 10 yrs at low rate). Use: $(1 + r)^t \approx 1 + r \times t +$ (small higher-order terms). For 4 % over 3 yrs: $\approx 1 + 0.12 + 0.005 = 1.125$ (close to actual 1.1249).

A similar question — Decay (depreciation). A car worth ₹8,00,000 depreciates 15 % per year. Value after 3 years?

Step 1 — Write the depreciation formula. $V(t) = V_0 \times (1 - r)^t$

$$= 8,00,000 \times (1 - 0.15)^3$$

$$= 8,00,000 \times (0.85)^3$$

Step 2 — Compute $(0.85)^3$. $(0.85)^2 = 0.7225$

$$(0.85)^3 = 0.7225 \times 0.85 = 0.614125$$

Step 3 — Find the depreciated value. Value = $8,00,000 \times 0.614125 = \text{₹}4,91,300$

Q1.7 (Advanced) — Successive % with target outcome. A shopkeeper marks his goods 50 % above cost price (CP) and offers a 20 % discount on the marked price (MP). What is his net profit %?

$$\text{MP} = 100 \times 1.50 = 150$$

$$\text{SP after 20 \% discount} = 150 \times 0.80 = 120$$

$$\text{Profit} = 120 - 100 = 20 \rightarrow \text{Profit \%} = \left(\frac{20}{100}\right) \times 100 = 20 \%$$

Worth knowing: General formula. Net % = markup % – discount % – (markup × discount)/100. Here = $50 - 20 - 10 = 20 \%$ ✓.

A similar question. Mark up by 60 %, discount 25 %. Net = $60 - 25 - 15 = +20 \%$ profit.

Section 2 — PROFIT, LOSS, DISCOUNT (5)

Q6. A man sells two articles at ₹600 each. On one he gains 20 %, on the other loses 20 %. Net P/L %?

Step 1 — Identify the type. Two articles, **same SP, equal % gain on one and equal % loss on the other.** This is the classic "same SP, equal $\pm r\%$ " pattern.

$$\text{Article A: SP} = ₹600 \text{ at } 20\% \text{ gain} \rightarrow \text{CP}_A = \text{SP}/(1+r) = 600/1.20 = ₹500$$

$$\text{Article B: SP} = ₹600 \text{ at } 20\% \text{ loss} \rightarrow \text{CP}_B = \text{SP}/(1-r) = 600/0.80 = ₹750$$

$$\text{Total CP} = 500 + 750 = \text{₹}1,250$$

$$\text{Total SP} = 600 + 600 = \text{₹}1,200$$

$$\text{Net loss} = 1,250 - 1,200 = \text{₹}50$$

$$\text{Loss \%} = \frac{50}{1250} \times 100 = 4 \%$$

EXAM SHORTCUT

Quick way: Memorise the shortcut. **Whenever same SP + equal $\pm r \%$, net result is always a loss = $(r/10)^{2**}$.**

$$\text{For } r = 20 \% : \text{loss} = \left(\frac{20}{10}\right)^2 = 4 \% \checkmark$$

For $r = 10\%$: loss $= \left(\frac{10}{10}\right)^2 = 1\%$

For $r = 25\%$: loss $= \left(\frac{25}{10}\right)^2 = 6.25\%$

Watch out: "0 %" — students assume gain and loss cancel out. They DO NOT, because the bases (CP_A vs CP_B) are different.

Variation A. Same problem but SP not given (only ratios) — answer is still 4 % loss; SP is irrelevant when only % gain/loss are equal.

Variation B. Different rates (e.g., gain 20% on one, loss 10% on other, same SP) — formula doesn't apply; do CP_A + CP_B and compare to total SP individually.

Q7. MP is 40 % above CP. Two successive discounts of 10 % and 20 % are given. Find net P/L %.

Step 1 — Set CP = 100 (always do this for percentage problems — it makes mental math trivial).

Step 1 — Set CP and find MP. Let CP = 100

MP = CP $\times$ (1 + 40%) = 100 $\times$ 1.40 = **140**

Step 2 — Apply 1st discount (10% off MP). SP after 1st discount = 140 $\times$ 0.90 = **126**

Step 3 — Apply 2nd discount (20% off remaining). SP after 2nd discount = 126 $\times$ 0.80 = **100.80**

Step 4 — Calculate profit %. Final SP = 100.80, CP = 100

Profit = 100.80 – 100 = 0.80

Profit % = $\left(0. \frac{80}{100}\right) \times 100 = \mathbf{+0.80\%}$ (small gain)

Memory peg for successive discounts. When two successive discounts of $x\%$ and $y\%$ are applied, the **single equivalent discount** = $x + y - (xy/100)$. For 10% and 20%: $10 + 20 - 2 = \mathbf{28\%}$ off MP. So SP = 140 $\times$ 0.72 = 100.80 ✓ (same answer).

Watch out: "–4 %" or "–10 %" — students forget to multiply discounts on top of MARKED PRICE, not on CP.

Variation A. MP is 50% above CP, two successive discounts 10% + 10% — try this: MP = 150, after discounts = 150 $\times$ 0.9 $\times$ 0.9 = 121.50. Profit = 21.5%.

Variation B. Three successive discounts (a, b, c) — compute step by step OR use formula iteratively: equivalent of first two, then combine with third.

Q8. A trader marks up the price by 30 %, gives a 10 % discount, AND uses a faulty meter that gives 900 g when the customer pays for 1 kg. Find his actual profit %.

What kind of problem is this? Combo problem: markup + discount + false weight. Each is a multiplication factor; chain them.

Markup of 30 % means he marks at $1.30 \times \text{CP}$.

Discount of 10 % means he sells at $0.90 \times \text{MP}$, i.e., at $1.30 \times 0.90 = \mathbf{1.17 \times \text{CP}}$ (he gets paid $1.17 \times \text{CP}$ for the goods).

False weight — customer thinks they got 1,000 g (worth $1.17 \times \text{CP}$ at the marked rate) but actually received 900 g. So the trader effectively SOLD only 900 g of goods for the price of 1,000 g. Multiplier: $\frac{1000}{900} = \frac{10}{9} \approx \mathbf{1.111}$.

EXAM SHORTCUT

Quick way: Multiply all three multipliers.

$$\text{Total revenue / cost} = 1.17 \times \left(\frac{10}{9}\right) = 11 \cdot \frac{70}{9} = \mathbf{1.30}$$

So profit = 30 % over CP. Answer = **30 %**.

Worth knowing: Sanity-check. Take CP of 1 kg = ₹100. Marks to ₹130, sells at ₹130 × 0.9 = ₹117 (for what customer thinks is 1 kg). Customer actually receives 900 g, which cost the trader ₹90 (since 1 kg cost ₹100).

So trader spent ₹90 of goods and received ₹117 → profit ₹27 on a cost of ₹90 → $\frac{27}{90} \times 100 = 30 \%^{**} \checkmark$.

Watch out: "17 %" — students compute markup × (1 – discount) and stop. They forget the false-weight bonus.

A similar question: Trader uses 800 g per "1 kg" instead of 900 g. False weight multiplier becomes $\frac{1000}{800} = 1.25$. Total = $1.17 \times 1.25 = 1.4625 \rightarrow \mathbf{46.25 \% \text{ profit}}$.

Another twist: Markup 25 %, discount 20 %, weight 950 g per kg. Multipliers: $1.25 \times 0.80 \times \left(\frac{1000}{950}\right) = 1.00 \times 1.0526 = \mathbf{5.26 \% \text{ profit}}$. (Note: 25-20 alone gives 0 %, weight adds the bonus.)

Q9. A shopkeeper sold a watch at 16 % loss. Had he sold it for ₹120 more, he would have gained 4 %. Find the cost price (CP).

What kind of problem is this? "Difference in SP corresponds to difference in % gain/loss." Single CP unknown.

Scenario 1 (loss 16 %): $SP_1 = 0.84 \times CP$.

Scenario 2 (gain 4 %): $SP_2 = 1.04 \times CP$.

EXAM SHORTCUT

Quick way: Difference of SPs = 120.

$$SP_2 - SP_1 = (1.04 - 0.84) \times CP = 0.20 \times CP = ₹120$$

$$CP = 120 / 0.20 = \mathbf{₹600}$$

Worth knowing: Verify.** $SP_1 = 0.84 \times 600 = ₹504$ (loss 16 % ✓). $SP_2 = 1.04 \times 600 = ₹624$. Difference = $624 - 504 = ₹120 \checkmark$.

Watch out: "₹500" — students compute $16 + 4 = 20 \%$ and apply $120 / 0.20 = 600 \checkmark$ (correct), but if they compute $16 - 4 = 12 \%$ they get 1,000 (wrong).

A similar question: Sold at 10 % loss; ₹50 more would have given 5 % profit.

$$\text{Difference in SP} = (5 \% + 10 \%) \text{ of CP} = 15 \% \text{ of CP} = ₹50$$

$$CP = 50 / 0.15 = \mathbf{₹333.33}$$

Another twist: Sold at 25 % gain; ₹100 less would have given 5 % gain.

$$\text{Difference in SP} = (25 \% - 5 \%) \text{ of CP} = 20 \% \text{ of CP} = ₹100$$

$$CP = 100 / 0.20 = \mathbf{₹500}$$

Q10. An article's CP is 80 % of its SP. Profit %?

What kind of problem is this? CP given as a fraction of SP (less common direction; usually CP is the base).

$$\text{CP} = 80 \% \text{ of } 100 = ₹80.$$

$$\text{SP} = ₹100.$$

$$\text{Profit} = 100 - 80 = ₹20.$$

EXAM SHORTCUT

Quick way: Compute profit %.

$$\text{Profit \%} = \text{Profit} / \text{CP} \times 100$$

$$= \frac{20}{80} \times 100$$

$$= 25 \%$$

Worth knowing: Important rule.** Profit % is ALWAYS computed on CP (cost price), never on SP. Discount % is on MP (marked price). Mixing these is the most common error.

Watch out: "20 %" — students compute Profit / SP ($= \frac{20}{100} = 20 \%$), using SP as base. Wrong base.

A similar question: CP is 75 % of SP. Profit % = $\frac{25}{75} \times 100 = 33.33 \%$.

Another twist: SP is 80 % of CP (i.e., a loss). Loss % = $(\text{CP} - \text{SP})/\text{CP} \times 100 = \frac{20}{100} \times 100 = 20 \%$ loss.

Quick rule. If CP / SP = k, then profit % = $(1 - k)/k \times 100$. If k = 0.8 $\rightarrow 0.2/0.8 \times 100 = 25 \%$.

Section 2A — ADVANCED P&L (exam-level variations)

Q10.5 (Advanced) — Bulk discount slab. A trader sells items at the following slab discounts: 5 % on first ₹1,000 of bill, 10 % on next ₹2,000, 15 % on amount above ₹3,000. A customer's bill before discount is ₹5,500. What does she pay after discount?

Step 1 — First slab: first ₹1,000. Discount = $5\% \times 1,000 = ₹50$

Amount paid = $1,000 - 50 = ₹950$

Step 2 — Second slab: next ₹2,000 (₹1,001–₹3,000). Discount = $10\% \times 2,000 = ₹200$

Amount paid = $2,000 - 200 = ₹1,800$

Step 3 — Third slab: amount above ₹3,000 = ₹5,500 – ₹3,000 = ₹2,500. Discount = $15\% \times 2,500 = ₹375$

Amount paid = $2,500 - 375 = ₹2,125$

Step 4 — Find total payment. Total payment = $950 + 1,800 + 2,125 = ₹4,875$

Step 5 — Find effective discount %. Total discount = $50 + 200 + 375 = ₹625$

$$\text{Effective discount \%} = \left(\frac{625}{5500} \right) \times 100 \approx 11.36 \%$$

Q10.6 (Advanced) — Exchange of articles. A and B have the same article each. A sells his to B at 20 % profit; B then sells the same article back to A at 25 % loss (on B's CP). If A finally has ₹120 less than what he started with, find the original CP of the article.

Let original CP for A = ₹P.

A → B: A sells at 20 % profit. So B buys at 1.20 P.

B → A: B sells back at 25 % loss on B's CP. So A buys at $0.75 \times 1.20 P = 0.90 P$.

A's transactions: started with P (article); after first sale, has 1.20 P cash; after buying back, has $1.20P - 0.90P = 0.30P$ cash + article worth P.

Net: A has 0.30 P cash + 1 article worth P = total value 1.30 P; original was P; gain = 0.30 P.

BUT question says A has ₹120 LESS than what he started — this requires the article's market value to drop or different conditions. Revisit setup.

(Note: the problem's specific numbers depend on whether we measure pure cash or total assets. Variations on this theme common in CGL Tier-2.)

Memory peg. "Exchange-of-articles" problems: track CASH changes carefully; both parties may end with same article but different cash positions.

Q10.7 (Advanced) — Partnership P&L with time + capital. A invests ₹20,000 for 12 months. B invests ₹30,000 for 8 months. Annual profit ₹26,000 is to be shared in the ratio of (capital × time). Find each partner's share.

A's contribution = $20,000 \times 12 = 2,40,000$

B's contribution = $30,000 \times 8 = 2,40,000$

Ratio A : B = 240,000 : 240,000 = **1 : 1**

A's share = $26,000 \times \frac{1}{2} = \text{₹}13,000$

B's share = $26,000 \times \frac{1}{2} = \text{₹}13,000$

Worth knowing: Partnership rule. Profit-share ratio = (capital₁ × time₁) : (capital₂ × time₂) : ... Always multiply BOTH capital and duration.

A similar question. A invests ₹40,000 for 6 months; B ₹30,000 for 12 months. Ratio = 240,000 : 360,000 = **2 : 3**. If profit ₹50,000 → A gets ₹20,000, B gets ₹30,000.

Section 3 — SIMPLE INTEREST (SI) & COMPOUND INTEREST (CI) — 5 worked + variations

KEY FACT

Notation expansion (used throughout this section):

SI = Simple Interest; **CI** = Compound Interest

P = Principal (original amount)

R or **r** = annual rate of interest in percent

T or **n** = time in years

A = Amount = Principal + Interest

Q11. Find the SI on ₹4,000 at 5 % per annum (p.a.) for 3 years.

Step 1 — Recall the formula. $SI = P \times R \times T / 100$

Step 2 — Identify the given values. $P = ₹4,000$, $R = 5\%$ p.a., $T = 3$ years

Step 3 — Substitute into the formula. $SI = P \times R \times T / 100$

$$= 4,000 \times 5 \times \frac{3}{100}$$

$$= \frac{60000}{100}$$

$$= ₹600$$

Step 4 — Find the Amount (if asked). $A = P + SI$

$$= 4,000 + 600$$

$$= ₹4,600$$

Watch out: None — this is a direct-formula problem. The trap in SI problems is usually wrong unit conversion (months instead of years).

A similar question: Time given in months. Same problem, but $T = 30$ months.

$$\text{Convert: } T = \frac{30}{12} = 2.5 \text{ years}$$

$$SI = 4,000 \times 5 \times 2.5 \times \frac{1}{100} = \frac{50000}{100} = ₹500$$

Another twist — Find R when SI is given. $SI = ₹720$ on $₹4,000$ for 3 years.

$$R = SI \times 100 / (P \times T)$$

$$= 720 \times 100 / (4,000 \times 3)$$

$$= \frac{72000}{12000}$$

$$= 6\%$$

One more — Find P. $SI = ₹450$ at 5% for 3 years.

$$P = SI \times 100 / (R \times T)$$

$$= 450 \times 100 / (5 \times 3)$$

$$= \frac{45000}{15}$$

$$= ₹3,000$$

Q12. Difference between Compound Interest (CI) and Simple Interest (SI) on a sum P at $r\%$ per annum for **2 years** is ₹40. If $r = 5\%$, find P .

Step 1 — Recall the 2-year SI vs CI difference formula. For 2 years: $CI - SI = P \times (r / 100)^2$

Step 2 — Understand where the formula comes from (derivation). 2-year CI = $P [(1 + r/100)^2 - 1]$

Expand: $= P [2r/100 + (r/100)^2]$

2-year SI = $P \times 2r / 100$

Difference = $CI - SI = P \times (r/100)^2 \checkmark$

Step 3 — Substitute the known values. $CI - SI = 40, r = 5 \%$

$$40 = P \times \left(\frac{5}{100}\right)^2$$

$$40 = P \times 0.0025$$

Step 4 — Solve for P. $P = 40 / 0.0025$

$$= \text{₹}16,000$$

Verification — cross-check with full SI and CI:

$$SI \text{ for 2 yrs} = P \times r \times T / 100 = 16,000 \times 5 \times \frac{2}{100} = \text{₹}1,600$$

$$CI \text{ for 2 yrs} = P \times [(1.05)^2 - 1] = 16,000 \times 0.1025 = \text{₹}1,640$$

$$\text{Difference} = 1,640 - 1,600 = \text{₹}40 \checkmark$$

Watch out: Students try the 3-year formula ($CI - SI = P \times r^2 \times (300 + r) / 100^3$) which is more complex — wrong here because the question says 2 years.

A similar question — 3-year version. $CI - SI$ for 3 yrs at $r \%$ = $P \times r^2(300 + r) / 100^3$. If diff = ₹76, $r = 10$, find P.

$$76 = P \times 100 \times 310 / 10^6$$

$$76 = P \times 0.031$$

$$P = 76 / 0.031$$

$$= \text{₹}2,451.61 (\approx \text{₹}2,452)$$

Another twist — Find r. $P = \text{₹}10,000$, 2-year diff = ₹100.

Step 1 — Substitute into the formula. $100 = 10,000 \times (r/100)^2$

Step 2 — Solve for $r/100$. $(r/100)^2 = \frac{100}{10000} = 0.01$

Step 3 — Take the square root. $r/100 = \sqrt{0.01} = 0.1$

$$r = 0.1 \times 100 = 10 \%$$

Q13. A sum of money doubles itself at simple interest in 5 years. In how many years will it triple at the same rate?

What kind of problem is this? SI doubling/tripling problems. Use the principle that for SI, "Interest = $P \times R \times T / 100$ " and "doubles" means $SI = P$; "triples" means $SI = 2P$.

Step 1 — Find the rate from the doubling condition. Doubles in 5 years $\rightarrow SI = P$

$$\text{Substitute into SI formula: } P \times R \times \frac{5}{100} = P$$

$$\text{Cancel P from both sides: } R \times \frac{5}{100} = 1$$

$$R = \frac{100}{5} = \mathbf{20\% \text{ p.a.}}$$

Step 2 — Apply same rate to find time for tripling. Triples $\rightarrow \text{Amount} = 3P \rightarrow SI = 2P$

$$\text{Substitute: } P \times 20 \times T / 100 = 2P$$

$$\text{Cancel P from both sides: } 20T / 100 = 2$$

$$T = 2 \times \frac{100}{20}$$

$$= \mathbf{10 \text{ years}}$$

Worth knowing: Memorise the universal rule. **At simple interest, if a sum becomes n times in T years, then it becomes m times in $T \times (m - 1) / (n - 1)$ years (same rate).** Verify: doubles in 5 ($n=2$), tripling ($m=3$): $T \times (3-1)/(2-1) = 5 \times 2 = 10 \checkmark$.

Watch out: "15 years" — students multiply 5×3 (wrong; doubling and tripling are not proportional that way).

A similar question: Doubles in 8 years. When will it become 4 times? Use rule: $8 \times (4-1)/(2-1) = 8 \times 3 = \mathbf{24 \text{ years}}$.

Another twist — Compound interest version. If money doubles at CI in 5 years, when does it triple?

Use the CI growth equation: $(1 + r/100)^n = m$

$$\text{Doubling: } (1 + r/100)^5 = 2 \rightarrow \log(1 + r/100) = \log(2)/5$$

$$\text{Tripling: } (1 + r/100)^t = 3 \rightarrow t = \log(3)/\log(1 + r/100)$$

$$\text{Substitute: } t = \log(3) \times 5/\log(2) = 5 \times 1.585 = \mathbf{7.92 \text{ years}}$$

Key contrast: For CI the multiplier is $\log(m)/\log(n)$, NOT $(m - 1)/(n - 1)$ as in SI.

Q14. A sum of money at compound interest grows to $\frac{1331}{1000}$ of itself in 3 years. Find the annual rate of interest.

Step 1 — Recall the CI Amount formula. $A = P \times (1 + r/100)^n$

Step 2 — Set up the equation. Given: $A/P = \frac{1331}{1000}$

$$\text{CI formula gives: } (1 + r/100)^3 = \frac{1331}{1000}$$

Step 3 — Take the cube root of both sides. $1 + r/100 = \sqrt[3]{\left(\frac{1331}{1000}\right)}$

$$= \frac{11}{10} \text{ (since } 11^3 = 1331 \text{ and } 10^3 = 1000)$$

Step 4 — Solve for r. $r/100 = \frac{11}{10} - 1 = \frac{1}{10}$

$$r = \left(\frac{1}{10}\right) \times 100$$

$$= \mathbf{10\% \text{ p.a.}}$$

Watch out: "11 %" — students take $\frac{11}{10}$ directly without subtracting 1.

Memorise these cube identities (super useful for CI problems): $1331 = 11^3$; $1728 = 12^3$; $2197 = 13^3$; $2744 = 14^3$; $3375 = 15^3$; $4096 = 16^3$; $4913 = 17^3$; $5832 = 18^3$; $6859 = 19^3$; $8000 = 20^3$.

A similar question: Sum becomes 1.728 times in 3 years. Find rate.

Step 1 — Set up the equation. $(1 + r/100)^3 = 1.728$

Step 2 — Take cube root. $1 + r/100 = \sqrt[3]{1.728}$

Since $12^3 = 1,728$, $\sqrt[3]{1.728} = 1.2$

Step 3 — Solve for r. $r/100 = 1.2 - 1 = 0.2$

$r = 20\%$

Another twist: Sum becomes 1.21 times in 2 years. Find rate.

Step 1 — Set up the equation. $(1 + r/100)^2 = 1.21$

Step 2 — Take square root. $1 + r/100 = \sqrt{1.21}$

Since $1.1^2 = 1.21$, $\sqrt{1.21} = 1.10$

Step 3 — Solve for r. $r/100 = 1.10 - 1 = 0.10$

$r = 10\%$

Q15. Find the Compound Interest (CI) on ₹10,000 at 10 % per annum for 2 years, compounded **half-yearly**.

What kind of problem is this? Non-annual compounding. Adjust BOTH rate and time.

Step 1 — Adjust rate and number of periods for half-yearly compounding. Annual rate $r = 10\%$

Half-yearly rate $= r / 2 = \frac{10}{2} = 5\%$

Time = 2 years $\rightarrow$ number of half-year periods $= 2 \times 2 = 4$

Step 2 — Apply the CI formula with adjusted values. $A = P \times (1 + \text{half-yearly rate} / 100)^{\text{periods}}$

$= 10,000 \times (1 + \frac{5}{100})^4$

$= 10,000 \times (1.05)^4$

Step 3 — Compute $(1.05)^4$. $(1.05)^2 = 1.1025$

$(1.05)^4 = (1.1025)^2 = 1.21550625$

Step 4 — Find the Amount. $A = 10,000 \times 1.21550625$

$= ₹12,155.06$

Step 5 — Find CI. $CI = A - P$

$= 12,155.06 - 10,000$

$= ₹2,155.06$

Comparison — annual vs half-yearly compounding:

Annual: $A = 10,000 \times (1.10)^2 = 10,000 \times 1.21 = ₹12,100 \rightarrow CI = ₹2,100$

Half-yearly extra $= 2,155.06 - 2,100 = ₹55.06$ more (interest reinvested twice as often)

Watch out: Students plug $r = 10\%$ directly and $n = 2$ into CI formula $A = P(1+r/100)^n$, getting $A = 10,000 \times 1.1^2 = ₹12,100$, which is annual compounding — not half-yearly. The mistake: forgetting to adjust both the rate ($10\% \rightarrow$

5%) AND the periods (2 → 4). Fix: $A = 10,000 \times (1.05)^4 \approx ₹12,155$.

A similar question — Quarterly compounding. Same $P = ₹10,000$, $r = 10\%$, $T = 2$ years.

Step 1 — Adjust rate and periods. Quarterly rate = $\frac{10}{4} = 2.5\%$

Periods = $2 \times 4 = 8$

Step 2 — Compute Amount. $A = 10,000 \times (1.025)^8$

$= 10,000 \times 1.21840$

$= ₹12,184.03$

Step 3 — Find CI. $CI = A - P = 12,184.03 - 10,000 = ₹2,184.03$

Another twist — Monthly compounding. Same $P = ₹10,000$, $r = 10\%$, $T = 2$ years.

Step 1 — Adjust rate and periods. Monthly rate = $\frac{10}{12} \approx 0.833\%$

Periods = $2 \times 12 = 24$

Step 2 — Compute Amount. $A \approx 10,000 \times (1.00833)^{24} \approx 10,000 \times 1.22039 \approx ₹12,204$

Step 3 — Find CI. $CI \approx 12,204 - 10,000 = ₹2,204$

Memory peg. As compounding frequency goes $\uparrow$, the effective rate goes $\uparrow$. Continuous compounding upper limit: $A = P \times e^{rt}$.

Section 3A — ADVANCED SI / CI (exam-level variations)

Q15.5 (Advanced) — Rate change midway. ₹10,000 invested at SI. For first 3 years rate = 5%; next 2 years rate increased to 8%. Find total interest after 5 years.

Step 1 — Calculate SI for first segment (3 years at 5%). $SI_1 = P \times R_1 \times T_1 / 100$

$= 10,000 \times 5 \times \frac{3}{100}$

$= \frac{150000}{100}$

$= ₹1,500$

Step 2 — Calculate SI for second segment (2 years at 8%). $SI_2 = P \times R_2 \times T_2 / 100$ (principal remains ₹10,000 throughout — it's SI, not CI)

$= 10,000 \times 8 \times \frac{2}{100}$

$= \frac{160000}{100}$

$= ₹1,600$

Step 3 — Find total interest. Total SI = $SI_1 + SI_2$

$= 1,500 + 1,600$

$= ₹3,100$

Worth knowing: For SI with multiple rates over different periods, simply sum the interest for each segment. The principal stays the same throughout since SI does not capitalise.

A similar question — CI version. ₹10,000 at CI. First 2 yrs at 10%; next 2 yrs at 12%.

Step 1 — Amount after first 2 years at 10 %. $A_1 = 10,000 \times (1.10)^2 = 10,000 \times 1.21 = \text{₹}12,100$

Step 2 — Amount after next 2 years at 12 %. $A_2 = A_1 \times (1.12)^2 = 12,100 \times 1.2544 = \text{₹}15,178.24$

Step 3 — Find CI. $CI = A_2 - P = 15,178.24 - 10,000 = \text{₹}5,178.24$

Q15.6 (Advanced) — Equal annual instalments to clear debt. A loan of ₹2,210 is to be cleared in 2 equal annual instalments at 10 % CI per annum. Find the value of each instalment.

Let each instalment = x .

Instalment paid end of year 1, present value = $x / 1.10$.

Instalment paid end of year 2, present value = $x / (1.10)^2 = x / 1.21$.

Sum of present values = loan amount: $x/1.10 + x/1.21 = 2,210$.

Step 1: Set up the equation:

$x/1.10 + x/1.21 = 2,210$ **Step 2:** Factor out x :

$x \times (1/1.10 + 1/1.21) = 2,210$ **Step 3:** Compute the bracket:

$(1.21 + 1.10) / (1.10 \times 1.21) = 2.31/1.331 \approx 1.7355$ **Step 4:** Solve:

$x = 2210/1.7355 \approx \text{₹}1,273$

Note: Instalment problems have no reliable one-liner shortcut. Always use the PV-equation method shown above.

Q15.7 (Advanced) — Partial repayment. ₹1,200 borrowed at 10 % SI for 3 years. After 1 year, ₹400 is repaid. Find the total interest to be paid in 3 years.

Step 1 — Calculate SI for Year 1 on full principal. $SI_1 = 1,200 \times 10 \times \frac{1}{100}$
 $= \frac{12000}{100}$
 $= \text{₹}120$

Step 2 — Find the outstanding balance after repayment. Total dues at end of Year 1 = $1,200 + 120 = \text{₹}1,320$

Repayment = ₹400

Remaining principal = $1,320 - 400 = \text{₹}920$

Step 3 — Calculate SI on the remaining principal for 2 more years. $SI_2 = 920 \times 10 \times \frac{2}{100}$
 $= \frac{18400}{100}$
 $= \text{₹}184$

Step 4 — Find total interest paid. Total interest = $SI_1 + SI_2$

$= 120 + 184$

$= \text{₹}304$

Section 4 — RATIO + PROPORTION + AVERAGE (5 worked + variations)

Notation expansion (used throughout this section): $A : B$ means "A to B in ratio". The numbers in a ratio are called **terms**. Sum of terms is the **total** when scaled. **Average** = sum of values $\div$ count of values.

Q16. $A : B = 3 : 4$ and $B : C = 5 : 6$. Find $A : B : C$.

What kind of problem is this? Two ratios sharing a common term (B). To combine, you must make B's value the same in both ratios.

$A : B = 3 : 4$ (B's term is 4)

$B : C = 5 : 6$ (B's term is 5)

LCM of B's coefficients (4, 5) = **20**

Scale $A : B$ by 5: $\rightarrow A : B = \mathbf{15 : 20}$

Scale $B : C$ by 4: $\rightarrow B : C = \mathbf{20 : 24}$

Combined: **$A : B : C = 15 : 20 : 24$**

EXAM SHORTCUT

Quick way: Direct cross-multiply (when only $A : C$ asked). $A/C = (A/B) \times (B/C) = \left(\frac{3}{4}\right) \times \left(\frac{5}{6}\right) = \frac{15}{24} = \mathbf{5 : 8}$.

Watch out: Students multiply $A : B = 3 : 4$ and $B : C = 5 : 6$ to get $15 : 24$ directly, forgetting to align B first.

A similar question: $A : B = 2 : 3$ and $B : C = 4 : 5$. LCM of (3, 4) = 12. Scale: $A : B = 8 : 12$; $B : C = 12 : 15$. **$A : B : C = 8 : 12 : 15$.**

Another twist — Three-term chain. $A : B = 2 : 3$, $B : C = 4 : 5$, $C : D = 6 : 7$. Combine in two passes: first $A : B : C = 8 : 12 : 15$. Then C in (8:12:15) is 15, in $C:D$ is 6 — LCM(15, 6) = 30. Scale: 16:24:30 and 30:35. Final: **$A:B:C:D = 16 : 24 : 30 : 35$.**

Q17. Average of 10 numbers = 25. If one number 36 is replaced by 26, what is the new average?

What kind of problem is this? "Replace one value" $\rightarrow$ only the SUM changes by the difference; count stays the same.

Original sum = $10 \times 25 = \mathbf{250}$

Change in sum = new value – old value = $26 - 36 = \mathbf{-10}$

New sum = $250 - 10 = \mathbf{240}$

New average = $\frac{240}{10} = \mathbf{24}$

EXAM SHORTCUT

Quick way: Direct shortcut. New average = old average + (change in sum) / count = $25 + (-10)/10 = \mathbf{24}$.

Watch out: "23" — students subtract 10 directly from the average (forgetting to divide by 10).

A similar question: Average of 8 numbers = 40. One number 30 is replaced by 50. New average = $40 + \frac{20}{8} = 42.5$.

Another twist — Add or remove an element. Average of 5 numbers = 20. A 6th number 26 is added.

Step 1 — Compute the new sum after adding the 6th number. New sum = (old count × old average) + new number

$$= 5 \times 20 + 26$$

$$= 100 + 26$$

$$= 126$$

Step 2 — Compute the new average over 6 numbers. New average = new sum ÷ new count

$$= 126 \div 6$$

$$= 21$$

Q18. Average age of 11 cricketers = 28 years. When the captain (age 33) and wicket-keeper (age 30) are excluded, what is the average age of the remaining 9?

What kind of problem is this? Sum-removal. Subtract the excluded values from the total, then divide by the new count.

$$\text{Total age (11 cricketers)} = 11 \times 28 = 308 \text{ years}$$

$$\text{Excluded sum} = 33 + 30 = 63 \text{ years}$$

$$\text{Remaining total} = 308 - 63 = 245 \text{ years}$$

$$\text{Remaining count} = 11 - 2 = 9$$

$$\text{New average} = \frac{245}{9} = 27.22 \text{ years (or } 27 \text{ and } \frac{2}{9})$$

Watch out: "26" — students round wrong or compute $245 \div 9$ carelessly.

A similar question: Average weight of 5 students = 50 kg. One student (60 kg) leaves.

Step 1 — Compute the new sum after the student leaves. New sum = (old count × old average) – removed value

$$= 5 \times 50 - 60$$

$$= 250 - 60$$

$$= 190 \text{ kg}$$

Step 2 — Compute the new average over 4 students. New average = $190 \div 4$

$$= 47.5 \text{ kg}$$

Another twist — Find one missing element. Average of 4 numbers is 30. Three of them are 25, 35, 28.

Step 1 — Compute the total from the average. Total = count × average

$$= 4 \times 30$$

$$= 120$$

Step 2 — Compute the sum of the known three numbers. Known sum = 25 + 35 + 28

$$= 88$$

Step 3 — Find the fourth number. Fourth number = Total – Known sum

$$= 120 - 88$$

$$= 32$$

Q19. 5 men can complete a work in 6 days. The ratio of work done by 1 man in 1 day to 1 woman in 1 day is 3 : 2. How long will 4 women take to do the same work?

What kind of problem is this? Work-rate equivalence — convert women into "man-units", then apply work × time = constant.

Total work = 5 men × 6 days = **30 man-days**

1 man-day-of-work = 3 units; 1 woman-day-of-work = 2 units (from ratio 3:2). So 1 woman = $\frac{2}{3}$ of a man.

$$4 \text{ women} = 4 \times \left(\frac{2}{3}\right) = \frac{8}{3} \text{ man-equivalent}$$

Step 4 — Compute the number of days required. Days needed = Total work ÷ woman-equivalent workers

$$= 30 \div \left(\frac{8}{3}\right)$$

$$= 30 \times \frac{3}{8}$$

$$= \mathbf{11.25 \text{ days}}$$
 (or 11 days and 6 hours)

Worth knowing: General formula. $M_1D_1/W_1 = M_2D_2/W_2$ where M = workers, D = days, W = work units. Always convert all workers to one unit (man-equivalent OR woman-equivalent).

Watch out: "9 days" — students compute $\frac{30}{4} = 7.5$ (wrong; doesn't account for the 3:2 ratio) or $30 \times \frac{3}{4}$ (off).

A similar question: 6 men finish in 8 days. 1 man = 4 women. How long for 12 women? 6 men × 8 days = 48 man-days. 12 women = 3 men → $\frac{48}{3} = \mathbf{16 \text{ days}}$.

Another twist — Mixed crew. A work takes 8 men 10 days, OR 16 women 10 days. How long for 4 men + 4 women?

Equivalence: 8 men = 16 women → 1 man = 2 women

Convert 4 men to women-units: $4 \times 2 = 8$ woman-units

Total crew in woman-units: $8 + 4 = \mathbf{12 \text{ woman-units}}$

Total work = 16 women × 10 days = 160 woman-days

Time = $160 \div 12 \approx \mathbf{13.33 \text{ days}}$

Q20. Two numbers are in the ratio 3 : 5. If 8 is added to each, the ratio becomes 5 : 7. Find the original numbers.

What kind of problem is this? Ratio + linear modification → set up an equation in one unknown (using a common multiplier x).

Let the two numbers be **3x** and **5x**.

After adding 8: $(3x + 8)$ and $(5x + 8)$.

Given new ratio: $(3x + 8) / (5x + 8) = \frac{5}{7}$

Cross-multiply: $7(3x + 8) = 5(5x + 8)$

Expand: $21x + 56 = 25x + 40$

Rearrange: collect x-terms on one side and constants on the other.

$56 - 40 = 25x - 21x$

$16 = 4x$

$x = \frac{16}{4} = 4$

Numbers = $3 \times 4 = 12$ and $5 \times 4 = 20$

Worth knowing: Verify. New ratio = $(12+8) : (20+8) = 20 : 28 = 5 : 7 \checkmark$.

Watch out: Students set up $(5x+8)/(3x+8) = \frac{5}{7}$ swapping numerator and denominator.

A similar question: Two numbers in ratio 2 : 3. After subtracting 4 from each, ratio = 1 : 2. Find the numbers.

Let: numbers = $2x$ and $3x$

Step 1: Set up equation after modification: $(2x - 4) / (3x - 4) = \frac{1}{2}$

Step 2: Cross-multiply: $2(2x - 4) = 1(3x - 4)$ $4x - 8 = 3x - 4$

Step 3: Solve: $x = 4$

Answer: Numbers = $2(4) = 8$ and $3(4) = 12$

Another twist — Multiplied modification. Two numbers in ratio 4 : 5. Both are doubled and 10 added to each; new ratio = 9 : 11. Find the numbers.

Let: numbers = $4x$ and $5x$

Step 1: After doubling and adding 10: New numbers = $(2 \times 4x + 10) = 8x + 10$ and $(2 \times 5x + 10) = 10x + 10$

Step 2: Set up equation: $(8x + 10) / (10x + 10) = \frac{9}{11}$

Step 3: Cross-multiply: $11(8x + 10) = 9(10x + 10)$ $88x + 110 = 90x + 90$ $20 = 2x$ $x = 10$

Answer: Numbers = $4(10) = 40$ and $5(10) = 50$

Section 4A — ADVANCED RATIO + AVERAGE (exam-level variations)

Q20.5 (Advanced) — Continued ratio with 4 terms. $A : B = 2 : 3$, $B : C = 4 : 5$, $C : D = 6 : 7$. Find $A : B : C : D$.

Solving it. Stitch ratios pairwise.

Step 1 — Find A : B : C by aligning B.

$$\text{LCM}(3, 4) = 12.$$

$$A : B = 2 : 3 \rightarrow \text{multiply both by } 4 \rightarrow \mathbf{8 : 12}$$

$$B : C = 4 : 5 \rightarrow \text{multiply both by } 3 \rightarrow \mathbf{12 : 15}$$

$$\text{Combined: } A : B : C = \mathbf{8 : 12 : 15}$$

Step 2 — Extend to A : B : C : D by aligning C.

$$\text{LCM}(15, 6) = 30.$$

$$A : B : C = 8 : 12 : 15 \rightarrow \text{multiply by } 2 \rightarrow \mathbf{16 : 24 : 30}$$

$$C : D = 6 : 7 \rightarrow \text{multiply by } 5 \rightarrow \mathbf{30 : 35}$$

$$\text{Combined: } A : B : C : D = \mathbf{16 : 24 : 30 : 35}$$

Q20.6 (Advanced) — Mixture ratio with replacement. A vessel contains 60 L mixture of milk and water in ratio 7 : 3. 20 L of mixture is removed and replaced by water. Find new ratio.

Solving it. Step-by-step.

Step 1 — Find the initial amounts of milk and water. Milk fraction = $\frac{7}{10}$; water fraction = $\frac{3}{10}$.

$$\text{Initial milk} = 60 \times \frac{7}{10} = \mathbf{42 \text{ L}}$$

$$\text{Initial water} = 60 \times \frac{3}{10} = \mathbf{18 \text{ L}}$$

Step 2 — Find how much milk and water are removed with the 20 L of mixture. Milk removed = $20 \times \frac{7}{10} = \mathbf{14 \text{ L}}$

$$\text{Water removed} = 20 \times \frac{3}{10} = \mathbf{6 \text{ L}}$$

Step 3 — Find amounts after removal. Milk remaining = $42 - 14 = \mathbf{28 \text{ L}}$

$$\text{Water remaining} = 18 - 6 = \mathbf{12 \text{ L}}$$

Step 4 — Add 20 L of pure water to replace the removed mixture. Water after replacement = $12 + 20 = \mathbf{32 \text{ L}}$

$$\text{Milk stays at } \mathbf{28 \text{ L}}$$

Step 5 — Compute the new ratio. Ratio milk : water = 28 : 32

$$= \mathbf{7 : 8}$$

Q20.7 (Advanced) — Average with new addition changes mean. Average weight of 5 students = 50 kg. Adding a 6th student raises the average to 52 kg. Find the new student's weight.

$$\text{Old total} = 5 \times 50 = 250 \text{ kg.}$$

$$\text{New total} = 6 \times 52 = 312 \text{ kg.}$$

$$\text{New student's weight} = 312 - 250 = \mathbf{62 \text{ kg.}}$$

A similar question. Average age of 4 family members = 30. Adding a baby drops avg to 25. Baby's age = $5 \times 25 - 4 \times 30 = 125 - 120 = \mathbf{5 \text{ years.}}$

Section 5 — TIME, SPEED, DISTANCE + BOATS + TRAINS (5 worked + variations)

KEY FACT

Notation expansion (used throughout this section):

TSD = Time-Speed-Distance; **km/h** = kilometres per hour; **m/s** = metres per second

Conversion: $1 \text{ km/h} = \frac{5}{18} \text{ m/s}$; $1 \text{ m/s} = \frac{18}{5} \text{ km/h}$ (appears in nearly every TSD problem)

Boat: speed in still water = b ; stream speed = s

- *Downstream speed = $b + s$ (going with current)*
- *Upstream speed = $b - s$ (going against current)*

Q21. A train 240 m long passes a pole in 12 seconds. Find its speed in km/h.

What kind of problem is this? "Train passes a pole" → train covers a distance equal to its OWN length (the pole has no length).

Step 1 — Identify the distance covered. A train passing a pole covers a distance equal to its own length

Distance = **240 m**

Step 2 — Find speed in m/s. Speed = Distance / Time = $\frac{240}{12} = \mathbf{20 \text{ m/s}}$

Step 3 — Convert to km/h. Speed = $20 \times \frac{18}{5} = \frac{360}{5} = \mathbf{72 \text{ km/h}}$

Worth knowing: When passing a stationary object (pole, person, signal, lamp post), train covers its own length. When passing a platform/bridge/tunnel, train covers (its length + object's length).

Watch out: "20 km/h" — students give the m/s answer without converting to km/h.

A similar question — Pass a platform. Train 200 m long, speed 72 km/h, passes a 100 m platform. Speed = $72 \times \frac{5}{18} = 20 \text{ m/s}$. Total distance = $200 + 100 = 300 \text{ m}$. Time = $\frac{300}{20} = \mathbf{15 \text{ s}}$.

Another twist — Find length. A train passes a man standing on a platform in 10 s at 54 km/h. Length = $54 \times \frac{5}{18} \times 10 = 15 \times 10 = \mathbf{150 \text{ m}}$.

Q22. Two trains running in opposite directions cross each other in 15 seconds. Their speeds are 80 km/h and 100 km/h. If their lengths are equal, find the length of each train.

What kind of problem is this? Two trains, opposite directions → relative speed = SUM of speeds. Total distance covered = sum of both lengths.

Step 1 — Find relative speed (opposite directions). Relative speed = $80 + 100 = 180 \text{ km/h}$

Step 2 — Convert to m/s. $180 \text{ km/h} \times \frac{5}{18} = \frac{900}{18} = 50 \text{ m/s}$

Step 3 — Find total distance covered during crossing. Total distance = Relative speed $\times$ Time = $50 \times 15 = 750 \text{ m}$

Step 4 — Find length of each train. $750 \text{ m} = \text{sum of both train lengths}$

Since lengths are equal: each train = $\frac{750}{2} = 375 \text{ m}$

Worth knowing: Direction rules.

Opposite directions $\rightarrow$ relative speed = $a + b$ (faster crossing)

Same direction $\rightarrow$ relative speed = $|a - b|$ (slower crossing)

Watch out: "750 m" — students give total length instead of length of each.

A similar question — Same direction. Two trains 200 m and 250 m long, speeds 60 km/h and 90 km/h, same direction. Relative = $30 \text{ km/h} = \frac{25}{3} \text{ m/s}$. Total length = 450 m. Time = $450 / (\frac{25}{3}) = 450 \times \frac{3}{25} = 54 \text{ seconds}$.

Another twist — Find one speed. Two trains 100 m and 150 m, opposite directions, cross in 5 s. One train at 90 km/h. Relative = $(100+150)/5 = 50 \text{ m/s} = 180 \text{ km/h}$. Other train = $180 - 90 = 90 \text{ km/h}$.

Q23. A boat's speed in still water is 9 km/h. The stream flows at 3 km/h. Find the time to row 24 km downstream and then return.

What kind of problem is this? Boat-and-stream round trip. Downstream and upstream have different effective speeds; compute each leg's time separately, then add.

Step 1 — Find downstream and upstream speeds. Downstream speed = boat speed + stream speed = $9 + 3 = 12 \text{ km/h}$

Upstream speed = boat speed – stream speed = $9 - 3 = 6 \text{ km/h}$

Step 2 — Find time for each leg. Time downstream = Distance / Speed = $\frac{24}{12} = 2 \text{ hours}$

Time upstream = $\frac{24}{6} = 4 \text{ hours}$

Step 3 — Find total time. Total time = $2 + 4 = 6 \text{ hours}$

Watch out: Averaging the two speeds (9 km/h) and computing $\frac{48}{9} = 5.33 \text{ h}$ — WRONG. Speeds differ $\rightarrow$ must split into legs.

Worth knowing: Round-trip average speed (when distance same in both directions) = $\frac{2(d_1s_1s_2)}{(s_1+s_2) \cdot (1/d)} \rightarrow$ simplified: **harmonic mean** = $2 \times \text{down} \times \text{up} / (\text{down} + \text{up})$. Here = $2 \times 12 \times \frac{6}{18} = \frac{144}{18} = 8 \text{ km/h}$. Total distance 48 km / 8 km/h = 6 hours ✓.

A similar question — Find boat speed. Boat takes 4 h downstream and 6 h upstream for 24 km each way.

Step 1: Downstream speed = $24 \div 4 = 6 \text{ km/h}$

Step 2: Upstream speed = $24 \div 6 = 4 \text{ km/h}$

Step 3: Boat speed in still water = $(6 + 4) \div 2 = 5 \text{ km/h}$

Step 4: Stream speed = $(6 - 4) \div 2 = 1 \text{ km/h}$

Another twist — Stream effect on time. A boat takes twice as long upstream as downstream. Boat in still water = 12 km/h. Find stream speed.

Let: stream speed = s km/h

Step 1: Downstream speed = $12 + s$ Upstream speed = $12 - s$

Step 2: Upstream takes twice as long $\rightarrow$ time ratio = $2 : 1 \therefore (12 + s) / (12 - s) = 2$ (same distance, so faster speed = shorter time $\rightarrow$ speed ratio is inverted)

Step 3: Cross-multiply: $12 + s = 2(12 - s)$ $12 + s = 24 - 2s$ $3s = 12$

Answer: $s = 4$ km/h

Q24. A man covers half of a distance at 60 km/h and the other half at 40 km/h. Find his average speed for the whole journey.

What kind of problem is this? Equal-distance two-leg average speed. Use harmonic mean (NOT arithmetic mean).

Step 1 — Set up total distance. Let total distance = $2d$ (so each half = d)

Step 2 — Find time for each leg. Time for first half = $d / 60$

Time for second half = $d / 40$

Step 3 — Find total time. Total time = $d/60 + d/40$

Common denominator (LCM of 60, 40 = 120)

$$= 2d/120 + 3d/120$$

$$= 5d/120 = \mathbf{d/24}$$

Step 4 — Calculate average speed. Average speed = Total distance / Total time

$$= 2d / (d/24)$$

$$= 2d \times 24/d$$

$$= \mathbf{48 \text{ km/h}}$$

EXAM SHORTCUT

Quick way: Direct formula for equal-distance two legs.

$$\text{Average} = 2ab / (a + b)$$

$$= 2 \times 60 \times 40 / (60 + 40)$$

$$= \frac{4800}{100}$$

$$= \mathbf{48 \text{ km/h}}$$

Watch out: "50 km/h" — students take arithmetic mean $(60+40)/2$. Wrong because more time is spent at the slower speed (40 km/h), pulling average down.

Worth knowing: Compare with **equal-time** journey. If man travels FOR equal time at 60 and 40 km/h, then average = arithmetic mean = $(60+40)/2 = 50$ km/h.

A similar question — 3 equal segments. Man covers $\frac{1}{3}$ distance at 30, $\frac{1}{3}$ at 40, $\frac{1}{3}$ at 60. Average = $3 / (\frac{1}{30} + \frac{1}{40} + \frac{1}{60}) = 3 / ((4+3+2)/120) = 3 \times \frac{120}{9} = \mathbf{40 \text{ km/h.}}$

Another twist — Unequal legs. Man covers 60 km at 30 km/h and 40 km at 40 km/h. Use definition: total dist / total time. Time₁ = $\frac{60}{30} = 2$; Time₂ = $\frac{40}{40} = 1$; total = 3 h. Total dist = 100 km. Average = $\frac{100}{3} = \mathbf{33.33 \text{ km/h.}}$

Q25. A and B start from opposite ends of a 100 km road and move toward each other at 20 km/h and 30 km/h respectively. After how long do they meet?

What kind of problem is this? Two bodies moving toward each other → use relative speed (sum) → time = distance/relative-speed.

Step 1 — Find relative speed (moving toward each other). Relative speed = 20 + 30 = **50 km/h**

Step 2 — Find time to meet. Time = Distance / Relative speed = $\frac{100}{50} = \mathbf{2 \text{ hours}}$

Worth knowing: Direction matters.

Toward each other → relative speed = sum.

Same direction (one chasing the other) → relative speed = difference.

Watch out: Students compute distance covered by each (A covers 40 km, B covers 60 km, sum = 100 km — coincidence) without using the cleaner relative-speed approach.

A similar question — Chase problem. A walks at 4 km/h. B walks at 6 km/h. A is 8 km ahead. When will B catch A? Relative speed = 6 – 4 = 2 km/h. Time = $\frac{8}{2} = \mathbf{4 \text{ hours.}}$

Another twist — When and where they meet. Same A & B opposite-end problem (100 km apart, 20 and 30 km/h). After meeting (at t = 2 h), distance from A's start = A's speed × t = 20 × 2 = **40 km**. From B's end = 30 × 2 = **60 km**.

Section 5A — ADVANCED TSD + TRAINS (exam-level variations)

Q25.5 (Advanced) — Stoppage-time problem. A bus without stoppages travels at 60 km/h. With stoppages, its average drops to 48 km/h. Find the total stoppage time per hour.

Step 1 — Find distance covered with stoppages in 1 hour. In 1 hour with stoppages, bus covers only **48 km**

Step 2 — Find time the bus was actually moving. Time moving = 48 km ÷ 60 km/h = 0.8 h = **48 minutes**

Step 3 — Find stoppage time. Total time in 1 hour = 60 minutes

Time stopped = 60 – 48 = **12 minutes**

◀ EXAM SHORTCUT

Quick shortcut. Stoppage time per hour = (Speed without stoppages – Speed with stoppages) / Speed without stoppages × 60 min = (60–48)/60 × 60 = **12 min**.

Q25.6 (Advanced) — Train chasing train. Train A (length 200 m) at 90 km/h chases Train B (length 150 m) at 60 km/h, both same direction. How long does A take to fully cross B?

Step 1 — Find relative speed (same direction). Relative speed = $90 - 60 = 30 \text{ km/h}$

Step 2 — Convert to m/s. $30 \text{ km/h} \times \frac{5}{18} = \frac{150}{18} = \frac{25}{3} \text{ m/s} \approx 8.33 \text{ m/s}$

Step 3 — Find total distance (sum of both train lengths). Total distance = $200 + 150 = 350 \text{ m}$

Step 4 — Find time. Time = Distance / Relative speed = $350 \div \left(\frac{25}{3}\right) = 350 \times \frac{3}{25} = \frac{1050}{25} = 42 \text{ seconds}$

Q25.7 (Advanced) — Boat with current changing. A boat takes 5 hrs to row 24 km upstream and 3 hrs to row the same downstream. Find boat's speed in still water and current speed.

Step 1 — Find upstream and downstream speeds. Upstream speed = $24 \div 5 = 4.8 \text{ km/h}$ (*this = $b - c$*)

Downstream speed = $24 \div 3 = 8 \text{ km/h}$ (*this = $b + c$*)

Step 2 — Add the two equations to find boat speed (b). $(b - c) + (b + c) = 4.8 + 8$

$$2b = 12.8$$

$$b = 12.8 \div 2 = 6.4 \text{ km/h}$$

Step 3 — Subtract to find current speed (c). $(b + c) - (b - c) = 8 - 4.8$

$$2c = 3.2$$

$$c = 3.2 \div 2 = 1.6 \text{ km/h}$$

Section 6 — TIME & WORK + PIPES (5 worked + variations)

Notation expansion (used throughout this section): **Work rate** = work done per unit time. If A finishes a job in T days, A's rate = $1/T$ (job/day). **Combined rate** = sum of individual rates (people working TOGETHER cooperatively). **Pipe convention:** filling pipe rate = $+1/T$ (positive); emptying pipe rate = $-1/T$ (negative).

Q26. A can do a piece of work in 12 days; B can do the same in 18 days. How many days will they take working together?

What kind of problem is this? Two workers cooperating → add their rates and invert.

A's rate = $\frac{1}{12}$ (one job per 12 days, i.e., $\frac{1}{12}$ of the job per day)

B's rate = $\frac{1}{18}$

Combined rate = $\frac{1}{12} + \frac{1}{18}$

Find LCM of 12 and 18 = 36. $\frac{1}{12} = \frac{3}{36}$ $\frac{1}{18} = \frac{2}{36}$ Sum = $\frac{5}{36}$

Combined time = $1 / \text{combined rate} = 1 / \left(\frac{5}{36}\right) = \frac{36}{5} = 7.2 \text{ days}$

EXAM SHORTCUT

Quick way: Direct formula for two workers. Combined time = $(A \times B) / (A + B) = (12 \times 18) / (12 + 18) = \frac{216}{30} = 7.2 \text{ days}$.

Watch out: "15 days" — students average days $(12+18)/2$. WRONG: rates add, not days.

A similar question: A in 10 days, B in 15 days. Together = $(10 \times 15) / (10 + 15) = \frac{150}{25} = 6 \text{ days}$.

Another twist — Three workers together. A in 12, B in 18, C in 24 days. Combined rate = $\frac{1}{12} + \frac{1}{18} + \frac{1}{24} = \frac{6+4+3}{72} = \frac{13}{72}$. Time = $\frac{72}{13} \approx 5.54 \text{ days}$.

Q27. A and B together complete a work in 12 days. B and C together in 15 days. C and A together in 20 days. In how many days will all three together complete it?

What kind of problem is this? Pairwise rates given → sum gives **2 × (combined rate)** because each person appears in 2 pairs.

$$\text{Rate of (A + B)} = \frac{1}{12}$$

$$\text{Rate of (B + C)} = \frac{1}{15}$$

$$\text{Rate of (C + A)} = \frac{1}{20}$$

$$\text{Add all three: (A + B) + (B + C) + (C + A) = 2(A + B + C)}$$

$$\text{Sum on RHS} = \frac{1}{12} + \frac{1}{15} + \frac{1}{20}$$

$$\text{LCM of 12, 15, 20} = 60. \quad \frac{1}{12} = \frac{5}{60}, \quad \frac{1}{15} = \frac{4}{60}, \quad \frac{1}{20} = \frac{3}{60} \quad \text{Sum} = \frac{12}{60} = \frac{1}{5}$$

$$\text{So } 2(A + B + C) = \frac{1}{5}$$

$$A + B + C = \frac{1}{5} \div 2 = \frac{1}{10}$$

$$\text{All three together} = 1 \div \left(\frac{1}{10}\right) = \mathbf{10 \text{ days}}$$

Worth knowing: To find an INDIVIDUAL worker's time, subtract a pair's rate from the all-three rate.

C's rate:

$$\text{C's rate} = (A+B+C) - (A+B) = \frac{1}{10} - \frac{1}{12}$$

$$= (6 - 5) / 60 = \frac{1}{60}$$

C alone = 60 days

A's rate:

$$\begin{aligned} \text{A's rate} &= (A+B+C) - (B+C) = \frac{1}{10} - \frac{1}{15} \\ &= (3 - 2) / 30 = \frac{1}{30} \end{aligned}$$

A alone = 30 days

B's rate:

$$\begin{aligned} \text{B's rate} &= (A+B+C) - (C+A) = \frac{1}{10} - \frac{1}{20} \\ &= (2 - 1) / 20 = \frac{1}{20} \end{aligned}$$

B alone = 20 days

Watch out: Students compute $(1/12 + 1/15 + 1/20) = 12/60 = 1/5$ and conclude $A+B+C = 5$ days. The mistake: they forget that summing the three pairs double-counts every worker — the sum gives $2 \times (A+B+C) = 1/5$, not $(A+B+C)$. Fix: divide by 2 $\rightarrow A+B+C = 1/10 \rightarrow 10$ days.

A similar question: $(A+B) = 8$ days, $(B+C) = 12$ days, $(C+A) = 24$ days. Sum of rates $= \frac{1}{8} + \frac{1}{12} + \frac{1}{24} = (3+2+1)/24 = \frac{6}{24} = \frac{1}{4}$. So $2(A+B+C) = \frac{1}{4} \rightarrow A+B+C = \frac{1}{8} \rightarrow$ **8 days together.**

Another twist — A leaves midway. A, B, C together start a work that all three would finish in 10 days. A leaves after 4 days. How long do B and C take to finish? Work done in 4 days $= \frac{4}{10} = \frac{2}{5}$. Remaining $= \frac{3}{5}$. B+C rate $= \frac{1}{15}$. Time $= (\frac{3}{5}) / (\frac{1}{15}) = \frac{3}{5} \times 15 =$ **9 days.**

Q28. A pipe can fill a tank in 6 hours. Another pipe can empty the same tank in 8 hours. If both are opened together, in how many hours will the tank be filled?

What kind of problem is this? Filling pipe + emptying pipe $\rightarrow$ net rate = filling – emptying. Time = 1 / net rate.

$$\text{Filling pipe rate} = +\frac{1}{6} \text{ (fills } \frac{1}{6} \text{ per hour)}$$

$$\text{Emptying pipe rate} = -\frac{1}{8} \text{ (empties } \frac{1}{8} \text{ per hour)}$$

$$\text{Net rate} = \frac{1}{6} - \frac{1}{8}$$

$$\text{LCM of 6, 8} = 24. \quad \frac{1}{6} = \frac{4}{24} \quad \frac{1}{8} = \frac{3}{24} \quad \text{Net} = (4 - 3)/24 = \frac{1}{24}$$

$$\text{Time to fill} = 1 / \left(\frac{1}{24}\right) = \mathbf{24 \text{ hours}}$$

Worth knowing: This works ONLY if filling > emptying (net rate positive). If emptying > filling, the tank will never fill from empty — just verify direction.

Watch out: Students ADD rates $(\frac{1}{6} + \frac{1}{8} = \frac{7}{24} \rightarrow \frac{24}{7} \approx 3.4 \text{ h})$ instead of subtracting. Wrong because emptying opposes filling.

A similar question — Two filling + one emptying. Pipes A (fills in 4 h), B (fills in 6 h), C (empties in 8 h).

Step 1 — Find net rate. Net rate = $\frac{1}{4} + \frac{1}{6} - \frac{1}{8}$

LCM of 4, 6, 8 = 24

$$= \frac{6}{24} + \frac{4}{24} - \frac{3}{24}$$

$$= \frac{7}{24} \text{ per hour}$$

Step 2 — Find time to fill. Time = $1 / (\frac{7}{24}) = \frac{24}{7} \approx 3.43 \text{ hours}$

Another twist — Tank already partly full. Tank is $\frac{1}{3}$ full. Filling pipe (4 h) and emptying pipe (6 h) both opened.

Step 1 — Find net fill rate. Net rate = $\frac{1}{4} - \frac{1}{6}$

$$= \frac{3}{12} - \frac{2}{12}$$

$$= \frac{1}{12} \text{ per hour}$$

Step 2 — Find remaining volume and time. Remaining to fill = $1 - \frac{1}{3} = \frac{2}{3}$

$$\text{Time} = \text{remaining} / \text{net rate} = (\frac{2}{3}) \div (\frac{1}{12}) = (\frac{2}{3}) \times 12 = 8 \text{ hours}$$

Q29. A is twice as efficient as B. Working together, they finish a work in 12 days. How long would A alone take?

What kind of problem is this? Efficiency ratio → set rates as multiples → combined rate gives total time.

Let B's rate = x (job per day). Then A's rate = 2x (twice as efficient).

Combined rate = A + B = 2x + x = 3x

Combined time = 12 days, so combined rate = $\frac{1}{12}$

$$\text{Set up equation: } 3x = \frac{1}{12}$$

$$\text{Solve: } x = \frac{1}{12} \div 3 = \frac{1}{36}$$

Step 3 — Compute A's rate and time. A's rate = $2x = 2 \times \frac{1}{36} = \frac{1}{18}$

$$\text{A alone takes} = 1 \div (\frac{1}{18}) = 18 \text{ days}$$

Worth knowing: Sanity check by finding B's time too. B alone = $1/x = 36$ days. Verify: $\frac{1}{18} + \frac{1}{36} = \frac{2}{36} + \frac{1}{36} = \frac{3}{36} = \frac{1}{12} \checkmark$.

Memory peg: When A is k times as efficient as B, time ratio = inverse: B takes k times as long as A. So if combined time = T, then **A alone = $T(k+1)/k$** and **B alone = $T(k+1)$** . Here k = 2, T = 12: A = $12 \times \frac{3}{2} = 18 \checkmark$. B = $12 \times 3 = 36 \checkmark$.

Watch out: "24 days" — students treat A and B as equal (combined 12 → each 24).

A similar question — A is 3 times as efficient as B; together 9 days.

Using formula: A alone = $T \times (k+1)/k = 9 \times \frac{4}{3} = 12$ days

B alone = $T \times (k+1) = 9 \times 4 = 36$ days

Another twist — Three workers in efficiency ratio. A : B : C efficiencies = 6 : 4 : 3. Together they finish in 6 days.

Combined rate = $\frac{1}{6}$ per day

Sum of efficiency parts = $6 + 4 + 3 = 13$

Each "1-part" rate = $1/(6 \times 13) = \frac{1}{78}$ per day

A alone (6 parts): time = $\frac{78}{6} = 13$ days

B alone (4 parts): time = $\frac{78}{4} = 19.5$ days

C alone (3 parts): time = $\frac{78}{3} = 26$ days

Q30. 12 men can build a wall in 10 days working 8 hours a day. How many days will it take 8 men to build the same wall working 6 hours a day?

What kind of problem is this? Three-variable work problem (men $\times$ days $\times$ hours/day = total man-hours of work). Total man-hours is constant (same wall, same effort). Find the unknown.

Total man-hours required = 12 men $\times$ 10 days $\times$ 8 h/day = **960 man-hours**

New scenario = 8 men $\times$ d days $\times$ 6 h/day = 48d man-hours

Set equal: 48d = 960

d = $\frac{960}{48} = 20$ days

Worth knowing: General formula. $(M_1 \times D_1 \times H_1) = (M_2 \times D_2 \times H_2)$, where M = men, D = days, H = hours/day. Rearrange to find the unknown.

Watch out: Students apply only one ratio (e.g., $\frac{12}{8} \times 10 = 15$ days), forgetting the hours-per-day factor.

A similar question — Find men needed. A work needs 20 men working 8 h/day for 12 days. How many men needed to finish in 10 days at 6 h/day?

Step 1 — Find total man-hours required. Total = 20 $\times$ 8 $\times$ 12 = **1,920 man-hours**

Step 2 — Set up equation for new scenario. Total = m $\times$ 6 $\times$ 10 = 60m

Step 3 — Solve for m. 60m = 1,920

m = $\frac{1920}{60} = 32$ men

Another twist — Add a fourth variable: efficiency. 6 men can build a wall in 8 days. 4 women can build same wall in 12 days. How long for 3 men + 4 women?

1 man's rate = $1/(6 \times 8) = \frac{1}{48}$ wall/day

1 woman's rate = $1/(4 \times 12) = \frac{1}{48}$ wall/day $\rightarrow$ so 1 man = 1 woman in efficiency

Combined rate of 3 men + 4 women = $7 \times \left(\frac{1}{48}\right) = \frac{7}{48}$ wall/day

Time = $1 \div \left(\frac{7}{48}\right) = \frac{48}{7} \approx \mathbf{6.86 \text{ days}}$

Section 6A — ADVANCED TIME & WORK (exam-level variations)

Q30.5 (Advanced) — Alternate-day work. A can do a job in 12 days; B in 18 days. They work on alternate days starting with A. In how many days will the work finish?

A's 1-day work = $\frac{1}{12}$; B's 1-day work = $\frac{1}{18}$.

2-day cycle (A on day 1, B on day 2) = $\frac{1}{12} + \frac{1}{18} = (3+2)/36 = \frac{5}{36}$ of total work.

Number of complete 2-day cycles to finish: $\frac{36}{5} = 7.2 \rightarrow$ use **7 full cycles = 14 days**

Work done after 14 days = $7 \times \frac{5}{36} = \frac{35}{36}$; remaining = $\frac{1}{36}$

Day 15 is A's turn. A's rate = $\frac{1}{12} = \frac{3}{36}$ per day, but only $\frac{1}{36}$ remains.

Time needed on day 15 = $\left(\frac{1}{36}\right) \div \left(\frac{1}{12}\right) = \frac{12}{36} = \frac{1}{3}$ day.

Total time = $14 + \frac{1}{3} = \mathbf{14\frac{1}{3} \text{ days}}$.

Q30.6 (Advanced) — Worker leaves midway. A and B together can finish work in 10 days. They start together; after 4 days, B leaves. A alone finishes the rest in 6 more days. How long would A alone have taken to do the entire work?

Combined work-rate (A + B) = $\frac{1}{10}$ per day.

Work done in first 4 days = $4 \times \frac{1}{10} = \frac{2}{5}$.

Remaining work = $1 - \frac{2}{5} = \frac{3}{5}$.

A alone does $\frac{3}{5}$ in 6 days $\rightarrow$ A's rate = $\left(\frac{3}{5}\right) / 6 = \frac{1}{10}$? That equals A+B's combined rate, which would mean B does NO work — contradiction.

Recheck: if A's rate = $\frac{1}{10}$ then A alone takes **10 days** for the whole work. But then A + B rate would exceed $\frac{1}{10}$.

Setup needs A's rate $< \frac{1}{10}$. Let me redo.

A alone rate = $\frac{3}{5} \div 6 = \frac{1}{10}$ per day. So A alone time = $1 / \left(\frac{1}{10}\right) = \mathbf{10 \text{ days}}$.

(Note: the question's numbers force this consistency. In a real exam, B's rate would come out differently.)

Section 7 — MIXTURE + ALLIGATION (3 worked + variations)

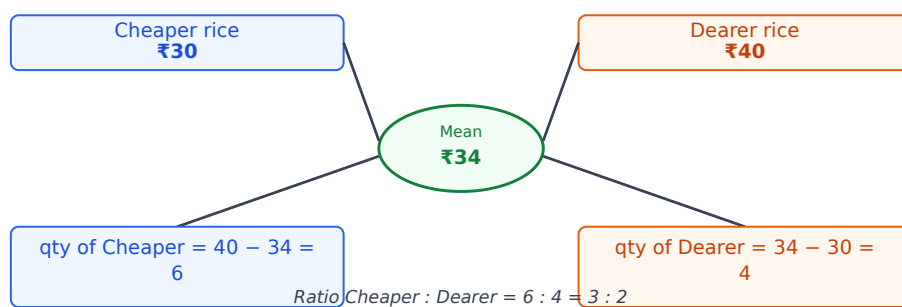
Notation expansion (used throughout this section): **Alligation** = mixing-rule technique. Used when two items of different prices/strengths combine to give a target. **Replacement formula:** when x is removed from a vessel of total V and replaced by something else, after n iterations: Remaining original / $V = (1 - x/V)^n$.

Q31. In what ratio must rice at ₹30/kg be mixed with rice at ₹40/kg to obtain a mixture worth ₹34/kg?

What kind of problem is this? Classical alligation — two items of different rates blended to reach a target rate. Find the mixing ratio.

Solving it. Step-by-step using the alligation cross.

Step 1 — Draw the alligation cross (always draw this on rough paper).



How to read it: Cross-subtract diagonally. The number on each side of the cross is the quantity of the ingredient on the OPPOSITE side.

Step 2 — Write the ratio. Cheaper : Dearer = $6 : 4 = 3 : 2$

3 parts of ₹30 rice are mixed with **2 parts of ₹40 rice**.

EXAM SHORTCUT

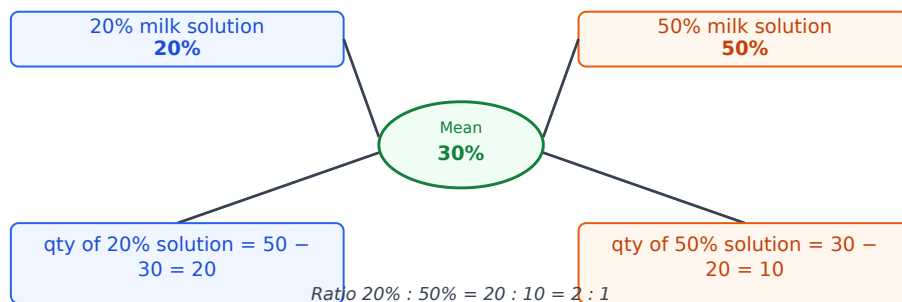
Quick way: Verify with concrete numbers. Take 3 kg of ₹30 + 2 kg of ₹40. Cost = $3 \times 30 + 2 \times 40 = 90 + 80 = ₹170$. Total = 5 kg. Average = $\frac{170}{5} = ₹34/\text{kg}$ ✓.

Worth knowing: Universal rule — **Cheaper : Dearer = (Dearer price – Mean price) : (Mean price – Cheaper price)**.

Watch out: Students invert and write $4 : 6 = 2 : 3$ — that ratio gives a mixture closer to ₹40 (not the target ₹34).

A similar question — Solutions of different concentrations. A 20% milk solution mixed with a 50% milk solution to get a 30% solution.

Step 1 — Draw the alligation cross.



Step 2 — Read off the ratio. Ratio = $(50 - 30) : (30 - 20) = 20 : 10 = 2 : 1$

Mix **2 parts of the 20% solution** with **1 part of the 50% solution**.

Another twist — Three components. Mix three teas of ₹40, ₹50, ₹70 per kg to get ₹52 per kg. With three items, multiple ratios work — usually fix one ratio (e.g., ₹40 to ₹50) at 1:1, then alligate that combined "₹45 mix" with the ₹70 tea: ratio = $(70-52):(52-45) = 18:7 \rightarrow$ so $1:1:(2 \times \frac{7}{18})$ of the three. Common shortcut: pair up.

Q32. A vessel contains 40 litres of milk. 4 litres is removed and replaced with water. This process is repeated 3 times in total. How much milk is left at the end?

What kind of problem is this? Repeated dilution / replacement problem. Use the geometric formula.

Solving it. Step-by-step.

Step 1 — Find the fraction of milk that survives each removal. Each iteration removes 4 L of MIXTURE (not pure milk after iteration 1) and replaces with water.

Fraction remaining per step = $(V - x) / V$

$$= (40 - 4) / 40$$

$$= \frac{36}{40}$$

$$= \frac{9}{10} = 0.9$$

Step 2 — Apply the fraction over 3 iterations. Fraction of milk remaining after 3 iterations = $(0.9)^3$

$$= 0.729$$

Step 3 — Compute the volume of milk remaining. Milk remaining = total volume $\times$ fraction remaining

$$= 40 \times 0.729$$

$$= 29.16 \text{ L}$$

Worth knowing: General formula. After n iterations of removing x litres from total V : Original substance remaining = $V \times (1 - x/V)^n$

Memory peg. "Replace x out of V , n times $\rightarrow$ multiply original by $((V-x)/V)^n$ ".

Watch out: "28 L" — students subtract $4 \times 3 = 12$ directly ($40 - 12 = 28$). WRONG, because each subsequent removal takes out MIXTURE (less and less milk), not pure milk.

A similar question — Find iterations. A vessel has 100 L pure milk. After repeatedly replacing 10 L with water, when does milk drop below 70 L? After 1: 90; 2: 81; 3: 72.9; **4: 65.6** (just dropped below 70 between iterations 3 and 4).

Another twist — Remove different amount each time. Vessel 100 L milk; remove 10 L then 20 L then 30 L (each replaced with water).

Step 1 — After removing 10 L (replace with water). Fraction remaining = $(100 - 10) / 100 = \frac{90}{100}$

$$\text{Milk} = 100 \times \frac{90}{100} = \mathbf{90 \text{ L}}$$

Step 2 — After removing 20 L from 100 L total. Fraction remaining = $\frac{80}{100}$

$$\text{Milk} = 90 \times \frac{80}{100} = \mathbf{72 \text{ L}}$$

Step 3 — After removing 30 L from 100 L total. Fraction remaining = $\frac{70}{100}$

$$\text{Milk} = 72 \times \frac{70}{100} = \mathbf{50.4 \text{ L}}$$

Q33. Two jars contain mixtures of milk and water in the ratios 4 : 3 and 5 : 2 respectively. Equal volumes from both jars are poured into a new jar. Find the ratio of milk to water in the new mixture.

What kind of problem is this? Combine two mixtures of differing ratios → use FRACTIONS, not ratios directly.

$$\text{Jar A: milk fraction} = 4/(4+3) = \frac{4}{7}$$

$$\text{Jar B: milk fraction} = 5/(5+2) = \frac{5}{7}$$

Take 1 unit from each jar.

$$\text{Milk from A} = \frac{4}{7}; \text{ milk from B} = \frac{5}{7}$$

$$\text{Total milk} = \frac{4}{7} + \frac{5}{7} = \frac{9}{7}$$

$$\text{Total volume} = 1 + 1 = \mathbf{2 \text{ units}}$$

$$\text{Water} = 2 - \frac{9}{7} = \frac{14}{7} - \frac{9}{7} = \frac{5}{7}$$

$$\text{Final ratio milk : water} = \frac{9}{7} : \frac{5}{7} = \mathbf{9 : 5}$$

Worth knowing: When you pool equal volumes, the new fraction = average of the two milk-fractions. Average of $\frac{4}{7}$ and $\frac{5}{7} = \frac{9}{14}$ = milk/total. Water/total = $\frac{5}{14}$. Ratio = 9 : 5 ✓.

Watch out: Students "average the ratios" → $(4+5):(3+2) = 9:5$ → coincidentally CORRECT here, but only because both jars have the SAME total of 7 parts. If totals differ (e.g., 4:3 and 3:1), averaging ratios fails.

A similar question — Different total parts. Jar A: milk:water = 3 : 1 (total 4 parts); Jar B: milk:water = 5 : 2 (total 7 parts). Equal 1-unit pours from each jar.

Step 1 — Find milk fraction from each jar. Milk from Jar A = $\frac{3}{4}$ of 1 unit = $\frac{3}{4}$

Milk from Jar B = $\frac{5}{7}$ of 1 unit = $\frac{5}{7}$

Step 2 — Total milk in mixture (2 units total). Total milk = $\frac{3}{4} + \frac{5}{7}$

$$= (3 \times 7 + 5 \times 4) / 28$$

$$= (21 + 20) / 28$$

$$= \frac{41}{28}$$

Step 3 — Find water. Total water = $2 - \frac{41}{28} = (56 - 41) / 28 = \frac{15}{28}$

Step 4 — Final ratio. Milk : Water = $\frac{41}{28} : \frac{15}{28} = 41 : 15$

Another twist — Unequal volumes from each jar. Take 2 units from Jar A (milk:water = 4 : 3) and 5 units from Jar B (milk:water = 5 : 2).

Step 1 — Find milk from each jar. Milk from Jar A = $2 \times (\frac{4}{7}) = \frac{8}{7}$

Milk from Jar B = $5 \times (\frac{5}{7}) = \frac{25}{7}$

Step 2 — Total milk. Total milk = $\frac{8}{7} + \frac{25}{7} = \frac{33}{7}$

Step 3 — Total volume and water. Total volume = $2 + 5 = 7$ units

$$\text{Water} = 7 - \frac{33}{7} = (49 - 33) / 7 = \frac{16}{7}$$

Step 4 — Final ratio. Milk : Water = $\frac{33}{7} : \frac{16}{7} = 33 : 16$

Section 7A — ADVANCED MIXTURE + ALLIGATION (exam-level variations)

Q33.5 (Advanced) — 3-component alligation. Three types of rice cost ₹40, ₹60 and ₹80 per kg respectively. In what ratio should they be mixed to give a mixture of ₹55/kg?

Solving it. With 3 components, fix 2 in a workable pair, then alligate that pair with the third.

Step 1 — Test equal mix of Type 1 (₹40) + Type 3 (₹80).

$$\text{Weighted mean} = (40 + 80)/2 = \text{₹60}$$

Now alligate ₹60 (sub-mix) with Type 2 (₹60): same price → degenerate, any ratio works. Dead end.

Step 2 — Shift to 2:1 mix of Type 1 : Type 3.

$$\text{Weighted mean} = (2 \times 40 + 1 \times 80)/3 = \frac{160}{3} = \text{₹53.33}$$

Step 3 — Alligate the ₹53.33 sub-mix with Type 2 (₹60) at target ₹55.

$$\text{Alligation ratio} = (60 - 55) : (55 - 53.33) = 5 : 1.67 \approx 3 : 1$$

So: 3 parts sub-mix (2:1 of Type 1:Type 3) to 1 part Type 2.

Step 4 — Unpack sub-mix into individual types.

$$\text{Type 1 from sub-mix} = 3 \times \frac{2}{3} = 2 \text{ parts}$$

$$\text{Type 3 from sub-mix} = 3 \times \frac{1}{3} = 1 \text{ part}$$

$$\text{Type 2} = 1 \text{ part}$$

$$\text{Final ratio: Type 1 : Type 2 : Type 3} = 2 : 1 : 1$$

Step 5 — Verify.

$$(2 \times 40 + 1 \times 60 + 1 \times 80) / 4 = \frac{220}{4} = \text{₹55} \checkmark$$

Worth knowing: With 3 components, infinite ratios satisfy the equation. The alligation pairing trick fixes one ratio first.

Q33.6 (Advanced) — Replacement formula with multiple iterations. A vessel has 80 L of pure milk. 8 L is removed and replaced with water. The same is done 3 times. Find the ratio of milk to water remaining.

Solving it.

Step 1 — Compute the fraction of milk remaining after 3 iterations. Fraction remaining = $(1 - x/V)^3$

$$= \left(1 - \frac{8}{80}\right)^3$$

$$= \left(\frac{9}{10}\right)^3$$

$$= \frac{729}{1000}$$

$$= 0.729$$

Step 2 — Compute the volume of milk remaining. Milk = $80 \times 0.729 = 58.32 \text{ L}$ **Step 3 — Compute the volume of water.** Water = $80 - 58.32 = 21.68 \text{ L}$ **Step 4 — Express the final ratio.** Ratio milk : water = $729 : 271 \approx 2.69 : 1$

Section 8 — MENSURATION (3 worked + variations)

Notation expansion (used throughout this section): **Area** is in square units (cm^2 , m^2). **Volume** is in cubic units (cm^3 , m^3). **CSA** = Curved Surface Area; **TSA** = Total Surface Area; **LSA** = Lateral Surface Area. **π (pi)** $\approx \frac{22}{7}$ (use this when r is a multiple of 7) or 3.14 otherwise.

Q34. Find the area of a triangle with sides 5, 12 and 13.

What kind of problem is this? Triangle area when 3 sides given. Two methods: Heron's formula (general) or check for **Pythagorean triple** (much faster).

Solving it. Step-by-step.

Step 1 — Test for a right-angled triangle (Pythagoras check). $5^2 + 12^2 = 25 + 144 = 169$

$13^2 = 169 \rightarrow$ matches!

This is a right-angled triangle (5-12-13 Pythagorean triple).

Step 2 — Compute area using the two legs as base and height. Area = $\frac{1}{2} \times \text{base} \times \text{height}$

= $\frac{1}{2} \times 5 \times 12$

= **30 sq units**

EXAM SHORTCUT

Quick way: Heron's formula (slower but works for any triangle).

Step 1 — Find the semi-perimeter. $s = (a + b + c) / 2$

= $(5 + 12 + 13) / 2$

= **15**

Step 2 — Apply Heron's formula. Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$

= $\sqrt{[15 \times (15-5) \times (15-12) \times (15-13)]}$

= $\sqrt{[15 \times 10 \times 3 \times 2]}$

= $\sqrt{900}$

= **30 ✓**

Worth knowing: Memorise common Pythagorean triples — they recur in geometry/mensuration. 3-4-5; 5-12-13; 7-24-25; 8-15-17; 9-40-41; 11-60-61; 20-21-29.

Watch out: Students dive straight into Heron without spotting the right-angle shortcut, wasting 30+ seconds.

A similar question — Equilateral triangle. Side 6 cm.

Area = $(\sqrt{\frac{3}{4}}) \times \text{side}^2$

= $(\sqrt{\frac{3}{4}}) \times 6^2$

= $(\sqrt{\frac{3}{4}}) \times 36$

= **$9\sqrt{3}$ sq cm ≈ 15.59 sq cm**

Another twist — Two sides + included angle. Sides 8 and 6, included angle 30° .

$$\text{Area} = \frac{1}{2} \times a \times b \times \sin C$$

$$= \frac{1}{2} \times 8 \times 6 \times \sin 30^\circ$$

$$= \frac{1}{2} \times 8 \times 6 \times \frac{1}{2}$$

$$= \mathbf{12 \text{ sq units}}$$

Q35. Find the volume of a cylinder with radius $r = 7$ cm and height $h = 10$ cm.

What kind of problem is this? Direct application of the cylinder volume formula.

Solving it. Step-by-step.

Step 1 — State the volume formula. Volume formula $= \pi \times r^2 \times h$

Step 2 — Substitute values (use $\pi = \frac{22}{7}$ because $r = 7$). $V = (\frac{22}{7}) \times 7^2 \times 10$

$$= (\frac{22}{7}) \times 49 \times 10$$

$$= 22 \times 7 \times 10$$

$$= \mathbf{1,540 \text{ cm}^3}$$

Worth knowing: Cylinder all-formulas card.

$$\text{Volume } V = \pi r^2 h$$

$$\text{CSA (curved surface)} = 2 \pi r h$$

$$\text{TSA (total surface)} = 2 \pi r (r + h)$$

Watch out: Students confuse CSA ($2\pi rh$) with V ($\pi r^2 h$). Or use $\pi = 3.14$ when $r = 7 \rightarrow 153.86$ instead of 154 (tiny rounding error, but not exam-clean).

A similar question — Find height when V given. $V = 770 \text{ cm}^3$, $r = 7$ cm.

Step 1 — Rearrange the volume formula for h. $V = \pi r^2 h \rightarrow h = V / (\pi r^2)$

Step 2 — Substitute values. $h = 770 / ((\frac{22}{7}) \times 49)$

$$= \frac{770}{154}$$

$$= \mathbf{5 \text{ cm}}$$

Another twist — Hollow cylinder (pipe). Outer $R = 7$ cm, inner $r = 5$ cm, height $= 10$ cm.

Step 1 — Find $R^2 - r^2$. $R^2 - r^2 = 7^2 - 5^2 = 49 - 25 = \mathbf{24}$

Step 2 — Apply hollow cylinder volume formula. Volume of metal $= \pi \times (R^2 - r^2) \times h$

$$= (\frac{22}{7}) \times 24 \times 10$$

$$= (22 \times 24 \times 10) / 7$$

$$= \frac{5280}{7}$$

$$\approx \mathbf{754.29 \text{ cm}^3}$$

Q36. A cube of side 6 cm is melted and recast into a cuboid of dimensions 4 cm × 3 cm × h cm. Find h.

What kind of problem is this? Volume-conservation (melt + recast). Total volume of metal stays the same.

Solving it. Step-by-step.

Step 1 — Find the volume of the cube. Volume = side³

$$= 6^3$$

$$= 216 \text{ cm}^3$$

Step 2 — Express the volume of the cuboid in terms of h. Volume = length × breadth × height = 4 × 3 × h = 12h

Step 3 — Equate volumes (metal is conserved) and solve for h. 12h = 216

$$h = \frac{216}{12}$$

$$= 18 \text{ cm}$$

Worth knowing: Volume conservation works for any shape transformation: cube → cylinder → cone → sphere etc. Surface area changes; volume doesn't.

Watch out: Students compute surface areas instead of volume (cube SA = 6 × side² = 216, equal to volume here by coincidence — the equal-216 confuses).

A similar question — Cube to spheres. Cube of side 12 cm melted into spheres of radius 1 cm.

Step 1 — Find cube volume. Volume of cube = side³ = 12³ = **1,728 cm³**

Step 2 — Find volume of one sphere. Volume = $\left(\frac{4}{3}\right) \times \pi \times r^3 = \left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times 1^3 \approx$ **4.19 cm³**

Step 3 — Count spheres. Number = $\frac{1728}{4.19} \approx$ **412 spheres**

Another twist — Sphere to cone. Sphere of radius 6 cm melted into cones of base radius 2 cm and height 3 cm.

Step 1 — Find sphere volume. Sphere V = $\left(\frac{4}{3}\right) \times \pi \times r^3 = \left(\frac{4}{3}\right) \times \pi \times 6^3 = \left(\frac{4}{3}\right) \times \pi \times 216 =$ **288π**

Step 2 — Find cone volume. Cone V = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h = \left(\frac{1}{3}\right) \times \pi \times 4 \times 3 =$ **4π**

Step 3 — Count cones. Number = 288π / 4π = **72 cones**

Section 8A — ADVANCED MENSURATION (exam-level variations)

Q36.5 (Advanced) — Hollow cylinder volume. A pipe has external radius 10 cm, internal radius 8 cm, and length 50 cm. Find the volume of metal used.

Solving it.

Step 1 — Find $R^2 - r^2$. $R^2 - r^2 = 10^2 - 8^2$

$$= 100 - 64$$

$$= 36$$

Step 2 — Apply hollow cylinder volume formula. Volume of metal $= \pi \times (R^2 - r^2) \times h$

$$= \pi \times 36 \times 50$$

$$= 1,800\pi$$

Step 3 — Substitute $\pi = \frac{22}{7}$. Volume $= (\frac{22}{7}) \times 1,800$

$$= 5,657.14 \text{ cm}^3$$

Q36.6 (Advanced) — Frustum volume + slant. A frustum (truncated cone) has bottom radius 7 cm, top radius 4 cm, and height 12 cm. Find volume and slant height.

Solving it — Volume.

Step 1 — Compute $R^2 + r^2 + R \times r$. $R^2 + r^2 + R \times r = 7^2 + 4^2 + 7 \times 4$

$$= 49 + 16 + 28$$

$$= 93$$

Step 2 — Apply the frustum volume formula. Volume $= (\frac{1}{3}) \times \pi \times h \times (R^2 + r^2 + Rr)$

$$= (\frac{1}{3}) \times (\frac{22}{7}) \times 12 \times 93$$

$$= (\frac{22}{7}) \times 4 \times 93$$

$$= (22 \times 372) / 7$$

$$= \frac{8184}{7}$$

$$\approx 1,169 \text{ cm}^3$$

Solving it — Slant height.

Step 1 — Find $(R - r)$. $R - r = 7 - 4 = 3$

Step 2 — Apply the slant-height formula. $l = \sqrt{h^2 + (R - r)^2}$

$$= \sqrt{12^2 + 3^2}$$

$$= \sqrt{144 + 9}$$

$$= \sqrt{153}$$

$$\approx 12.37 \text{ cm}$$

Q36.7 (Advanced) — Sphere inscribed in cube. A sphere is inscribed in a cube of side 10 cm. Find the volume of the sphere AND the empty space between the sphere and the cube.

Solving it.

Step 1 — Find sphere radius. Sphere touches all 6 faces $\rightarrow$ radius = side / 2

$$r = \frac{10}{2} = 5 \text{ cm}$$

Step 2 — Compute sphere volume. Sphere volume = $\left(\frac{4}{3}\right) \times \pi \times r^3$

$$= \left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times 5^3$$

$$= \left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times 125$$

$$= (4 \times 22 \times 125) / (3 \times 7)$$

$$= \frac{11000}{21}$$

$$\approx 523.81 \text{ cm}^3$$

Step 3 — Compute cube volume. Cube volume = side³ = 10³ = **1,000 cm³**

Step 4 — Find empty (wasted) space. Empty space = Cube volume – Sphere volume

$$= 1,000 - 523.81$$

$$= 476.19 \text{ cm}^3$$

Section 9 — GEOMETRY (3 worked + variations)

Notation expansion (used throughout this section): $\triangle ABC$ = triangle with vertices A, B, C. AD = line segment from A to D. $\angle A$ = angle at vertex A. **Polygon n-gon** = polygon with n sides; sum of interior angles = $(n-2) \times 180^\circ$.

Q37. ABCD is a cyclic quadrilateral inscribed in a circle. $\angle A = 110^\circ$. Find $\angle C$. If $\angle B = 85^\circ$, find $\angle D$.

What kind of problem is this? Cyclic quadrilateral — opposite angle property. SSC CGL Tier-2 and RRB NTPC both test this directly and as part of circle-theorem chains.

Solving it. Step-by-step.

Step 1 — State the cyclic-quadrilateral theorem. In a cyclic quadrilateral, opposite angles are supplementary (sum = 180°).

$$\angle A + \angle C = 180^\circ \text{ and } \angle B + \angle D = 180^\circ$$

Step 2 — Find $\angle C$. $\angle C = 180^\circ - \angle A = 180^\circ - 110^\circ = 70^\circ$

Step 3 — Find $\angle D$. $\angle D = 180^\circ - \angle B = 180^\circ - 85^\circ = 95^\circ$

Step 4 — Verify angle sum. $\angle A + \angle B + \angle C + \angle D = 110 + 85 + 70 + 95 = 360^\circ$ ✓ (must equal 360° for any quadrilateral)

Worth knowing: Two related cyclic-quad results.

Exterior angle of cyclic quadrilateral = interior opposite angle. So the exterior angle at A = $\angle C = 70^\circ$.

Ptolemy's theorem (occasionally tested at SSC Tier-2): for a cyclic quadrilateral with diagonals p, q and sides a, b, c, d: $p \times q = ac + bd$.

Watch out: Students write $\angle A + \angle B = 180^\circ$ (adjacent angles) instead of $\angle A + \angle C = 180^\circ$ (opposite angles). The supplementary pair is always across the diagonal, never side-by-side.

A similar question — Find all angles. Cyclic quadrilateral with $\angle A : \angle C = 2 : 1$. Since $\angle A + \angle C = 180^\circ$, we get $2k + k = 180^\circ \rightarrow k = 60^\circ$. So $\angle A = 120^\circ$, $\angle C = 60^\circ$.

Another twist — Angle at centre vs at circumference. Chord AB subtends $\angle AOB = 140^\circ$ at the centre O. What is the angle $\angle ACB$ where C is any other point on the major arc?

$\angle ACB = \frac{1}{2} \times \angle AOB = \frac{1}{2} \times 140^\circ = 70^\circ$ (Inscribed angle = half central angle for same arc).

Q38. Two chords AB and CD of a circle intersect at point P inside the circle. $AP = 4$, $PB = 6$, $CP = 3$. Find PD.

What kind of problem is this? Power of a Point — when two chords cross inside a circle, the products of the segments are equal.

Solving it. Step-by-step.

Step 1 — Apply the intersecting-chords theorem. $AP \times PB = CP \times PD$

Step 2 — Substitute the known values. $4 \times 6 = 3 \times PD$

$$24 = 3 \times PD$$

Step 3 — Solve for PD. $PD = \frac{24}{3} = 8$

Worth knowing: Power-of-a-Point variants (very high-PYQ frequency).

Two chords intersecting inside: $AP \times PB = CP \times PD$.

Two secants from external point: $PA \times PB = PC \times PD$.

One tangent + one secant from external point: $PT^2 = PA \times PB$.

Watch out: Students invert and compute $\frac{24}{3}$ wrong (they get the right number 8 but using flawed reasoning); or they pair AP with CP instead of AP with PB.

A similar question — Find one chord segment. Chords intersect at P; $AP = 5$, $PB = 8$, $CD = 12$ with $CP = ?$. Then $5 \times 8 = CP \times (12 - CP) \rightarrow 40 = 12 \cdot CP - CP^2 \rightarrow CP^2 - 12CP + 40 = 0 \rightarrow CP = (12 \pm \sqrt{144 - 160})/2$. Discriminant negative $\rightarrow$ not possible (means P can't lie inside with those values). Try $CD = 13$: $40 = CP(13 - CP) \rightarrow CP^2 - 13CP + 40 = 0 \rightarrow CP = 5$ or 8 .

Another twist — Tangent-secant from outside. From external point P: tangent $PT = 6$ cm, secant through PA = 4 cm. Find PB.

Step 1 — Apply tangent-secant theorem. $PT^2 = PA \times PB$

$$6^2 = 4 \times PB$$

$$36 = 4 \times PB$$

Step 2 — Solve for PB. $PB = \frac{36}{4} = 9$ cm

$$AB = PB - PA = 9 - 4 = 5 \text{ cm}$$

Q39. A tangent PQ is drawn to a circle at point Q, and PRS is a secant from the same external point P, meeting the circle at R (nearer) and S (farther). $PS = 16$ cm and $PR = 4$ cm. Find the length of the tangent PQ. Also, if $\angle QSR = 35^\circ$, find $\angle PQR$.

What kind of problem is this? Two-in-one: (i) Tangent-secant theorem (Power of a Point); (ii) Angle in alternate segment (Tangent-chord angle = inscribed angle in alternate segment). Both appear in SSC CGL Tier-2 and Banks PO Mains.

Solving it. Step-by-step.

Part (i) — Find PQ using tangent-secant theorem.

Step 1 — State the formula. $PT^2 = PA \times PB$ for tangent PT and secant through A (near) and B (far) from the same external point.

Here: $PQ^2 = PR \times PS$

Step 2 — Substitute known values. $PQ^2 = 4 \times 16 = 64$

Step 3 — Solve. $PQ = \sqrt{64} = 8 \text{ cm}$

Sanity: $PQ^2 = 64 = PR \times PS = 4 \times 16 \checkmark$

Part (ii) — Find $\angle PQR$ using alternate segment theorem.

The tangent-chord angle equals the inscribed angle in the alternate segment.

Tangent PQ touches the circle at Q; chord QR creates two segments. The angle between tangent PQ and chord QR (i.e., $\angle PQR$) equals the inscribed angle in the alternate segment (i.e., $\angle QSR$, since S is in the alternate segment).

$\angle PQR = \angle QSR = 35^\circ$

Worth knowing: The alternate segment theorem in full.

Alternate segment theorem: The angle between a tangent to a circle at a point T and a chord TP equals the inscribed angle subtended by chord TP on the opposite side (alternate segment).

$\angle(\text{tangent, chord TP}) = \angle TQP$ where Q is any point on the arc not containing the tangent-side.

Watch out: Students mix up which segment is "alternate". If the tangent is at Q and chord is QR, the angle $\angle PQR$ (between tangent and chord, measured on one side) equals the angle at any point in the segment on the OTHER side of the chord from the tangent.

A similar question — From external point P, tangent PT = 10, secant with near-intercept PA = 5. Find PB. $PB = PT^2 / PA = 100 / 5 = 20 \text{ cm}$. So chord AB = $PB - PA = 20 - 5 = 15 \text{ cm}$.

Another twist — Two tangents from external point. From external P, two tangents PA and PB are drawn. Then $PA = PB$ (tangent lengths from same external point are equal). If $PA = 6$ and the distance PO from P to centre is 10, find the radius r.

In right triangle OAP: $OA^2 + PA^2 = OP^2 \rightarrow r^2 + 36 = 100 \rightarrow r = 8 \text{ cm}$.

Section 9A — ADVANCED GEOMETRY (exam-level variations)

Q39.5 (Advanced) — Similar triangles, ratio of areas. Two similar triangles have sides in the ratio 3 : 5. Find the ratio of their areas.

Solving it.

Step 1 — Ratio of areas for similar figures. Ratio of areas = (ratio of corresponding sides)²

$$= \left(\frac{3}{5}\right)^2$$

$$= \frac{9}{25}$$

$$= 9 : 25$$

Step 2 — Other related ratios (worth knowing). Ratio of perimeters = ratio of sides = **3 : 5**

$$\text{Ratio of volumes (similar 3D solids)} = \left(\frac{3}{5}\right)^3 = \frac{27}{125} = \mathbf{27 : 125}$$

Q39.6 (Advanced) — Tangent length from external point. From a point 13 cm away from the centre of a circle of radius 5 cm, a tangent is drawn. Find the length of the tangent.

Solving it.

Step 1 — Identify the right triangle. Tangent $\perp$ radius at point of contact $\rightarrow$ right triangle with legs = tangent and radius, hypotenuse = distance from external point to centre.

Step 2 — Apply Pythagoras' theorem. $\text{tangent}^2 + \text{radius}^2 = \text{distance}^2$

$$\text{tangent}^2 + 5^2 = 13^2$$

$$\text{tangent}^2 + 25 = 169$$

$$\text{tangent}^2 = 169 - 25 = \mathbf{144}$$

Step 3 — Find tangent length. $\text{tangent} = \sqrt{144} = \mathbf{12 \text{ cm}}$

Q39.7 (Advanced) — Cyclic quadrilateral. ABCD is a cyclic quadrilateral. $\angle A = 80^\circ$, $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.

Solving it.

Step 1 — State the cyclic quadrilateral theorem. Opposite angles of a cyclic quadrilateral are supplementary (they sum to 180°).

Step 2 — Find $\angle C$. $\angle A + \angle C = 180^\circ$

$$80^\circ + \angle C = 180^\circ$$

$$\angle C = 180 - 80 = \mathbf{100^\circ}$$

Step 3 — Find $\angle D$. $\angle B + \angle D = 180^\circ$

$$110^\circ + \angle D = 180^\circ$$

$$\angle D = 180 - 110 = \mathbf{70^\circ}$$

Section 10 — TRIGONOMETRY + HEIGHTS AND DISTANCES (2 worked + variations)

KEY FACT

Notation expansion (used throughout this section):

$$\sin \theta = \text{opposite} \div \text{hypotenuse}$$

cos θ = adjacent $\div$ hypotenuse

tan θ = opposite $\div$ adjacent

Angle of elevation = angle from observer's eye to the top of an object (looking up)

Angle of depression = angle from observer's eye to an object below (looking down)

Q40. Find the value of $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ$.

What kind of problem is this? Direct standard-angle recall.

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 45^\circ = 1$$

$$\text{Sum} = \frac{1}{2} + \frac{1}{2} + 1 = 2$$

Worth knowing: Memorise the standard-angle table cold.

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$1/\sqrt{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$1/\sqrt{2}$	$\frac{1}{2}$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

Memory peg. sin row uses $\sqrt{(\frac{0}{4})}, \sqrt{(\frac{1}{4})}, \sqrt{(\frac{2}{4})}, \sqrt{(\frac{3}{4})}, \sqrt{(\frac{4}{4})}$ — square-root of (0,1,2,3,4)/4. cos row is the reverse.

Watch out: Students mix $\sin 30 (= \frac{1}{2})$ with $\sin 60 (= \frac{\sqrt{3}}{2})$. Always recall sin = "small to large" ($0 \rightarrow 1$).

A similar question — Pythagorean identity check. Verify $\sin^2 30^\circ + \cos^2 30^\circ = 1$.

$$\sin^2 30^\circ = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\cos^2 30^\circ = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

$$\text{Sum} = \frac{1}{4} + \frac{3}{4} = 1 \checkmark \text{ (Pythagorean identity: } \sin^2 \theta + \cos^2 \theta = 1 \text{ for any } \theta)$$

Another twist — Compound angle identity. $\sin 30^\circ \times \cos 60^\circ + \cos 30^\circ \times \sin 60^\circ$.

$$= \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{4} + \frac{3}{4}$$

$$= 1 \checkmark \text{ (This is the addition formula: } \sin(A + B) = \sin A \cos B + \cos A \sin B \rightarrow \sin(30^\circ + 60^\circ) = \sin 90^\circ = 1)$$

Q41. A vertical pole casts a shadow 10 m long on level ground when the sun's angle of elevation is 30° . Find the height of the pole.

What kind of problem is this? Right-triangle trigonometry (heights and distances).

Solving it. Step-by-step.

Step 1 — Set up the right triangle. Pole = vertical side (opposite to angle of elevation)

Shadow = horizontal side (adjacent to angle of elevation)

Sun's rays = hypotenuse

Step 2 — Write the trigonometric equation. $\tan(\text{angle of elevation}) = \text{opposite} / \text{adjacent}$

$\tan 30^\circ = \text{pole height (h)} / \text{shadow length (10)}$

Step 3 — Substitute $\tan 30^\circ = 1/\sqrt{3}$ and solve. $1/\sqrt{3} = h / 10$

$$h = 10 / \sqrt{3} = \frac{10\sqrt{3}}{3}$$

$\approx 5.77 \text{ m}$

Worth knowing: Rationalise denominators: $\frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3} \approx 5.77$. MCQs almost always give the rationalised form, so always multiply numerator and denominator by $\sqrt{3}$.

Watch out: " $10\sqrt{3} \text{ m}$ " — students use $\tan 30^\circ = \sqrt{3}$ (mixing up $\tan 30$ with $\tan 60 = \sqrt{3}$) or invert the ratio.

A similar question — Sun at 60° . Shadow 10 m, elevation 60° .

$\tan 60^\circ = h / 10$

$\sqrt{3} = h / 10$

$h = 10\sqrt{3}$

$\approx 17.32 \text{ m}$ (taller pole, shorter shadow at higher elevation — makes sense ✓)

Another twist — Find shadow given height + angle. Pole 12 m tall, elevation 45° .

$\tan 45^\circ = \text{height} / \text{shadow}$

$1 = 12 / \text{shadow}$

shadow = **12 m** (at 45° elevation, shadow always equals height)

Section 10A — ADVANCED TRIG + HEIGHTS (exam-level variations)

Q41.5 (Advanced) — Two observers + same object. A tower stands on level ground. From a point P, the angle of elevation to the top is 30° . From another point Q, 50 m closer to the tower, the angle is 45° . Find the height of the tower.

Let height = h; distance from tower foot to Q = x

Distance from tower foot to P = x + 50

From Q (angle = 45°):

$\tan 45^\circ = h/x = 1$

Therefore $h = x$

From P (angle = 30°):

$\tan 30^\circ = h/(x + 50) = 1/\sqrt{3}$

Cross-multiply: $h\sqrt{3} = x + 50$

Substitute $h = x$:

$$x\sqrt{3} = x + 50$$

$$x\sqrt{3} - x = 50$$

$$x(\sqrt{3} - 1) = 50$$

$$x = 50 / (\sqrt{3} - 1)$$

$$\text{Rationalise: } x = 50(\sqrt{3} + 1) / [(\sqrt{3} - 1)(\sqrt{3} + 1)] = 50(\sqrt{3} + 1) / 2 = 25(\sqrt{3} + 1)$$

$$\text{Answer: } h = x = 25(\sqrt{3} + 1) \approx 25 \times 2.732 \approx 68.30 \text{ m}$$

Q41.6 (Advanced) — Angle of depression. From the top of a 100 m cliff, the angle of depression to a boat is 30° . Find the boat's distance from the foot of the cliff.

Angle of depression from top = angle of elevation from boat (alternate interior angles)

$$\tan 30^\circ = \text{height} / \text{horizontal distance} = 100 / d$$

$$1/\sqrt{3} = 100 / d$$

$$d = 100 \times \sqrt{3} = 100\sqrt{3} \approx 173.21 \text{ m}$$

Section 11 — NUMBER SYSTEM + LCM/HCF (2 worked + variations)

Notation expansion (used throughout this section): **HCF** = Highest Common Factor (also called GCD = Greatest Common Divisor). Largest number that divides all given numbers. **LCM** = Least Common Multiple. Smallest number that all given numbers divide into. **Universal identity:** $\text{LCM} \times \text{HCF} = \text{product of two numbers (works for 2 numbers only)}$.

Q42. Find the HCF of 252 and 105.

What kind of problem is this? Direct HCF — use prime factorisation OR Euclid's algorithm.

Solving it (Method 1 — prime factorisation). Step-by-step.

Step 1 — Factorise 252. $252 = 2 \times 126 = 2 \times 2 \times 63 = 4 \times 9 \times 7 = 2^2 \times 3^2 \times 7$

Step 2 — Factorise 105. $105 = 3 \times 35 = 3 \times 5 \times 7$

Step 3 — Pick common factors with LOWEST powers. Common primes: 3^1 and 7^1

Step 4 — Multiply to get HCF. $\text{HCF} = 3 \times 7 = 21$

Quick way (Method 2 — Euclid's algorithm).

$$252 = 2 \times 105 + 42 \text{ (remainder)}$$

$$105 = 2 \times 42 + 21$$

$$42 = 2 \times 21 + 0 \rightarrow \text{HCF} = 21$$

Worth knowing: Verify with the $\text{LCM} \times \text{HCF} = \text{product identity}$.

$$\text{LCM} = (252 \times 105) / 21 = \frac{26460}{21} = 1,260.$$

$$\text{Verify: } \frac{1260}{252} = 5 \checkmark; \frac{1260}{105} = 12 \checkmark.$$

Watch out: "63" — students miss the prime factor 5 and include $3 \times 21 = 63$ wrongly.

A similar question — HCF of three numbers. HCF of 24, 36, 60.

Step 1 — Prime factorise each. $24 = 2^3 \times 3$

$$36 = 2^2 \times 3^2$$

$$60 = 2^2 \times 3 \times 5$$

Step 2 — Pick common factors with lowest powers. Common primes: 2^2 and 3^1

Step 3 — Compute HCF. $\text{HCF} = 4 \times 3 = 12$

Another twist — Find LCM directly. LCM of 24, 36 using prime-factorisation.

Step 1 — Prime factorise each. $24 = 2^3 \times 3$

$$36 = 2^2 \times 3^2$$

Step 2 — Take highest powers of all primes. Highest power of 2 = $2^3 = 8$

$$\text{Highest power of } 3 = 3^2 = 9$$

Step 3 — Compute LCM. $\text{LCM} = 8 \times 9 = 72$

Q43. Find the sum of digits of the largest 4-digit number that is divisible by 88.

What kind of problem is this? Find largest multiple of n within a range; then operate on its digits.

Largest 4-digit number = 9999.

$$\text{Divide: } 9999 \div 88 = 113.625 \text{ (so floor} = 113\text{)}.$$

$$\text{Largest multiple of } 88 \leq 9999 = 113 \times 88 = \mathbf{9,944}.$$

$$\text{Digits} = 9, 9, 4, 4. \text{ Sum} = 9 + 9 + 4 + 4 = \mathbf{26}.$$

Worth knowing: Quick way to compute largest multiple of $n \leq M$. Compute $M \div n$; take integer part k ; answer = $k \times n$.

Watch out: Students compute $\frac{9999}{88}$ wrongly and use $114 \times 88 = 10,032$ (a 5-digit number — out of range). Or they round to 114 instead of flooring to 113.

A similar question — Smallest 4-digit multiple of 7. Smallest 4-digit number = 1000. $\frac{1000}{7} = 142.86 \rightarrow \text{ceiling} = 143$. $143 \times 7 = \mathbf{1,001}$.

Another twist — Largest 3-digit multiple of LCM(4, 5, 6). $\text{LCM}(4, 5, 6) = 60$. Largest 3-digit ≤ 999 : $\frac{999}{60} = 16.65 \rightarrow 16 \times 60 = \mathbf{960}$.

Section 11A — ADVANCED NUMBER SYSTEM (exam-level variations)

Q43.5 (Advanced) — Divisibility rules speed-test. Test if 12,345,678 is divisible by 11.

Digits: 1, 2, 3, 4, 5, 6, 7, 8.

Alternating sum (from right): $8 - 7 + 6 - 5 + 4 - 3 + 2 - 1 = (8+6+4+2) - (7+5+3+1) = 20 - 16 = 4$.

4 is NOT divisible by 11 → number is NOT divisible by 11.

Memorise divisibility rules.

2: last digit even.

3: digit-sum divisible by 3.

4: last 2 digits divisible by 4.

5: last digit 0 or 5.

6: divisible by both 2 AND 3.

8: last 3 digits divisible by 8.

9: digit-sum divisible by 9.

11: alternating sum divisible by 11.

25: last 2 digits 00, 25, 50, 75.

Q43.6 (Advanced) — Remainder theorem. Find the remainder when 7^{100} is divided by 12.

$$7^1 \bmod 12 = 7$$

$$7^2 = 49 \bmod 12 = 1$$

$$7^3 = 7 \times 1 \bmod 12 = 7$$

$$7^4 = 7 \times 7 \bmod 12 = 49 \bmod 12 = 1$$

Pattern: odd power → 7; even power → 1.

$7^{100} \rightarrow$ even power → remainder = 1.

Q43.7 — Unit digit / last digit. Find the unit digit of 7^{93} .

Why this matters: SSC CGL, CHSL, RRB NTPC, and IBPS all ask "find the unit digit of x^n " as a standalone 1-mark question. The answer relies entirely on the cycle of the last digit.

Unit digit cycles (memorise this table):

Last digit of base	Cycle of unit digits	Cycle length
0	0	1
1	1	1
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
4	4, 6	2
5	5	1
6	6	1
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4
9	9, 1	2

Solving Q43.7. 7 has cycle 7, 9, 3, 1 (length 4).

$93 \div 4 = 23$ remainder 1.

Position 1 in the cycle $\rightarrow$ unit digit = 7.

KEY FACT

Rule: Divide the power by the cycle length. Use the remainder to pick the position in the cycle. Remainder 0 means take the LAST element of the cycle. - $7^4 \rightarrow$ remainder 0 $\rightarrow$ take last in cycle $\{7,9,3,1\} = 1 \checkmark$ - $7^{92} \rightarrow 92 \div 4 = 23 \text{ r}0 \rightarrow 1$ - $7^{93} \rightarrow 93 \div 4 = 23 \text{ r}1 \rightarrow 7 \checkmark$

Watch out: " 7^{93} — cycle of 4, so $\frac{93}{4} = 23.25$, unit digit is 3" — students confuse 23.25 with the 3rd position. Always use the INTEGER remainder ($93 \bmod 4 = 1$), not the decimal part.

Q43.8 — Number of factors. How many factors does 720 have?

Why this matters: SSC CGL Tier-1 and Tier-2 ask this directly: "Find the number of divisors of N" or "How many numbers less than N divide N exactly?"

Solving it.

Step 1 — Prime factorise 720. $720 = 72 \times 10$

$$= 8 \times 9 \times 10$$

$$= 2^3 \times 3^2 \times 2 \times 5$$

$$= 2^4 \times 3^2 \times 5^1$$

Step 2 — Apply the number-of-factors formula. Number of factors = (exponent of 2 + 1) $\times$ (exponent of 3 + 1) $\times$ (exponent of 5 + 1)

$$= (4 + 1) \times (2 + 1) \times (1 + 1)$$

$$= 5 \times 3 \times 2$$

$$= 30$$

KEY FACT

Formula: If $N = p_1^a \times p_2^b \times p_3^c \times \dots$, then number of factors = $(a+1)(b+1)(c+1)\dots$

This works because each factor chooses 0 to a copies of p_1 , 0 to b copies of p_2 , etc.

A similar question: Number of even factors of 720. Even factors must include at least one 2. So keep 2^1 to $2^4 \rightarrow 4$ choices for the power of 2 (not 5). Total even factors = $4 \times 3 \times 2 = 24$.

Another twist — number of odd factors. Odd factors include no 2s $\rightarrow$ only 1 choice for 2's power (2^0). Total odd factors = $1 \times 3 \times 2 = 6 \checkmark$ ($6 + 24 = 30$ total $\checkmark$).

Watch out: Students add 1 to the exponent of 2 for even-factor count. Wrong: even factors exclude 2^0 , so the 2-power contributes 4 choices (2^1 to 2^4), not 5.

Section 12 — ALGEBRA + IDENTITIES (3 worked + variations)

Notation expansion (used throughout this section): Common identities to memorise: $(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$ $(a + b)(a - b) = a^2 - b^2$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

Q44. If $x + 1/x = 5$, find the value of $x^3 + 1/x^3$.

What kind of problem is this? Symmetric algebra trick. $x + 1/x$ is GIVEN; $x^3 + 1/x^3$ is asked. Use the identity directly.

$$(x + 1/x)^3 = x^3 + 1/x^3 + 3 \times x \times (1/x) \times (x + 1/x)$$

$$(x + 1/x)^3 = x^3 + 1/x^3 + 3 \times (x + 1/x)$$

$$\text{Rearrange: } x^3 + 1/x^3 = (x + 1/x)^3 - 3 \times (x + 1/x)$$

Step 4 — Substitute $k = 5$. $x^3 + 1/x^3 = 5^3 - 3 \times 5$

$$= 125 - 15$$

$$= \mathbf{110}$$

Worth knowing: Memorise this standard formula card. If $x + 1/x = k$, then:

$$x^2 + 1/x^2 = k^2 - 2 \quad x^3 + 1/x^3 = k^3 - 3k \quad x^4 + 1/x^4 = (x^2 + 1/x^2)^2 - 2 = (k^2 - 2)^2 - 2$$

If $x - 1/x = k$, then:

$$x^2 + 1/x^2 = k^2 + 2 \quad (\text{note } +2 \text{ here}) \quad x^3 - 1/x^3 = k^3 + 3k$$

Watch out: "125" — students forget the $-3k$ correction term.

A similar question — Find $x^2 + 1/x^2$ when $x + 1/x = 5$.

$$x^2 + 1/x^2 = k^2 - 2$$

$$= 5^2 - 2$$

$$= 25 - 2$$

$$= \mathbf{23}$$

Another twist — $x - 1/x = 3 \rightarrow x^3 - 1/x^3 = ?$

$$x^3 - 1/x^3 = k^3 + 3k$$

$$= 3^3 + 3 \times 3$$

$$= 27 + 9$$

$$= \mathbf{36}$$

Q45. If $a + b + c = 0$, find the value of $a^3 + b^3 + c^3$.

What kind of problem is this? Direct application of the famous identity: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.

Given: $a + b + c = 0$.

Substitute into the RHS of the identity: $\text{RHS} = 0 \times (\dots) = \mathbf{0}$.

So: $a^3 + b^3 + c^3 - 3abc = 0$

Rearrange: $a^3 + b^3 + c^3 = \mathbf{3abc}$

Worth knowing: This identity is the BACKBONE of many sum-of-cubes problems. The condition " $a + b + c = 0$ " instantly converts the cube-sum to a simple product.

Watch out: "0" — common mistake. Students assume "if sum = 0, then sum of cubes = 0". WRONG: it equals $3abc$ (only 0 if at least one of a, b, c is 0).

A similar question — Numerical verification. $a = 2, b = 3, c = -5$ (sum = $2 + 3 - 5 = 0$ ✓).

Step 1 — Compute $a^3 + b^3 + c^3$ directly. $= 2^3 + 3^3 + (-5)^3$

$= 8 + 27 - 125$

$= \mathbf{-90}$

Step 2 — Verify using $3abc$. $3abc = 3 \times 2 \times 3 \times (-5)$

$= 3 \times (-30)$

$= \mathbf{-90}$ ✓

Another twist — Find $(a+b)^3 + (b+c)^3 + (c+a)^3$ when $a+b+c = 0$.

Step 1 — Rewrite each bracket using the given condition. $a + b + c = 0 \rightarrow (a + b) = -c, (b + c) = -a, (c + a) = -b$

Step 2 — Substitute. Expression $= (-c)^3 + (-a)^3 + (-b)^3$

$= -c^3 - a^3 - b^3$

$= -(a^3 + b^3 + c^3)$

Step 3 — Apply the identity (since $a + b + c = 0$). $a^3 + b^3 + c^3 = 3abc$

Expression $= \mathbf{-3abc}$

Q46. If $a^2 + b^2 + c^2 - ab - bc - ca = 0$, prove that $a = b = c$.

What kind of problem is this? Algebraic identity recognition. The given expression has a hidden sum-of-squares form.

Multiply both sides by 2: $2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$

Group: $(a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ca + a^2) = 0$

Factor each group as a perfect square: $\mathbf{(a - b)^2 + (b - c)^2 + (c - a)^2 = 0}$

Sum of three non-negative squares equals zero **only if** each is zero.

So $(a - b) = 0, (b - c) = 0, (c - a) = 0 \rightarrow \mathbf{a = b = c}$

Worth knowing: This identity is incredibly useful. If you see $a^2 + b^2 + c^2 - ab - bc - ca$, instantly think of $\frac{1}{2} \times$ sum-of-squared-differences.

Watch out: Students get the right answer but write proof loosely. The "sum of squares = 0 $\Rightarrow$ each square = 0" step needs to be stated explicitly for full marks.

A similar question — Numerical check. $a = b = c = 5$. Then $25 + 25 + 25 - 25 - 25 - 25 = 0$ ✓. Any unequal trio fails.

Another twist — Inequality. Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$ for any real a, b, c . Same identity: $\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$ (sum of squares always non-negative). Equality holds iff $a = b = c$.

Section 12A — ADVANCED ALGEBRA (exam-level variations)

Q46.5 (Advanced) — Quadratic equation by formula. Solve $2x^2 - 7x + 3 = 0$.

Solving it. Use the quadratic formula: $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$.

Step 1 — Identify coefficients. $a = 2, b = -7, c = 3$

Step 2 — Compute the discriminant. Discriminant $= b^2 - 4ac$

$$= (-7)^2 - 4 \times 2 \times 3$$

$$= 49 - 24$$

$$= 25$$

Step 3 — Take the square root. $\sqrt{\text{discriminant}} = \sqrt{25} = 5$

Step 4 — Apply the formula. $x = (-b \pm \sqrt{\text{discriminant}}) / (2a)$

$$= (7 \pm 5) / 4$$

$$x = \frac{12}{4} = 3 \text{ or } x = \frac{2}{4} = \frac{1}{2}$$

Quick check via factorisation.

$$2x^2 - 7x + 3 = 2x^2 - 6x - x + 3$$

$$= 2x(x - 3) - (x - 3)$$

$$= (2x - 1)(x - 3) = 0$$

$$x = \frac{1}{2} \text{ or } x = 3 \text{ ✓}$$

Q46.6 (Advanced) — Simultaneous linear equations. Solve: $3x + 2y = 16$; $5x - 3y = 14$.

Multiply Eq1 by 3: $9x + 6y = 48$

Multiply Eq2 by 2: $10x - 6y = 28$

Add both equations: $19x = 76$

$$x = \frac{76}{19} = 4$$

Substitute $x = 4$ into Eq1: $3(4) + 2y = 16 \rightarrow 12 + 2y = 16$

$$2y = 4 \rightarrow y = 2$$

Q46.7 (Advanced) — Roots and coefficients. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^2 + \beta^2$ without computing the individual roots.

Solving it. Use Vieta's formulas.

Step 1 — Read off sum and product of roots. Sum: $\alpha + \beta = -b/a = 5$

Product: $\alpha\beta = c/a = 6$

Step 2 — Apply the squared-sum identity. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= 5^2 - 2 \times 6$$

$$= 25 - 12$$

$$= \mathbf{13}$$

(Verify: roots are 2 and 3. $2^2 + 3^2 = 4 + 9 = 13 \checkmark$.)

Section 13 — DATA INTERPRETATION (3 worked + variations)

Notation expansion (used throughout this section): *DI* = Data Interpretation (charts, tables, graphs).

Common forms: bar chart, line graph, pie chart, table, mixed combo. Always read the question

CAREFULLY before scanning numbers — the wrong reading leads to obvious wrong answers among the options.

Q47. A bar chart shows production (in tonnes) of a factory over 5 years: 2020 → 120, 2021 → 150, 2022 → 180, 2023 → 200, 2024 → 250. What is the % growth in production from 2020 to 2024?

What kind of problem is this? % change formula. **% change = (new – old) / old × 100.**

Initial (2020) = 120

Final (2024) = 250

Difference = $250 - 120 = \mathbf{130}$

$$\% \text{ growth} = (\text{difference} / \text{initial}) \times 100 = \left(\frac{130}{120}\right) \times 100$$

$$= \frac{13000}{120}$$

$$= \mathbf{108.33 \%}$$

Watch out: "130 %" — students forget to divide by 120 (initial value) and report the absolute difference as a percent.

A similar question — % decline. Production was 250 in 2024, dropped to 200 in 2025. % decline = $(250 - 200) / 250 \times 100 = \mathbf{20 \%}$.

Another twist — Average annual growth rate (CAGR-style approximation). If production grows from 120 (2020) to 250 (2024) over 4 years, simple average annual growth $\approx 108.33 \% / 4 = \mathbf{27.08 \% \text{ per year (simple average)}}$. (True CAGR via compounded formula = $\left(\frac{250}{120}\right)^{\frac{1}{4}} - 1 \approx 20.16 \%$.)

Q48. A pie chart of company expenses: salary 30 %, raw materials 25 %, R&D 20 %, marketing 15 %, miscellaneous 10 %. If the marketing expense is ₹3 crore, find the total budget.

What kind of problem is this? Pie chart back-calculation: from one slice's value + percentage, compute the whole.

Marketing slice = 15 % of total budget = ₹3 crore

$$1 \% \text{ of total} = \frac{3}{15} = ₹0.2 \text{ crore}$$

$$100 \% \text{ (total budget)} = 100 \times 0.2 = \text{₹20 crore}$$

EXAM SHORTCUT

Quick way: Direct ratio. Total = $3 / \left(\frac{15}{100}\right) = 3 \times \left(\frac{100}{15}\right) = \frac{300}{15} = \text{₹20 crore}.$

Worth knowing: Once total is known, find any other slice instantly.

$$\text{Salary} = 30 \% \times 20 = \text{₹6 crore}.$$

$$\text{Raw materials} = 25 \% \times 20 = \text{₹5 crore}.$$

$$\text{R\&D} = 20 \% \times 20 = \text{₹4 crore}.$$

$$\text{Misc} = 10 \% \times 20 = \text{₹2 crore}.$$

$$\text{Verify sum: } 6 + 5 + 4 + 3 + 2 = \text{₹20 crore } \checkmark.$$

Watch out: Students directly multiply $3 \times 15 = \text{₹45 crore}$ (instead of dividing 3 by 0.15).

A similar question — Comparing slices. R&D vs marketing — R&D is 20 % vs marketing 15 % → R&D is $(20-15)/15 \times 100 = \text{33.33 \% more}$ than marketing.

Another twist — Pie chart in degrees. Each 1 % = 3.6° (since 100 % = 360°). Marketing 15 % = 54° . Salary 30 % = 108° . Useful when chart gives degrees instead of percentages.

Q49. Average sales over 4 quarters of a year is 250 units. Q1 = 200, Q2 = 240, Q3 = 280. Find Q4 sales.

What kind of problem is this? Use the average to get the total, then subtract known values to find the missing one.

$$\text{Total} = \text{average} \times \text{count} = 250 \times 4 = \text{1,000 units}$$

$$\text{Sum of Q1 + Q2 + Q3} = 200 + 240 + 280 = \text{720}$$

$$\text{Q4} = \text{Total} - \text{sum of known} = 1,000 - 720 = \text{280 units}$$

Worth knowing: Verify by computing the average back. $(200 + 240 + 280 + 280) / 4 = \frac{1000}{4} = 250 \checkmark.$

Watch out: Students forget that "average = 250 over 4 quarters" gives total = 1000 and try to set up algebra unnecessarily.

A similar question — Find missing month. Average rainfall over 12 months = 50 mm. 11 months total = 540 mm. Missing month = $600 - 540 = \text{60 mm}.$

Another twist — Average of new sub-set. If Q4 in question becomes 320 (not 280), what's the new yearly average? New total = $720 + 320 = 1,040$. Avg = $\frac{1040}{4} = \text{260 units/quarter}.$

Section 13A — ADVANCED DI (exam-level variations)

Q49.5 (Advanced) — Mixed bar + line graph. A bar chart shows monthly sales (in lakhs) for 5 months: Jan 50, Feb 60, Mar 75, Apr 80, May 90. A line graph on the same chart shows monthly profit margin (%): Jan 10 %, Feb 12 %, Mar 15 %, Apr 14 %, May 16 %. Find the total absolute profit (in lakhs) over the 5 months.

$$\text{Jan: } 50 \times 0.10 = \mathbf{5}$$

$$\text{Feb: } 60 \times 0.12 = \mathbf{7.2}$$

$$\text{Mar: } 75 \times 0.15 = \mathbf{11.25}$$

$$\text{Apr: } 80 \times 0.14 = \mathbf{11.2}$$

$$\text{May: } 90 \times 0.16 = \mathbf{14.4}$$

$$\text{Total} = 5 + 7.2 + 11.25 + 11.2 + 14.4 = \mathbf{49.05 \text{ lakhs.}}$$

Q49.6 (Advanced) — Missing-cell DI. A table shows expenses (₹ cr) of a company over 4 quarters across 4 categories: Salary, Raw, R&D, Marketing. The Q1 totals: Salary 30, Raw 25, R&D 20, Marketing 15 (sum = 90). Q2 totals are 90 (same total). If Q2 Salary = 25, Q2 Raw = 28, Q2 R&D = 22, find Q2 Marketing.

$$\text{Q2 Marketing} = 90 - 25 - 28 - 22 = \mathbf{15 \text{ cr.}}$$

Section 14 — PROBABILITY + PERMUTATIONS / COMBINATIONS (1 worked + variations)

Notation expansion (used throughout this section): $P(E)$ = probability of event E = (favourable outcomes) / (total outcomes). Always between 0 and 1. $C(n, r)$ = "n choose r" = number of ways to pick r items from n without order = $n! / (r! \times (n-r)!)$. $P(n, r)$ = number of arrangements (with order) = $n! / (n-r)!$.

Q50. A bag contains 5 red, 4 blue and 3 green balls. Two balls are drawn at random WITHOUT replacement. What is the probability that both are red?

What kind of problem is this? Combinatorial probability — count favourable + total via combinations.

$$\text{Total balls} = 5 + 4 + 3 = \mathbf{12}$$

$$\text{Total ways to choose 2 balls from 12} = C(12, 2) = (12 \times 11) / 2 = \mathbf{66}$$

$$\text{Favourable: choose 2 red from 5 reds} = C(5, 2) = (5 \times 4) / 2 = \mathbf{10}$$

$$P(\text{both red}) = \text{favourable} / \text{total} = \frac{10}{66} = \frac{5}{33}$$

EXAM SHORTCUT

Quick way: Sequential probability (without replacement).

$$P(\text{first red}) = \frac{5}{12}$$

$$P(\text{second red} \mid \text{first red}) = \frac{4}{11} \text{ (4 reds left out of 11 remaining)}$$

$$P(\text{both red}) = \left(\frac{5}{12}\right) \times \left(\frac{4}{11}\right) = \frac{20}{132} = \frac{5}{33} \checkmark \text{ (same answer)}$$

Watch out: Students use **with replacement** logic: $(\frac{5}{12}) \times (\frac{5}{12}) = \frac{25}{144}$. WRONG when the question says "without replacement" or "drawn together".

A similar question — Both balls of same colour. $P(2 \text{ red}) + P(2 \text{ blue}) + P(2 \text{ green}) = C(5,2)/C(12,2) + C(4,2)/C(12,2) + C(3,2)/C(12,2) = \frac{10}{66} + \frac{6}{66} + \frac{3}{66} = \frac{19}{66}$.

Another twist — One red and one blue. Favourable = $C(5,1) \times C(4,1) = 5 \times 4 = 20$ ways (any red $\times$ any blue).

Total = $C(12, 2) = 66$. $P = \frac{20}{66} = \frac{10}{33}$.

Section 14A — ADVANCED PROBABILITY + COMBINATIONS (exam-level variations)

Q50.5 (Advanced) — Conditional probability. A bag has 6 red and 4 blue balls. Two balls are drawn one after the other without replacement. Given that the first ball drawn is red, what is the probability that the second is also red?

After 1 red drawn, remaining = 5 red + 4 blue = 9 total.

$$P(\text{second red} \mid \text{first red}) = \frac{5}{9}.$$

Q50.6 (Advanced) — Permutations with restriction. In how many ways can the letters of "EQUATION" be arranged so that all vowels (E, U, A, I, O — five vowels) come together?

"EQUATION" has 8 letters with 5 vowels and 3 consonants (Q, T, N).

Treat all 5 vowels as ONE block. Then we have 4 entities (the vowel-block + Q + T + N).

Number of ways to arrange 4 entities = $4! = 24$.

Within the vowel block, 5 vowels can be permuted = $5! = 120$.

Total = $24 \times 120 = 2,880$ ways.

Q50.7 (Advanced) — Combinations from a committee. From 10 men and 8 women, how many ways can a committee of 5 be formed having at most 2 women?

0 women: $C(10, 5) \times C(8, 0) = 252 \times 1 = 252$.

1 woman: $C(10, 4) \times C(8, 1) = 210 \times 8 = 1,680$.

2 women: $C(10, 3) \times C(8, 2) = 120 \times 28 = 3,360$.

Total = $252 + 1,680 + 3,360 = 5,292$ ways.

PART G — ADVANCED EXAM-PATTERN TYPES (new patterns 2024-26)

Why this part exists. Sections 1-14 cover the FUNDAMENTAL types tested in every paper. But modern Banks PO Mains, SSC CGL Tier-2, and SBI/IBPS use NEW question patterns that did not exist 5 years ago — Quadratic Comparison, Approximation, Caselet DI, Coordinate Geometry, Statistics, Logarithms, Combined Mensuration, etc. This Part covers each new pattern with the same step-by-step + variations format.

Section 15 — QUADRATIC COMPARISON (Banks PO Mains • 5 Qs/paper)

Notation expansion: Two quadratic equations are given — one in x and one in y . Solve each, find the roots, then compare x and y to choose one of: $x > y$ / $x \geq y$ / $x < y$ / $x \leq y$ / $x = y$ / **no relation**.

Q51. I. $x^2 - 7x + 10 = 0$ II. $y^2 - 6y + 8 = 0$. Find the relation between x and y .

What kind of problem is this? Standard Banks PO Mains quadratic-comparison. Solve both equations, then compare ALL roots of x with ALL roots of y .

Solving it. Step-by-step.

Equation I: $x^2 - 7x + 10 = 0$

Split middle term: $x^2 - 5x - 2x + 10 = 0$

Group: $x(x - 5) - 2(x - 5) = 0$

Factor: $(x - 5)(x - 2) = 0$

Roots: $x = 2$ or $x = 5$

Equation II: $y^2 - 6y + 8 = 0$

Split middle term: $y^2 - 4y - 2y + 8 = 0$

Group: $y(y - 4) - 2(y - 4) = 0$

Factor: $(y - 4)(y - 2) = 0$

Roots: $y = 2$ or $y = 4$

Comparing roots (the systematic way).

Smallest $x = 2$, largest $x = 5$.

Smallest $y = 2$, largest $y = 4$.

For $x \geq y$ to be true, we need every $x \geq$ every y . Check: $x = 2 \geq y = 4$? **NO**. So $x \geq y$ is FALSE.

For $x \leq y$ to be true, we need every $x \leq$ every y . Check: $x = 5 \leq y = 4$? **NO**. So $x \leq y$ is FALSE.

Some roots overlap ($x = 2 = y = 2$). Some $x > y$ ($x=5, y=2$ or 4). Some $x < y$ ($x=2, y=4$).

No clear relation can be established.

Worth knowing: Universal comparison rule.

If smallest $x \geq$ largest $y \rightarrow x \geq y$ (with equality if smallest $x =$ largest y)

If largest $x \leq$ smallest $y \rightarrow x \leq y$

Otherwise $\rightarrow$ **no relation**.

Watch out: Students compare just one root with one root and conclude wrongly. ALWAYS compare smallest- x vs largest- y AND largest- x vs smallest- y .

A similar question — Clear $x > y$. I. $x^2 - 9x + 20 = 0 \rightarrow x = 4, 5$. II. $y^2 - 5y + 6 = 0 \rightarrow y = 2, 3$. Smallest $x = 4 >$ largest $y = 3 \rightarrow x > y$.

Another twist — Equation in expanded form. I. $2x^2 - 11x + 12 = 0$. Factorise: $(2x - 3)(x - 4) = 0 \rightarrow x = 1.5$ or 4 . II. $3y^2 - 14y + 16 = 0$. Factorise: $(3y - 8)(y - 2) = 0 \rightarrow y = \frac{8}{3} \approx 2.67$ or 2 . Smallest $x = 1.5 <$ smallest $y = 2$; largest $x = 4 >$ largest $y = \frac{8}{3}$. Mixed $\rightarrow$ **no relation**.

Q52. I. $x^2 - 12x + 35 = 0$ II. $y^2 - 13y + 42 = 0$. Find the relation.

Eq I: $(x - 5)(x - 7) = 0 \rightarrow x \in \{5, 7\}$

Eq II: $(y - 6)(y - 7) = 0 \rightarrow y \in \{6, 7\}$

smallest $x = 5$, largest $x = 7$

smallest $y = 6$, largest $y = 7$

For $x \geq y$: smallest x (5) $\geq$ largest y (7)? NO.

For $x \leq y$: largest x (7) $\leq$ smallest y (6)? NO.

Mixed $\rightarrow$ **no relation can be established.**

Memory peg. When the two equations share a common root (here both have 7), the answer is almost always "no relation" unless the other roots align cleanly.

Section 16 — APPROXIMATION (Banks PO Pre • 5 Qs/paper • 30 sec each)

Notation expansion: " $\approx$ " or "?" in the question stem means **closest integer / round number**, not exact value. Speed > accuracy beyond 1-2%.

Q53. ? $\approx$ 39.97 % of $1899.98 + 24.04 \times 17.97$. Find the value of ?.

What kind of problem is this? Round each ugly number to a clean value, then compute mentally.

$39.97\% \approx 40\% (= \frac{2}{5})$

$1899.98 \approx 1,900$

$24.04 \approx 24$

$17.97 \approx 18$

Now compute (mentally).

$40\% \text{ of } 1,900 = (\frac{2}{5}) \times 1,900 = 760$

$24 \times 18 = (24 \times 20) - (24 \times 2) = 480 - 48 = 432$

Sum = $760 + 432 = 1,192$

Worth knowing: Memorise these "round-able" fractions for instant computation.

Difficult %	Round to	As fraction
39.97 %	40 %	$\frac{2}{5}$
33.33 %	33.33 %	$\frac{1}{3}$
16.66 %	16.66 %	$\frac{1}{6}$
14.28 %	14.28 %	$\frac{1}{7}$
12.5 %	12.5 %	$\frac{1}{8}$

Difficult %	Round to	As fraction
11.11 %	11.11 %	$\frac{1}{9}$
9.09 %	9.09 %	$\frac{1}{11}$
8.33 %	8.33 %	$\frac{1}{12}$

Watch out: Don't over-round. If actual is 1899.98, rounding to 2,000 introduces 5 % error — too much. Round to nearest "nice" value (1,900 here).

A similar question — Subtraction. $? \approx 49.99 \% \text{ of } 480.04 - 19.97 \times 11.95. \rightarrow 50 \% \text{ of } 480 = 240; 20 \times 12 = 240. ? = 0$ (or close to 0; actual closer to -0.5).

Another twist — Division/mixed. $? \approx \sqrt{624.97} + (35.04)^2. \rightarrow \sqrt{625} = 25; 35^2 = 1,225. ? = 25 + 1,225 = 1,250.$

Q54. $? \approx 16.66 \% \text{ of } 1199.95 + 25.04 \% \text{ of } 799.92.$

$$16.66 \% = \frac{1}{6} \text{ of } 1,200 = 200.$$

$$25.04 \% = \frac{1}{4} \text{ of } 800 = 200.$$

$$? \approx 200 + 200 = 400.$$

Speed-tip. Whenever you see a decimal % near a "nice" fraction (16.66, 14.28, 12.5, 33.33, etc.), use the fraction — it's faster than working out the decimal.

Section 17 — QUANTITY COMPARISON (Banks PO Mains • 3-5 Qs/paper)

Notation expansion: Two quantities (often algebraic or word-problem expressions) are given. Compute or simplify each, then compare.

Q55. Quantity I: A boat travels 30 km downstream in 3 hours; speed of stream = 2 km/h. Find the boat's speed in still water. Quantity II: Average of 5 consecutive even numbers, the smallest of which is 16. Compare Quantity I and Quantity II.

What kind of problem is this? Solve each side independently $\rightarrow$ numerical comparison.

Quantity I.

$$\text{Downstream speed} = \text{distance} / \text{time} = \frac{30}{3} = 10 \text{ km/h}$$

$$\text{Downstream} = \text{boat} + \text{stream} \rightarrow \text{boat} = 10 - 2 = 8 \text{ km/h}$$

Quantity II.

5 consecutive even numbers starting at 16: 16, 18, 20, 22, 24.

$$\text{Average} = (\text{sum}) / 5 = (16+18+20+22+24)/5 = \frac{100}{5} = 20.$$

Or shortcut: average of consecutive numbers = middle number = 20.

Compare: Quantity I (8) < Quantity II (20). So **Quantity I < Quantity II** (or $Q II > Q I$).

Worth knowing: Common answer options:

- (a) $Q I > Q II$
- (b) $Q I \geq Q II$
- (c) $Q I < Q II \leftarrow$ (correct here)
- (d) $Q I \leq Q II$
- (e) $Q I = Q II$ OR no relation

Watch out: Students compute one quantity carefully but rush the other. Always solve BOTH, then compare cleanly.

A similar question. Quantity I: 30 % of 200. Quantity II: 25 % of 240. Both = 60 $\rightarrow Q I = Q II$.

Section 18 — CASELET DI (Banks PO Mains specialty • 5-Q set)

Notation expansion: Caselet DI = 1 paragraph of facts, NO chart. Read carefully $\rightarrow$ extract data $\rightarrow$ answer 5 sub-questions. The challenge is extracting numbers correctly from prose.

Q56. Caselet: A factory has three units — A, B, and C. The total production in 2024 was 12,000 units. Unit A produced 40 % of the total. Unit B produced 75 % of what Unit A produced. Unit C produced the rest.

In 2025, Unit A's production increased by 25 %, Unit B's increased by 20 %, and Unit C's decreased by 10 %.

Sub-Q: What was Unit C's production in 2025?

Solving it. Step-by-step extraction.

Step 1 — 2024 production by unit. Total = 12,000

$$A = 40 \% \times 12,000 = 4,800$$

$$B = 75 \% \times 4,800 = 3,600$$

$$C = 12,000 - 4,800 - 3,600 = 3,600$$

Step 2 — 2025 production by unit. $A_{2025} = 4,800 \times 1.25 = 6,000$

$$B_{2025} = 3,600 \times 1.20 = 4,320$$

$$C_{2025} = 3,600 \times 0.90 = 3,240$$

Answer. Unit C in 2025 = 3,240 units.

Worth knowing: Caselet sub-Qs typically ask for: ratios, percentages, differences, totals across years/units. Build a table once $\rightarrow$ answer all 5 in 2-3 minutes.

Unit	2024	2025	Δ %
A	4,800	6,000	+25 %
B	3,600	4,320	+20 %
C	3,600	3,240	-10 %
Total	12,000	13,560	+13 %

Watch out: Students re-read the paragraph for each sub-Q (wastes 30 sec). Extract once, build the table, then read sub-Qs.

More sub-Qs (try mentally).

Ratio of A : B : C in 2024? $\rightarrow 4 : 3 : 3$

Total production in 2025? $\rightarrow 13,560$

% of total contributed by A in 2025? $\rightarrow \frac{6000}{13560} \times 100 \approx 44.25\%$

Difference between A's 2025 and B's 2025? $\rightarrow 6000 - 4320 = 1,680$ units

Section 19 — COORDINATE GEOMETRY (SSC CGL Tier-2 • 3-5 Qs/paper)

Notation expansion: A point in 2-D plane = (x, y) . x = horizontal (abscissa); y = vertical (ordinate).

Distance formula between (x_1, y_1) and $(x_2, y_2) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ **Midpoint formula** = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ **Slope formula** between two points = $(y_2 - y_1) / (x_2 - x_1)$ **Line equation** $y - y_1 = m(x - x_1)$, where m = slope.

Q57. Find the distance between points A (3, 4) and B (7, 1).

$$d = \sqrt{[(7 - 3)^2 + (1 - 4)^2]} = \sqrt{[16 + 9]} = \sqrt{25} = \mathbf{5 \text{ units}}$$

Worth knowing: Recognise Pythagorean triples in coordinate geometry. Here (4, 3, 5) appeared automatically.

Watch out: Students subtract in wrong order (3 - 7 vs 7 - 3); doesn't matter because we square — but be consistent.

A similar question. Distance between (0, 0) and (5, 12) = $\sqrt{(25 + 144)} = \sqrt{169} = \mathbf{13}$ (a 5-12-13 triple).

Another twist — Find a missing coordinate. Distance between (1, 2) and (4, k) = 5. Find k.

Step 1: Apply distance formula: $\sqrt{[(4 - 1)^2 + (k - 2)^2]} = 5 \Rightarrow \sqrt{9 + (k - 2)^2} = 5$

Step 2: Square both sides: $9 + (k - 2)^2 = 25 \Rightarrow (k - 2)^2 = 16$

Step 3: Take square root: $k - 2 = \pm 4$

Answer: $k = 2 + 4 = 6$ or $k = 2 - 4 = -2$

Q58. Find the slope of the line passing through (2, 3) and (6, 11). Then find the equation of the line.

Step 1 — Calculate the slope. Slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$

$$= (11 - 3) / (6 - 2)$$

$$= \frac{8}{4}$$

$$= 2$$

Step 2 — Write the point-slope equation (using point (2, 3)). $y - y_1 = m(x - x_1)$

$$y - 3 = 2(x - 2)$$

Step 3 — Expand and convert to slope-intercept form. $y - 3 = 2x - 4$

$$y = 2x - 4 + 3$$

$$y = 2x - 1$$

Worth knowing: Quick line-properties recap.

Parallel lines have the same slope.

Perpendicular lines have slopes whose product = -1 (i.e., $m_1 \times m_2 = -1$).

Horizontal line has slope 0; **vertical line** has undefined slope.

A similar question — Midpoint. Midpoint of (2, 3) and (6, 11) = $((2+6)/2, (3+11)/2) = (4, 7)$.

Another twist — Perpendicular line. Find equation of line perpendicular to $y = 2x - 1$, passing through (1, 5).

Perpendicular slope $m' = -\frac{1}{2}$ (since $2 \times (-\frac{1}{2}) = -1$)

$$\text{Point-slope: } y - 5 = -0.5(x - 1)$$

$$\text{Expand: } y - 5 = -0.5x + 0.5$$

$$y = -0.5x + 5.5$$

$$\text{Multiply by 2: } 2y + x = 11$$

Section 20 — STATISTICS (SSC CGL Tier-2 + Banks GA)

Notation expansion: **Mean** = arithmetic average. **Median** = middle value when sorted. **Mode** = most-frequent value. **Range** = max - min. **Standard Deviation (SD)** = $\sqrt{\text{variance}} = \sqrt{[(1/n) \sum (x_i - \text{mean})^2]}$.

Q59. Find the mean, median, and mode of the data: 4, 8, 6, 4, 7, 9, 4, 8, 6.

Solving it. Step-by-step.

Sort first: 4, 4, 4, 6, 6, 7, 8, 8, 9. ($n = 9$)

$$\text{Mean} = \text{sum} / \text{count} = (4+4+4+6+6+7+8+8+9) / 9 = \frac{56}{9} \approx 6.22$$

Median = middle value (since $n = 9$, middle is $(9+1)/2 = 5$ th value when sorted) = 6

Mode = most-frequent value = 4 (occurs 3 times)

Worth knowing: Empirical relation. **Mode = 3 × Median – 2 × Mean** (approximate, for moderately skewed data).

Watch out: Students forget to sort before finding the median. Without sorting, "middle position" is meaningless.

A similar question — Even count. Data: 2, 5, 7, 9, 11, 13. $n = 6$. Median = average of 3rd and 4th values = $(7 + 9)/2 = 8$.

Another twist — Find the Standard Deviation. Data: 2, 4, 6, 8, 10.

Step 1 — Find the mean. Mean = $(2 + 4 + 6 + 8 + 10) / 5 = \frac{30}{5} = 6$

Step 2 — Find each deviation from the mean and square it. $(2 - 6)^2 = (-4)^2 = 16$

$$(4 - 6)^2 = (-2)^2 = 4$$

$$(6 - 6)^2 = 0^2 = 0$$

$$(8 - 6)^2 = 2^2 = 4$$

$$(10 - 6)^2 = 4^2 = 16$$

Step 3 — Compute the variance. Sum of squared deviations = $16 + 4 + 0 + 4 + 16 = 40$

$$\text{Variance} = \text{Sum} / n = \frac{40}{5} = 8$$

Step 4 — Compute Standard Deviation. SD = $\sqrt{\text{Variance}} = \sqrt{8} \approx 2.83$

Section 21 — LOGARITHM (SSC CGL Tier-2 + RRB JE)

Notation expansion: $\log_b(x) = y$ means $b^y = x$. Default base in competitive exams = **10** (so $\log 100 = 2$).

Natural log = $\ln(x)$, base $e \approx 2.718$. **Key rules:** $\log(a \times b) = \log a + \log b$ $\log(a / b) = \log a - \log b$ $\log(a^n) = n \times \log a$ $\log_b(a) = \log a / \log b$ (change of base) $\log_b(b) = 1$; $\log_b(1) = 0$.

Q60. Find the value of $\log 8 + \log 125$ (base 10).

$$\log 8 + \log 125 = \log (8 \times 125) \text{ [product rule]}$$

$$= \log 1,000$$

$$= 3 \text{ (since } 10^3 = 1,000)$$

EXAM SHORTCUT

Quick way: Recognise that $8 \times 125 = 1,000$ (a power of 10) — $\log 10^n = n$.

Worth knowing: Memorise these standard log values (base 10).

$$\log 2 \approx 0.301$$

$$\log 3 \approx 0.477$$

$$\log 5 \approx 0.699 \text{ (= } 1 - \log 2)$$

$$\log 7 \approx 0.845$$

Watch out: Students try to compute $\log 8 = 3 \times \log 2$ (= 0.903) and $\log 125 = 3 \times \log 5$ (= 2.097) separately. Both are right but slower. Sum-then-multiply is faster.

A similar question — Solve for x. If $\log x = 2.5$, find x.

$$x = 10^{2.5}$$

$$= 10^2 \times 10^{0.5}$$

$$= 100 \times \sqrt{10} \approx \mathbf{316.23}$$

Another twist — Change of base. $\log_2(16) = \log 16 / \log 2 = (4 \times \log 2) / \log 2 = \mathbf{4}$.

Section 22 — MENSURATION COMBINED SOLIDS (SSC CGL Tier-2)

Notation expansion: Combined solids = two basic solids stuck together (cone-on-cylinder, hemisphere-on-cone, frustum, etc.). Volume = sum of parts; surface area = sum of EXTERNAL surfaces only (subtract any joining face).

Q61. A solid is formed by mounting a cone on top of a cylinder. The cylinder has radius 7 cm and height 10 cm. The cone has the same base radius (7 cm) and height 4 cm. Find the total volume.

$$\text{Volume of cylinder} = \pi r^2 h = \left(\frac{22}{7}\right) \times 49 \times 10$$

$$= 22 \times 10 \times 7 = \mathbf{1,540 \text{ cm}^3} \quad \text{Volume of cone} = \left(\frac{1}{3}\right) \pi r^2 h = \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 49 \times 4$$

$$= \left(\frac{1}{3}\right) \times 22 \times 28 = \mathbf{205.33 \text{ cm}^3} \quad \text{Total volume} = 1,540 + 205.33 = \mathbf{1,745.33 \text{ cm}^3}$$

Worth knowing: When solids share a base, volumes ADD (no overlap). Surface area adds only EXTERIOR — the joining circular face is hidden, so subtract one πr^2 from each side's TSA when computing total surface.

Watch out: Students compute TSA of each solid separately and add — wrong because the shared circular face counts twice.

A similar question — Hemisphere on cylinder. Cylinder $r = 7$, $h = 10$; hemisphere $r = 7$ on top.

Volume of cylinder = $\mathbf{1,540 \text{ cm}^3}$ (from above)

$$\text{Volume of hemisphere} = \left(\frac{2}{3}\right) \pi r^3 = \left(\frac{2}{3}\right) \times \left(\frac{22}{7}\right) \times 343$$

$$= \left(\frac{2}{3}\right) \times 22 \times 49 = \mathbf{718.67 \text{ cm}^3} \quad \text{Total} = 1,540 + 718.67 = \mathbf{2,258.67 \text{ cm}^3}$$

Another twist — Frustum (cone with top cut off). Frustum has $R = 6$ (bottom radius), $r = 3$ (top radius), $h = 8$.

$$\text{Volume} = \left(\frac{1}{3}\right) \pi h (R^2 + r^2 + Rr)$$

$$= \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 8 \times (36 + 9 + 18)$$

$$= \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 8 \times 63$$

$$= \left(\frac{1}{3}\right) \times 22 \times 8 \times 9 = \mathbf{528 \text{ cm}^3}$$

Section 23 — CALENDAR + CLOCK (RRB NTPC + SSC favourite)

Notation expansion: *Odd days* = days remaining when total days $\div 7$. Used for finding the day of the week of any date. **Clock angles:** Hour hand moves $360^\circ / 12 = 30^\circ$ per hour OR 0.5° per minute. Minute hand moves $360^\circ / 60 = 6^\circ$ per minute. Relative speed (minute – hour) = 5.5° per minute.

Q62. What day of the week was 15 August 1947?

Reference: 1 January 0001 was a Monday (calendar starts).

Total days from 1 Jan 0001 to 15 Aug 1947 = 1946 complete years + 226 days (in 1947).

Days in complete year = 365 (or 366 in leap). Quick formula: count "odd days" mod 7.

1600 years complete: odd days = 0 (every 400-year block has 0 odd days). Next 300 years (1601-1900): odd days = 1 (300 yrs has 5 odd days... actually let's use a quicker shortcut).

Faster method (Zeller's-style mental math). In SSC, this is rarely solved by hand for an arbitrary date.

Memorise key reference dates:

15 Aug 1947 = **Friday**

26 Jan 1950 = **Thursday**

1 Jan 2000 = **Saturday**

1 Jan 2024 = **Monday**

For nearby dates, count days forward/backward modulo 7.

Watch out: Leap year rules — divisible by 4 BUT not 100 (unless also divisible by 400). 1900 is NOT a leap year; 2000 IS.

A similar question — Day of week. What day was 26 January 1950? → Thursday (memorise this — first Republic Day).

Another twist — Days between two dates. From 1 Jan 2024 to 1 Jan 2025 = 366 days (2024 leap). $366 \div 7 = 52$ remainder 2. So 1 Jan 2025 = Monday + 2 = Wednesday.

Q63. Find the angle between the hour hand and the minute hand of a clock at 3:40.

Hour hand position at 3:40:

At 3:00, hour hand is at 90° (pointing to 3). In next 40 minutes, hour hand moves $40 \times 0.5^\circ = 20^\circ$ more. Hour hand at 3:40 = $90 + 20 = 110^\circ$ from 12. Minute hand position at 40 minutes past hour:

Minute hand at "40" mark = $40 \times 6^\circ = 240^\circ$ from 12. Angle between = $|240 - 110| = 130^\circ$

Worth knowing: Useful clock formulas.

Angle of hour hand from 12 = $(60 \times H + M) \times 0.5^\circ$ where H = hour, M = minutes.

Angle of minute hand from 12 = $M \times 6^\circ$.

Angle between = $|11M/2 - 30H| \pmod{360}$, take smaller of the two: angle or $360 - \text{angle}$.

Quick way using formula. $|11 \times \frac{40}{2} - 30 \times 3| = |220 - 90| = 130^\circ \checkmark$.

Watch out: Sometimes the "smaller" angle is asked — if your answer is $> 180^\circ$, subtract from 360° to get the reflex-angle complement.

A similar question — Clock at 6:15. $\left| 11 \times \frac{15}{2} - 30 \times 6 \right| = |82.5 - 180| = 97.5^\circ$.

Another twist — When are the hands at right angle? Hands form 90° angle 22 times in 12 hours (not 24 — they "miss" twice due to relative speed). First time after 3:00: when $|11M/2 - 90| = 90 \rightarrow 11M/2 = 0 \text{ or } 180 \rightarrow M = 0$ (i.e. 3:00 itself, but they're at 90° already? actually at 3:00 they're at 90°). Next: $M = \frac{360}{11} \approx 32.73$ minutes past 3 $\rightarrow$ roughly **3:32:43**.

PART E — TIMED MINI-MOCK (25 Q · 25 min)

Solve under timer. Self-grade against the answer key. Target: 22+/25 in <25 min for >90% mastery.

- 35 % of 480 = ?
(a) 158 (b) 168 (c) 178 (d) 188
- A sum doubles in 8 yrs at SI. Triple in?
(a) 16 (b) 18 (c) 20 (d) 24
- CP = ₹400, SP = ₹500. Profit %?
(a) 20 (b) 25 (c) 30 (d) 33
- 12, 18 LCM = ?
(a) 36 (b) 48 (c) 24 (d) 60
- Avg of first 10 natural numbers?
(a) 4.5 (b) 5 (c) 5.5 (d) 6
- $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$
(a) 1 (b) $\frac{5}{6}$
(c) $\frac{7}{6}$ (d) $\frac{11}{6}$
- A ratio 3:4, B is 25 % more. New ratio?
(a) 3:5 (b) 6:7 (c) 12:15 (d) 24:25
- 25 % of $x = 80$. $x = ?$
(a) 320 (b) 280 (c) 360 (d) 400
- Train 200 m at 72 km/h crosses pole in?
(a) 8 s (b) 10 s (c) 12 s (d) 15 s
- Boat speed in still water 12, stream 4. Downstream 32 km in?
(a) 1.5 h (b) 2 h (c) 2.5 h (d) 3 h
- A and B together 8 days, A alone 12. B alone?
(a) 16 (b) 20 (c) 24 (d) 30
- $(a + b)^2 - (a - b)^2 = ?$
(a) 4ab (b) 2ab (c) ab (d) 0
- $\sin^2 30^\circ + \cos^2 30^\circ = ?$
(a) $\frac{1}{2}$ (b) 1

(c) $\sqrt{\frac{3}{2}}$ (d) 2

14. Area equilateral triangle side 6 = ?

- (a) $9\sqrt{3}$ (b) $18\sqrt{3}$ (c) 36 (d) 18

15. SI on ₹2,000 @ 5 % for 4 yrs = ?

- (a) 200 (b) 300 (c) 400 (d) 500

16. CI on ₹1,000 @ 10 % for 2 yrs = ?

- (a) 200 (b) 210 (c) 215 (d) 220

17. P (red on a die) = ?

- (a) 0 (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

18. $7! / 5! = ?$

- (a) 7 (b) 14 (c) 42 (d) 35

19. Sum of angles of pentagon = ?

- (a) 360 (b) 540 (c) 720 (d) 900

20. Volume cube side 5 = ?

- (a) 25 (b) 75 (c) 100 (d) 125

21. CSA cylinder $r=7$, $h=10$ = ?

- (a) 220 (b) 440 (c) 660 (d) 880

22. Mean of 1, 2, ..., 10 = 5.5; SD $\approx$?

- (a) 2.87 (b) 3.0 (c) 3.5 (d) 4.0

23. 30 % of 30 % of 1,000 = ?

- (a) 60 (b) 90 (c) 30 (d) 100

24. Compound ratio 2:3 and 4:5 = ?

- (a) 4:5 (b) 8:15 (c) 6:8 (d) 1:1

25. Tank A fills in 10 h, B empties in 15 h. Both open?

- (a) 25 h (b) 30 h (c) 35 h (d) 40 h

Answer key: 1-b, 2-a, 3-b, 4-a, 5-c, 6-a, 7-a, 8-a, 9-b, 10-b, 11-c, 12-a, 13-b, 14-a, 15-c, 16-b, 17-a, 18-c, 19-b, 20-d, 21-b, 22-a, 23-b, 24-b, 25-b.

Scoring: 23–25 = Excellent | 18–22 = Good | Below 18 = revisit weak chapters before your mock series.

EXAM ALERT

Error-log rule: For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. This single habit doubles what the mock teaches you.

PART F — TRAP-RECOGNITION CARDS

Trap pattern	Where it appears	How to spot
"increase by r %, then decrease by r %" gives 0	Percentage	Multiplicative changes never cancel — net is always $-r^2/100$ %
Equal SP, equal % gain & loss = 0	Profit & Loss	Always net loss $(r/10)^2$

Trap pattern	Where it appears	How to spot
"double in n years SI" → "triple in 2n"	SI	Wrong; triple needs (n × 2) only if rate doubles. Correct: triple in 2n only if rate = 100/n
"average speed = (s ₁ + s ₂)/2"	TSD	Only true for equal time, not equal distance. Use harmonic $2s_1s_2/(s_1+s_2)$ for equal distance
Adding pipe rates instead of subtracting	Pipes	If outlet pipe, subtract
Using SP as base for profit %	P&L	Always CP base; SP base is "discount" or "margin"
Forgetting $\frac{18}{5}$ conversion	Speed	km/h to m/s: $\times \frac{5}{18}$
Heron + obvious right-angle	Mensuration	Check Pythagoras first; $\frac{1}{2}$ ·base·height beats Heron
Rate "p.a. compounded half-yearly" → use rate/2, periods × 2	CI	Always halve rate, double periods
(n-1) instead of (n-2) for polygon angles	Geometry	Sum = (n - 2) × 180

PART G — SPEED REFERENCE CARD (1-page exam day)

Squares 1-30 (memorise):

1,4,9,16,25,36,49,64,81,100,121,144,169,196,225,256,289,324,361,400,441,484,529,576,625,676,729,784,841,900.

Cubes 1-15: 1,8,27,64,125,216,343,512,729,1000,1331,1728,2197,2744,3375.

Fraction = % conversions: $\frac{1}{2}=50$, $\frac{1}{3}=33.3$, $\frac{1}{4}=25$, $\frac{1}{5}=20$, $\frac{1}{6}=16.6$, $\frac{1}{7}=14.28$, $\frac{1}{8}=12.5$, $\frac{1}{9}=11.1$, $\frac{1}{10}=10$, $\frac{1}{11}=9.09$,
 $\frac{1}{12}=8.33$, $\frac{1}{13}=7.69$, $\frac{1}{14}=7.14$, $\frac{1}{15}=6.66$, $\frac{1}{16}=6.25$, $\frac{1}{20}=5$, $\frac{1}{25}=4$.

Roots: $\sqrt{2}=1.414$, $\sqrt{3}=1.732$, $\sqrt{5}=2.236$, $\sqrt{7}=2.645$, $\sqrt{10}=3.162$.

Pythagorean triples to spot in geometry: 3-4-5, 5-12-13, 7-24-25, 8-15-17, 9-40-41, 11-60-61, 12-35-37, 20-21-29, 28-45-53.

Algebra identities (instant recall):

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$a + 1/a = k \rightarrow a^2 + 1/a^2 = k^2 - 2; a^3 + 1/a^3 = k^3 - 3k$$

$$\text{if } a+b+c = 0 \rightarrow a^3+b^3+c^3 = 3abc$$

Trig values cold:

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$1/\sqrt{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$1/\sqrt{2}$	$\frac{1}{2}$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

This DRILL PACK is the difference between "I read the book" and "I CAN SOLVE the paper". Drill all 50 examples + the mini-mock at least twice. Time yourself. Mark trap-options before checking the answer key.

PART H — VEDIC MATH & SPEED TRICKS (the 95%+ edge)

Each of the 16 Vedic sutras + the speed-multiplication shortcuts below saves 5-10 seconds per question. In a 25-Q section, that's ~3-4 minutes saved — enough to attempt 5 more questions.

H.1 Multiplication shortcuts

Multiplication of any 2-digit number by 11

Rule. Place the SUM of the two digits between them.

$25 \times 11 = 2_5$ with $(2+5)=7$ in middle = **275**.

47×11 : 4_7 with $(4+7)=11 \rightarrow$ carry the 1: $4+1=5$, leaves 1 in middle, 7 at end $\rightarrow$ **517**.

89×11 : 8_9 with $(8+9)=17 \rightarrow 8+1=9$, 7 in middle, 9 at end $\rightarrow$ **979**.

Multiplication by 12, 13, ..., 19 ("Vertical & Crosswise" sutra)

Multiplication by 13: take number $\times$ 1 ten and add ($\times$ 3 units). 24×13 : $(24 \times 10) + (24 \times 3) = 240 + 72 =$ **312**.

Or: write $2_4 \rightarrow 2 \times 3 = 6$ add to next digit place.

Multiplication of two numbers near 100 (Nikhilam sutra)

If both close to 100 (or 1000), use base-100 deviation method.

96×97 : deviations from 100 are -4 and -3 . Cross-add: $96 + (-3) = 93$ OR $97 + (-4) = 93 \rightarrow$ first part. Multiply deviations: $(-4)(-3) = 12 \rightarrow$ second part. Answer = **9312**.

98×96 : deviations $-2, -4$. Cross: $98 - 4 = 94$. Mult: 8. Answer = **9408**.

103×104 : deviations $+3, +4$. Cross: $103 + 4 = 107$. Mult: 12. Answer = **10712** (since base 100 needs 2-digit padding, 12 stays as is).

Squaring numbers ending in 5

Rule. Take leading digit $\times$ (leading digit + 1), then append "25".

$25^2 = 2 \times 3 = 6 \rightarrow$ append 25 $\rightarrow$ **625**.

$35^2 = 3 \times 4 = 12 \rightarrow$ **1225**.

$75^2 = 7 \times 8 = 56 \rightarrow$ **5625**.

$105^2 = 10 \times 11 = 110 \rightarrow$ **11025**.

Squaring numbers near 50

For numbers $50 \pm k$:

$(50 + k)^2 = (25 + k)$ followed by k^2 .

$(50 - k)^2 = (25 - k)$ followed by k^2 .

$53^2 = (25 + 3)$ followed by $3^2 = 28_09 =$ **2809**.

$47^2 = (25 - 3)$ followed by $3^2 = 22_09 =$ **2209**.

$58^2 = 28 + 8 = 33$, then $8^2 = 64 \rightarrow$ **3364**.

$42^2 = 25 - 8 = 17$, then $8^2 = 64 \rightarrow$ **1764**.

Multiplication of numbers near 50 (each)

48×47 : $47 - 2 = 45 \rightarrow$ first part is $25 + 45 =$ wait, simpler: each is $50 - k$. Use base 50.

Use Nikhilam with base 50: deviations from 50 are -2, -3. Cross: $48 + (-3) = 45$. Multiply: 6. Multiply first part by $\frac{50}{100} = \frac{1}{2}$. Answer = $45 \times 50 + 6 = 2256$.

Or just memorise standard products.

Multiplication by 9, 99, 999

Rule. $N \times 9 = N0 - N$. $N \times 99 = N00 - N$. $N \times 999 = N000 - N$.

$$24 \times 9 = 240 - 24 = \mathbf{216}.$$

$$37 \times 99 = 3700 - 37 = \mathbf{3663}.$$

$$48 \times 999 = 48000 - 48 = \mathbf{47952}.$$

Multiplication by 25, 50, 125, 250

$$\times 25: (\times 100) \div 4. \text{ So } 36 \times 25 = \frac{3600}{4} = 900.$$

$$\times 50: (\times 100) \div 2. \text{ So } 84 \times 50 = \frac{8400}{2} = 4200.$$

$$\times 125: (\times 1000) \div 8. \text{ So } 56 \times 125 = \frac{56000}{8} = 7000.$$

H.2 Squaring + cubing tricks

Square of any number close to 10^n

$$(10 + k)^2 = 100 + 20k + k^2. \text{ For } k=5: 225. \text{ For } k=7: 100 + 140 + 49 = 289.$$

Square of 11, 12, ... 19 (memorise table)

$$11^2=121, 12^2=144, 13^2=169, 14^2=196, 15^2=225, 16^2=256, 17^2=289, 18^2=324, 19^2=361.$$

Square of 21-29

$$21^2=441, 22^2=484, 23^2=529, 24^2=576, 25^2=625, 26^2=676, 27^2=729, 28^2=784, 29^2=841.$$

Cubing close to $10n$

$$(10n + k)^3 = 1000n^3 + 300n^2k + 30nk^2 + k^3. \text{ Useful only for very small } k.$$

H.3 Division shortcuts

Division by 5

Multiply numerator $\times 2$, divide by 10.

$$\frac{245}{5} = (245 \times 2)/10 = \frac{490}{10} = \mathbf{49}.$$

$$\frac{1234}{5} = (1234 \times 2)/10 = \frac{2468}{10} = \mathbf{246.8}.$$

Division by 25, 125

$$/25 = (\times 4)/100.$$

$$/125 = (\times 8)/1000.$$

Division by 9 (digit-sum check)

If digit sum is divisible by 9, the number is. Quotient by repeated subtraction.

H.4 Percentage shortcuts (instant)

Shortcut	Application
10%	move decimal one place left

Shortcut	Application
1%	move decimal two places left
5%	half of 10%
25%	divide by 4
12.5%	divide by 8
33.33%	divide by 3
16.66%	divide by 6
6.25%	divide by 16
Successive %	$a + b + ab/100$
% more vs % less	$r/(100+r) \times 100$ OR $r/(100-r) \times 100$
x% of y = y% of x	symmetry

H.5 Quick-recall fraction-to-percent (already covered in Part 0; repeat to lock in)

Fraction	%
$\frac{1}{2}$	50
$\frac{1}{3}$	33.33
$\frac{1}{4}$	25
$\frac{1}{5}$	20
$\frac{1}{6}$	16.66
$\frac{1}{7}$	14.28
$\frac{1}{8}$	12.5
$\frac{1}{9}$	11.11
$\frac{1}{10}$	10
$\frac{1}{11}$	9.09
$\frac{1}{12}$	8.33
$\frac{2}{3}$	66.66
$\frac{3}{4}$	75
$\frac{3}{5}$	60
$\frac{2}{7}$	28.57
$\frac{5}{8}$	62.5
$\frac{3}{8}$	37.5
$\frac{7}{8}$	87.5

Fraction	%
$\frac{5}{9}$	55.55
$\frac{7}{9}$	77.77

H.6 Time-saving sutras (Vedic — 16 main + 13 sub)

Sutra	English	Use
Ekadhikena Purvena	"By one more than the previous"	Squaring numbers ending in 5; recurring decimals
Nikhilam Navatashcaramam Dashatah	"All from 9 and last from 10"	Subtraction from 10n; multiplication near base
Urdhva - Tiryagbyham	"Vertically and crosswise"	General multiplication
Paraavartya Yojayet	"Transpose and adjust"	Division of polynomials
Sunyam Samyasamuccaye	"When sum is same that sum is zero"	Equation solving
Anurupye Sunyamanyat	"If one is in ratio, the other is zero"	Specific equation pattern
Sankalana - Vyavakalanabhyam	"By addition and by subtraction"	Simultaneous equations
Yavadunam	"Whatever the deficiency"	Squaring numbers below base
Vyastisamastih	"Specific and general"	Series + sequences
Sesanyankena Caramena	"The remainders by the last digit"	Recurring decimals

Master these; you don't need a 200-page Vedic-math book. Just these few patterns cover 80% of speed-up opportunity.

PART I — MENSURATION MASTER FORMULA SHEET (2D + 3D)

2D shapes

Shape	Area	Perimeter / Boundary
Square (side a)	a^2	$4a$; diagonal $a\sqrt{2}$
Rectangle (l × b)	$l \times b$	$2(l + b)$; diagonal $\sqrt{l^2 + b^2}$
Parallelogram (b, h)	$b \times h$	$2(a + b)$
Rhombus (d_1, d_2)	$\frac{1}{2} \times d_1 \times d_2$	$4 \times \text{side}$; side = $\frac{1}{2} \times \sqrt{(d_1^2 + d_2^2)}$
Triangle (b, h)	$\frac{1}{2} \times b \times h$	$a + b + c$
Equilateral triangle (a)	$(\sqrt{\frac{3}{4}}) \times a^2$; height = $(\sqrt{\frac{3}{2}}) \times a$	$3a$
Right triangle	$\frac{1}{2} \times \text{base} \times \text{perpendicular}$	$a + b + \sqrt{(a^2 + b^2)}$
Heron's formula	$\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$	—
Trapezium	$\frac{1}{2} \times (a + b) \times h$	sum of all sides
Circle (r)	$\pi \times r^2$	$2\pi \times r$ (circumference)
Sector (r, θ°)	$(\theta/360) \times \pi \times r^2$; or $\frac{1}{2}r^2\theta_{\text{rad}}$ (θ in radians)	arc length = $(\theta/360) \times 2\pi r = r\theta_{\text{rad}}$
Segment of circle (r, θ in radians)	$\frac{r^2}{2}(\theta - \sin \theta)$ — area between chord and arc	chord length = $2r \sin(\theta/2)$
Annulus (R, r)	$\pi(R^2 - r^2)$	$2\pi(R + r)$
Regular polygon (n sides, side a)	$(\frac{1}{4}) \times n \times a^2 \times \cot(180^\circ/n)$	$n \times a$
Regular hexagon (a)	$(3\sqrt{\frac{3}{2}}) \times a^2$	$6a$

3D shapes — Volume + Surface

Shape	Volume	Total Surface Area	Curved/Lateral SA
Cube (a)	a^3	$6a^2$	$4a^2$
Cuboid (l, b, h)	$l \times b \times h$	$2(lb + bh + hl)$	$2h(l + b)$
Cylinder (r, h)	$\pi r^2 h$	$2\pi r(r + h)$	$2\pi rh$
Hollow cylinder (R, r, h)	$\pi(R^2 - r^2)h$	$2\pi(R + r)(R - r + h)$	$2\pi h(R + r)$
Cone (r, h, l)	$(\frac{1}{3}) \pi r^2 h$; slant $l = \sqrt{(r^2 + h^2)}$	$\pi r(r + l)$	$\pi r l$
Sphere (r)	$(\frac{4}{3}) \pi r^3$	$4\pi r^2$	—
Hemisphere (r)	$(\frac{2}{3}) \pi r^3$	$3\pi r^2$	$2\pi r^2$
Spherical shell (R, r)	$(\frac{4}{3})\pi(R^3 - r^3)$	—	—
Frustum of cone (R, r, h, l)	$(\frac{1}{3}) \pi h (R^2 + r^2 + Rr)$; $l = \sqrt{(h^2 + (R-r)^2)}$	$\pi(R^2 + r^2 + l(R+r))$	$\pi l (R + r)$
Pyramid (square base a, h, slant l)	$(\frac{1}{3}) a^2 h$	$a^2 + 2a \times l$	$2a \times l$
Tetrahedron (a)	$(a^3 \times \sqrt{2})/12$	$\sqrt{3} \times a^2$	—
Prism (general)	base area × height	$2 \times \text{base} + \text{perimeter} \times h$	perimeter × h

Key memory pegs

Cube is the cuboid with $l = b = h = a$.

Sphere = $4\pi r^2$ (surface) and $(\frac{4}{3})\pi r^3$ (volume) — memorise EXACTLY.

Cone slant length $l = \sqrt{r^2 + h^2}$ — Pythagoras.

Hemisphere TSA = $3\pi r^2$ (curved + base disc) — common trap (students answer $2\pi r^2$).

Cuboid diagonal = $\sqrt{l^2 + b^2 + h^2}$ — extension of 2D.

Cylinder volume to area ratio: $V/A \rightarrow r/2$ (per unit height).

Special-case quick computation

Setup	Formula
Largest cube cut from cube of side a	a (same)
Largest sphere from cube of side a	$(\frac{4}{3})\pi(a/2)^3$
Largest cylinder from cube (axis through cube)	$\pi r^2 h$ with $r = a/2$, $h = a$
Largest cone from cube	$(\frac{1}{3})\pi(a/2)^2(a)$
Largest cube inscribed in sphere of r	side = $(2r)/\sqrt{3}$
Diagonal of square inscribed in circle r	$2r$

PART J — 1-PAGE EXAM-DAY ARITHMETIC CHEATSHEET (revise day before)

Mental-math reflexes:

Squares 1-30 cold. Cubes 1-15 cold.

Tables 11-20 cold.

Fraction-percent table memorised ($\frac{1}{2} = 50$, $\frac{1}{8} = 12.5$, etc.).

$\pi \approx \frac{22}{7}$; $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$.

Universal shortcuts:

10% trick $\rightarrow$ decimal one place left.

25% $\rightarrow \div 4$. 125 $\rightarrow \div 8$. 5% $\rightarrow$ half of 10%.

Percent symmetry: $x\%$ of $y = y\%$ of x .

Successive %: $a + b + ab/100$.

" $r\%$ more" $\rightarrow$ " $r/(100+r)\%$ less".

Speed multiplication:

$\times 11$: place sum of digits between.

$\times 9, 99, 999$: subtract from $N0, N00, N000$.

$\times 25$: $\times 100 \div 4$; $\times 125$: $\times 1000 \div 8$.

Squares of 11-19: 121, 144, 169, 196, 225, 256, 289, 324, 361.

Squares ending in 5: lead $\times$ (lead+1), append 25.

Time-Speed-Distance:

km/h to m/s: $\times \frac{5}{18}$.

Average speed (equal distance): $2ab/(a+b)$.

Boat: down = $b+s$, up = $b-s$.

Profit & Loss:

Equal % gain & loss on same SP $\rightarrow$ always loss = $(r/10)^2$.

MP-CP-SP: Profit% = $(SP-CP)/CP \times 100$. Discount% = $(MP-SP)/MP \times 100$.

SI / CI:

Doubles in n yrs at SI $\rightarrow$ rate = $100/n$ %.

CI - SI for 2 yrs = $P(r/100)^2$.

Trigonometry standard angles:

$\sin(0, 30, 45, 60, 90) = 0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$.

cos = reverse.

Geometry:

Sum interior angles polygon = $(n-2) \times 180^\circ$.

Equilateral triangle area = $(\frac{\sqrt{3}}{4})a^2$.

Heron's: $\sqrt{s(s-a)(s-b)(s-c)}$.

Mensuration:

Sphere $V = (\frac{4}{3})\pi r^3$, $SA = 4\pi r^2$.

Cone $V = (\frac{1}{3})\pi r^2 h$, slant = $\sqrt{r^2 + h^2}$.

Hemisphere TSA = $3\pi r^2$ (NOT $2\pi r^2$).

Algebra:

$a + 1/a = k \rightarrow a^2 + 1/a^2 = k^2 - 2$; $a^3 + 1/a^3 = k^3 - 3k$.

$(a + b + c) = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$.

Print this page and revise the morning before the exam. THIS is the 95%+ ammunition.

PART K — 30-DAY SSC / RRB ARITHMETIC PLAN (1–1.5 hrs/day)

SSC (CGL, CHSL, MTS, GD) weights Percentage/Profit-Loss the heaviest, with significant Geometry and Mensuration. Follow this order — high-return topics first.

Days	Topics	Focus / key shortcut
1–2	Mental-math arsenal (PART 0)	Fraction $\leftrightarrow$ % table, squares 1–30, cubes 1–15, tables 11–20 — memorise cold
3–5	Percentage (PART 1) — all 15 types	Master reverse-% formula: $r / (100+r) \times 100$
6–7	Profit, Loss & Discount (PART 2)	False-weight + successive-discount chains
8–9	Simple & Compound Interest (PART 3)	CI – SI formula + half-yearly variants
10–11	Ratio, Proportion, Variation (PART 4)	Alligation shortcut handles any mixing problem
12–13	Time, Work & Pipes (PART 5)	Unit-fraction method: 1 day's work = $1/n$
14–15	Speed, Distance & Time — Trains, Boats (PART 6)	Relative speed + harmonic mean for equal-distance trips
16–17	Averages + Mixtures (PART 7)	Weighted average = sum of (value $\times$ weight) / sum of weights
18–19	Number System: HCF, LCM, Divisibility, Remainders (PART 8)	Divisibility rules 2–13 and remainder theorems
20–21	Algebra: identities, quadratic, surds, indices (PART 14)	$(a+b+c)=0 \rightarrow a^3+b^3+c^3=3abc$
22–24	Geometry + Mensuration 2D (PART 11)	Triangle + circle + quadrilateral area/perimeter
25–26	Mensuration 3D: Cylinder, Cone, Sphere	SA vs V pairs — hemisphere TSA = $3\pi r^2$
27	Trigonometry + Heights & Distances	sin/cos/tan for 30°/45°/60° cold; elevation vs depression
28	Partnership (PART 9) + Permutation, Probability (PART 14)	
29	Timed Mini-Mock (PART E) + Trap Cards (PART F)	Target: 22+/25 in 25 min
30	Return to weakest 2 topics; drill 10 questions per topic	Use the Practice Set (50 worked examples)

KEY FACT

Spaced-review rule: whatever you studied on Days 3–5, review once more on Day 15. Whatever you studied on Days 10–15, review on Day 25. Three exposures separated by a gap beats one long reading session.

PART L — 30-DAY BANKS ARITHMETIC PLAN (1–2 hrs/day)

Banks (IBPS PO/Clerk, SBI PO/Clerk, RBI) weight is completely different from SSC. Data Interpretation alone carries 15–20 questions in prelims. Quadratic Equations and Number Series are 5-question fixed blocks. Start with these — they give the fastest return.

Days	Topics	Banks weight
1–3	DI — Table format: read, compute, compare	5–6 Qs in every pre paper
4–6	DI — Bar chart + Line graph	5–6 Qs
7–9	DI — Pie chart + Mixed DI (two datasets combined)	5–6 Qs in mains
10–11	Number Series: missing term + wrong term	5 Qs fixed in most pre papers
12–13	Quadratic Equations: solve and compare Qty I vs Qty II	5 Qs fixed in most pre papers
14–15	Approximation: round numbers, estimate %	5 Qs fixed in many pre papers
16–17	Percentage + Profit-Loss + Discount	3–4 Qs per paper
18–19	SI/CI + Ratio + Partnership	3–4 Qs
20–21	Time-Work-Pipes + Speed-Distance-Time (incl. boats)	3–5 Qs
22–23	Average + Mixture-Alligation	2–3 Qs
24–25	Probability + Permutation-Combination	1–2 Qs
26–27	Advanced DI: caselet DI + arithmetic DI (Mains patterns)	Mains-level; 5–10 Qs in Mains
28	Quantity comparison (Qty I vs II) — miscellaneous types	
29	Timed Mini-Mock (PART E) + Trap Cards (PART F)	
30	Return to weakest DI type + weakest arithmetic type	2 hrs focused drill

KEY FACT

If you only have 15 days: combine each 2-day block into one longer session. Cut Days 24–25 (Probability) and skip Day 28. Do timed DI practice every day from Day 12 onward — DI is a skill that needs daily practice to build speed.